



Data Sheet

NT35510

One-chip Driver IC with internal GRAM
for 16.7M colors 480RGB x 864 a-Si TFT LCD
with CPU / RGB / MIPI / MDDI Interface
or without internal CGRAM
for 16.7M colors 480RGB x 1024 a-Si TFT LCD
with RGB Interface

V0.03

Preliminary

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REVISION HISTORY

Version	Contents	Prepared by	Checked by	Approved by	Date
0.00	Original	Kevin	SW	Dennis	2010/02/12
0.01	- Page 9, remove 320RGB x 480 - Page 10, Features, remove 320RGB x 480 and MUX description VGHO VGLO for gate control signals, remove VDDIM/VSSIM - Page11, update power voltage range - Page12, Block diagram - Page 13 to 22 : - Add : VDD_DET, DIOPWR, PSWAP, DSWAP, VGHO, VGLO, VRGH, VREFCP, CSP, CSN, LVGL, C61P, C61N, VRGH, VREF, GOUT, - Remove : VDDIM/VSSIM, VDDEL - Update : MVDDL, VGL, VGH, Test pins - Page23, update IF table - Page 51 to 66 : update SPI, IM3= 1 setting in figure - Page102,103, change DSIM, DSIG bit Reg to 0xB100 - Page115,124 Add WRPFD 50h on table - Page201, modified to 480x864 memory - Page202, Remove 320x480 - Page204, update whole Frame memory table - Page205, TE map to 480 lines, DOPCTR change to B100h - Page207, tvdI TBD - Page225,226, update VDD in figure - Page227, Modes to 7 - Page232, Sout update to Gout - Page235, Add chip attachment Detection section - Page237, update Gamma Structure - Page255,270, update FOSC, Example - Page266, update KB_CLED - Page272, Add inversion section - Page273,274, Power Architecture - Page275, update DIOPWR, VREFCP, VGMP1, VGLO - Page276, update C61P/N, LVGL, VGLO, VRGH, VREFCP, DIOPWR, VGMP1/2, VGMN, VGSP, VGSN, - Page291, change name to RAMKP - Page306 to 312, remove 320x 480 resolution setting - Page337, 5400h Cmd add A and G bit - Page385, Absolute Max Rating for MV HV, remove VDDIM - Page386, VDDIM remove - Page387, Vdev value modified - Page402,403, Remove MVDDI in note - Page406, Remove 320 x 480, update 360x640 Sout sequence - Page173 to 181, MDDI windowless packet - Page377,379, A1,A8 cmd update - Page387 to 396, VDDI to 3.3V - Page362 to 376 70h to 7Eh cmd default value - Page28,29,30,40,41,42 MPU figure update - Page 12,274 Block and power architecture update	Kevin	SW	Dennis	2010/03/17

0.02	- Page 10,remove 360RGB x 640, Add 480RGBx720 - Page 11, update GPO[3:0] - Page 12, update VGHO, VGLO - Page 13, update Block Diagram - Page 18, update IM, GPO, VSEL, and EXB1T - Page 20, update VGLO,LVGL - Page 21, update VGLX,VGL_REG, Remove CP6_P/N - Page 23, update VDD_BC - Page 24, update CONTACT1~4, VSSIDUM - Page 25, update IF description table - Page 207, update Address Counter - Page 235, update Resolution Data - Page 252, remove CLED_VOL - Page 271, remove KB_CLED_VOL - Page 277, add 4 dot inversion - Page 306,308,313,326 resolution update, remove nHD, add 480x720 - Page 384, update absolute voltage - Page 385, update DC spec - Page 386, update Note3,Note5 - Page 405, update resolution - Page 406, update Alignment Mark	Kevin	SW	Dennis	2010/04/06
0.03	- Page 10,11,205,206,234,305,307,312,325,404, update resolution - Page 13, update Block diagram - Page 17-24, update pin description(MDDI not support DSWAP, Update TE_R,TE_L, DSTB_SEL, RESX,VSEL,VREF_PWR, I2C_SDA remove VDD_BD, ENDIOV) - Page 104,121, remove generic data type 0x24 - Page 134, update EoTP Option - Page 175, update MDDI support type - Page 176,177, update sub frame header, link shut down packet - Page 179,180, update skew calibration packet, client capability packet - Page 184, update packet type is 20 - Page 209,214, update TE off,output is low, tering effect bus trigger - Page 241, update gamma to 10 bits setting - Page 276, update 3-dots inversion - Page 384, update VIH,VIL,VOH,VOL - Page 388, update hibernation wake up - Page 390,392 update Note2 - Remove pad chapter to application note	Kevin	SW	Dennis	2010/05/18

1 DESCRIPTION

1.1 Purpose of this Document

This document has been created to provide complete reference specifications for the NT35510. IC design engineers should refer to these specifications when designing ICs, test engineers when testing the compliance of manufactured ICs to guarantee their performance, and application engineers when helping customers to make sure they are using this IC properly.

1.2 General Description

The NT35510 device is a single-chip solution for a-Si TFT LCD that incorporates gate drivers and is capable of 480RGBx864, 480RGBx854, 480RGBx800, 480RGBx720, 480RGBx640, 480RGBx360 with internal CGRAM and 480RGB x 1024 by pass internal CGRAM. It includes a 9,953,280 bits internal memory, a timing controller with glass interface level-shifters and a glass power supply circuit..

The NT35510 supports MDDI interface, MIPI Interface, 16/18/24 bits RGB interface, 8/16/24-bit system interfaces, serial peripheral interfaces (SPI) and I2C interface. The specified window area can be updated selectively, so that moving pictures can be displayed simultaneously independent of the still picture area. The 480RGB x 1024 by pass CGRAM application is used for RGB interface only.

The NT35510 is also able to make gamma correction settings separately for RGB dots to allow benign adjustments to panel characteristics, resulting in higher display qualities. The IC possesses internal GRAM that stores 480-RGB x 864-dot 16.77M-color images. A deep standby mode is also supported for lower power consumption.

This LSI is suitable for small or medium-sized portable mobile solutions requiring long-term driving capabilities, including bi-directional pagers, digital audio players, cellular phones and handheld PDA..

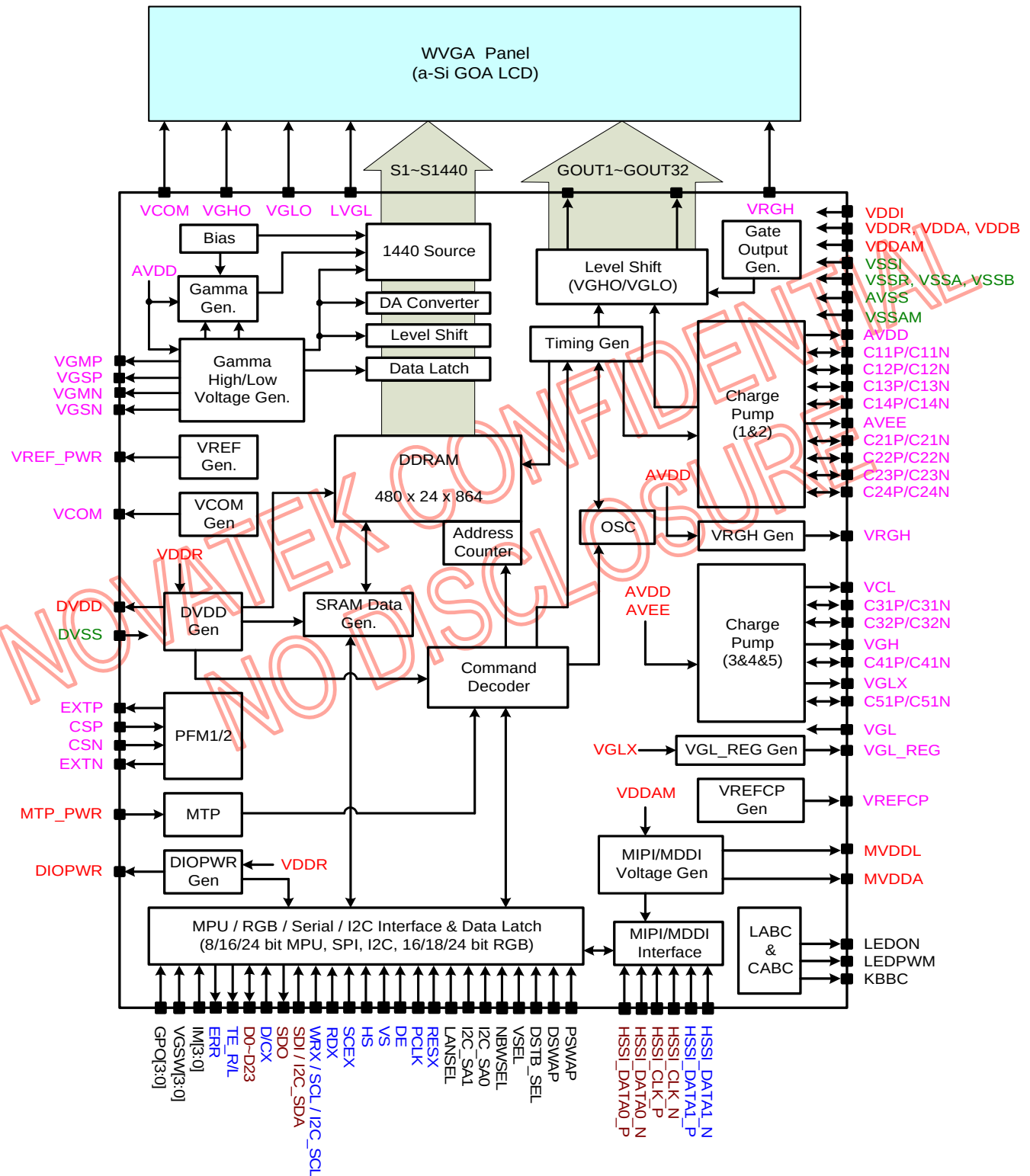
2 FEATURES

- ◆ Single chip WVGA a-Si TFT LCD Controller/driver with Display RAM.
- ◆ Display resolution option
 - 480RGB x 1024 by pass GRAM
 - 480RGB x 864 with 480x24-bitsx 864 GRAM
 - 480RGB x 854 with 480x24-bitsx 854 GRAM
 - 480RGB x 800 with 480x24-bitsx 800 GRAM
 - 480RGB x 720 with 480x24-bitsx 720 GRAM
 - 480RGB x 640 with 480x24-bitsx 640 GRAM
 - 480RGB x 360 with 480x24-bitsx 360 GRAM
- ◆ Display data RAM (frame memory): 480 x 864 x 24-bits = 9,953,280 bits
- ◆ Display mode (Color mode)
 - Full color mode: 16.7M-colors
 - Reduce color mode: 262K colors
 - Reduce color mode: 65K colors
 - Idle mode: 8-colors
- ◆ Interface
 - 8-/16-/24-bits 80-series MPU interface
 - 16-bit serial peripheral interface
 - I2C interface
 - 16-/18-/24-bits RGB interface (DE mode and SYNC mode with polarity of HS/VS can be set by register)
 - MIPI Display Serial Interface (DSI V1.01 r11 and D-PHY V1.0, 1 clock and 1 or 2 data lane pairs)
 - Mobile Display Digital Interface (MDDI V1.2, 1 strobe and 1 or 2 data lane pairs)
- ◆ Display features
 - Window address functions for specifying a rectangular area on the internal RAM to write data
 - Individual gamma correction setting for RGB dots
 - Deep standby function
- ◆ On chip
 - VGHO/VGLO voltage generator for gate control signal and panel
 - Oscillator for display clock
 - Supports gate control signals to gate driver in the panel
 - On module color characteristics
 - On module checksums checking
 - Four GPO (General Purpose Output) pins for external control
- ◆ Supply voltage range
 - I/O supply voltage range for VDDI to VSSI: 1.65V ~ 3.3V (VDDI) or 1.1 ~ 1.3V (VDDIL)
 - Analog supply voltage range for VDDDB/VDDA/VDDR to VSSB/VSSA/VSSR: 2.3V ~ 4.8V
 - MIPI/MDDI regulator supply voltage range for VDDAM to VSSAM: 2.3V ~ 4.8V

◆ Output voltage levels

- Positive gate driver voltage range for VGH: $AVDD + VDDB \sim 2 \times AVDD - AVEE$
- Negative gate driver voltage range for VGLX: $AVEE + VCL \sim 2 \times AVEE - AVDD$
- Step-up 1 output voltage range for AVDD: 4.5 ~ 6.5V
- Step-up 2 output voltage range for AVEE: -4.5 ~ -6.5V
- Positive gamma high voltage range for VGMP: 3.0 ~ 6.3V ($AVDD - 0.3V$)
- Positive gamma low voltage range for VGSP: 0.0, 0.3 ~ 3.7V
- Negative gamma high voltage range for VGMPN: -3.0 ~ -6.3V ($AVEE + 0.3V$)
- Negative gamma low voltage range for VGSPN: 0.0, -0.3 ~ -3.7V
- Common electrode voltage range for VCOM: 0.0 ~ -3.5V ($VCL + 0.3V$)
- Panel voltage range for VRGH: 1.0V ~ 6.0V ($AVDD - 0.3V$)

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NO DISCLOSURE

3 BLOCK DIAGRAM


4 PIN DESCRIPTION

4.1 Power Supply Pins

Symbol	Name	Description
Vddb	DC/DC Power	Power supply for DC/DC converter Vddb, Vdda and Vddr should be the same input voltage level
Vdda	Analog Power	Power supply for analog system Vddb, Vdda and Vddr should be the same input voltage level
Vddr	Regulator Power	Power supply for regulator system Vddb, Vdda and Vddr should be the same input voltage level
VDD_DET	Detection Power	Connect to Vddb/Vdda/Vddr for detection.
VDDAM	MIPI Power	Power supply for MIPI/MDDI analog regulator system
VDDI	I/O Power	Power supply for interface system except MIPI/MDDI interface
DVDD	Digital Voltage	Regulator output for logic system power (1.5V typical) Connect a capacitor for stabilization.
DIOPWR	Dual I/O Voltage	Regulator output for dual I/O voltage system (1.2V/1.8V typical). Connect a capacitor for stabilization.
MVDDA	MIPI/MDDI Voltage	Regulator output for internal MIPI/MDDI analog system (1.5V typical) Connect a capacitor for stabilization. If not use MIPI/MDDI interface, please open this pin.
MVDDL	MIPI Voltage	Regulator output for internal MIPI low power system (1.2V typical) Connect a capacitor for stabilization. If not use MIPI interface, please open this pin
VSSB	DC/DC GND	System ground for DC/DC converter
VSSA	Analog GND	System ground for analog system
VSSR	Regulator GND	System ground for regulator system
VSSAM	MIPI GND	System ground for internal MIPI/MDDI analog system
VSSI	I/O GND	System ground for interface system except MIPI/MDDI interface
DVSS	Digital GND	System ground for internal digital system
AVSS	Source OP GND	System ground for source OP system.
MTP_PWR	MTP Power	MTP programming power supply pin (7.4 to 7.6V and 7.5V typical) Must be left open or connected to DVSS in normal condition.

4.2 80-System Interface Pins

Symbol	I/O	Description
CSX	I	Chip select input pin ("Low" enable) in 80-series MPU I/F and SPI I/F. This pin is not used for MIPI or MDDI I/F, please connect to VSSI this pin.
WRX / SCL / I2C_SCL	I	Writes strobe signal to write data when WRX is "Low" in 80-series MPU I/F. Serial interface clock in SPI and I2C I/F. This pin is not used for MIPI I/F, please connect to VSSI this pin.
RDX	I	Reads strobe signal to write data when RDX is "Low" in 80-series MPU interface. This pin is not used for RGB+SPI, MIPI or MDDI I/F, please connect to VSSI this pin.
D/CX	I	Display data / command selection in 80-series MPU I/F. D/CX = "0" : Command D/CX = "1" : Display data or Parameter This pin is not used for RGB+3-SPI, MIPI or MDDI I/F, please connect to VSSI this pin.
D[23:0]	I/O	24-bit bi-directional data bus for 80-series MPU I/F and 24-bit input data bus for RGB I/F. For 8080-series MPU I/F: 8-bit interface: D[7:0] are used, D[23:8] should be connected to VSSI 16-bit interface: D[15:0] are used, D[23:16] should be connected to VSSI 24-bit interface: D[23:0] are used These pins are not used for MIPI or MDDI I/F, please connect to VSSI these pins.

NOTE: "1" = VDDI level, "0" = VSSI level.

4.3 SPI /I2C Interface Pins

Symbol	I/O	Description
CSX	I	Chip select input pin ("Low" enable) in 80-series MPU I/F and SPI I/F. This pin is not used for MIPI or MDDI I/F, please connect to VSSI this pin.
WRX / SCL / I2C_SCL	I	Writes strobe signal to write data when WRX is "Low" in 80-series MPU I/F. Serial interface clock in SPI I/F. This pin is not used for MIPI I/F, please connect to VSSI this pin.
SDI / I2C_SDA	I/O	Serial input signal in SPI I/F. The data is input on the rising/falling edge of the SCL signal. Serial input/output signal in I2C I/F. The data is input/output on the rising edge of the SCL signal. This pin is not used for 80-series MPU or MIPI I/F, please connect to VSSI this pin.
SDO	O	Serial output signal in SPI I/F. The data is output on the rising/falling edge of the SCL signal. If the host places the SDI line into high-impedance state during the read interval, the SDI and SDO can be tied together. This pin is not used for 80-series MPU, MIPI or MDDI I/F, please open this pin.

NOTE: "1" = VDDI level, "0" = VSSI level.

4.4 RGB Interface Pins

Symbol	I/O	Description
PCLK	I	Pixel clock signal in RGB I/F. This pin is not used for 80-series MPU, MIPI or MDDI I/F, please connect to VSSI this pin.
VS	I	Vertical sync. Signal in RGB I/F. This pin is not used for 80-series MPU, MIPI or MDDI I/F, please connect to VSSI this pin.
HS	I	Horizontal sync. Signal in RGB I/F. This pin is not used for 80-series MPU, MIPI or MDDI I/F, please connect to VSSI this pin.
DE	I	Data enable signal in RGB I/F mode 1. This pin is not used for RGB mode 2, 80-series MPU, MIPI or MDDI I/F, please connect to VSSI this pin.
D[23:0]	I/O	24-bit bi-directional data bus for 80-series MPU I/F and 24-bit input data bus for RGB I/F.. For RGB I/F: 16-bit/pixel: D[20:16]=R[4:0], D[13:8]=G[5:0] and D[4:0]=B[4:0], connect unused pins to VSSI 18-bit/pixel: D[21:16]=R[5:0], D[13:8]=G[5:0] and D[5:0]=B[5:0], connect unused pins to VSSI 24-bit/pixel: D[23:16]=R[7:0], D[15:8]=G[7:0] and D[7:0]=B[7:0] These pins are not used for MIPI or MDDI I/F, please connect to VSSI these pins.

NOTE: "1" = VDDI level, "0" = VSSI level.

4.5 MIPI/MDDI Interface Pins

Symbol	I/O	Description																																			
HSSI_CLK_P HSSI_CLK_N	I	-These pins are DSI-CLK+/- differential clock signals if MIPI interface is used. -These pins are MDDI_STB_P/M differential strobe signals if MDDI interface is used. -HSSI_CLK_P/N are differential small amplitude signals. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. -If not used, please connect these pins to VSSAM.																																			
HSSI_D0_P HSSI_D0_N	I/O	-These pins are DSI-D0+/- differential data signals if MIPI interface is used. -These pins are MDDI_DATA0_P/M differential strobe signals if MDDI interface is used. -HSSI_D0_P/N are differential small amplitude signals. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. -If not used, please connect these pins to VSSAM.																																			
HSSI_D1_P HSSI_D1_N	I	-These pins are DSI-D1+/- differential data signals if MIPI interface is used. -These pins are MDDI_DATA1_P/M differential strobe signals if MDDI interface is used. -HSSI_D1_P/N are differential small amplitude signals. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. -If not used, please connect these pins to VSSAM.																																			
ERR	O	CRC and ECC error output pin for MIPI interface. This pin is output low when it is not activated. When this pin is activated, it output high if CRC/ECC error found. If not used, please open this pin.																																			
LANSEL	I	Input pin to select 1 data lane or 2 data lanes in MIPI/MDDI interface. <table><tr><td>LANSEL</td><td>Data Lane of MIPI/MDDI</td></tr><tr><td>0</td><td>1 data lane</td></tr><tr><td>1</td><td>2 data lanes</td></tr></table> If not used, please connect to VSSI.	LANSEL	Data Lane of MIPI/MDDI	0	1 data lane	1	2 data lanes																													
LANSEL	Data Lane of MIPI/MDDI																																				
0	1 data lane																																				
1	2 data lanes																																				
DSWAP PSWAP	I	Input pin to select HSSI_D0/D1 data lane sequence and polarity in high speed interface only. For MIPI interface, both DSWAP and PSWAP function are available. For MDDI interface, only PSWAP function is available. i.e. DSWAP=1 in below table is invalid. <table><tr><td>Pin Name</td><td>HSSI_D0_P</td><td>HSSI_D0_N</td><td>HSSI_CLK_P</td><td>HSSI_CLK_N</td><td>HSSI_D1_P</td><td>HSSI_D1_N</td></tr><tr><td>DSWAP=0 PSWAP=0</td><td>DSI-D0+</td><td>DSI-D0-</td><td>DSI-CLK+</td><td>DSI-CLK-</td><td>DSI-D1+</td><td>DSI-D1-</td></tr><tr><td>DSWAP=0 PSWAP=1</td><td>DSI-D0-</td><td>DSI-D0+</td><td>DSI-CLK-</td><td>DSI-CLK+</td><td>DSI-D1-</td><td>DSI-D1+</td></tr><tr><td>DSWAP=1 PSWAP=0</td><td>DSI-D1+</td><td>DSI-D1-</td><td>DSI-CLK+</td><td>DSI-CLK-</td><td>DSI-D0+</td><td>DSI-D0-</td></tr><tr><td>DSWAP=1 PSWAP=1</td><td>DSI-D1-</td><td>DSI-D1+</td><td>DSI-CLK-</td><td>DSI-CLK+</td><td>DSI-D0-</td><td>DSI-D0+</td></tr></table> If not used, please connect to VSSI.	Pin Name	HSSI_D0_P	HSSI_D0_N	HSSI_CLK_P	HSSI_CLK_N	HSSI_D1_P	HSSI_D1_N	DSWAP=0 PSWAP=0	DSI-D0+	DSI-D0-	DSI-CLK+	DSI-CLK-	DSI-D1+	DSI-D1-	DSWAP=0 PSWAP=1	DSI-D0-	DSI-D0+	DSI-CLK-	DSI-CLK+	DSI-D1-	DSI-D1+	DSWAP=1 PSWAP=0	DSI-D1+	DSI-D1-	DSI-CLK+	DSI-CLK-	DSI-D0+	DSI-D0-	DSWAP=1 PSWAP=1	DSI-D1-	DSI-D1+	DSI-CLK-	DSI-CLK+	DSI-D0-	DSI-D0+
Pin Name	HSSI_D0_P	HSSI_D0_N	HSSI_CLK_P	HSSI_CLK_N	HSSI_D1_P	HSSI_D1_N																															
DSWAP=0 PSWAP=0	DSI-D0+	DSI-D0-	DSI-CLK+	DSI-CLK-	DSI-D1+	DSI-D1-																															
DSWAP=0 PSWAP=1	DSI-D0-	DSI-D0+	DSI-CLK-	DSI-CLK+	DSI-D1-	DSI-D1+																															
DSWAP=1 PSWAP=0	DSI-D1+	DSI-D1-	DSI-CLK+	DSI-CLK-	DSI-D0+	DSI-D0-																															
DSWAP=1 PSWAP=1	DSI-D1-	DSI-D1+	DSI-CLK-	DSI-CLK+	DSI-D0-	DSI-D0+																															

4.6 Interface Logic Pins

Symbol	I/O	Description						
RESX	I	This signal will reset the device and must be applied to properly initialize the chip. Signal is active low. The input voltage range for RESX pin is related to DSTB_SEL and VSEL pins.						
		Input Voltage Level (DSTB_SEL="0")	Min.	Max.	Unit			
		VDDI=1.65~3.3V	Logic High level input voltage	0.7xVDDI	VDDI	V		
			Logic Low level input voltage	VSSI	0.3xVDDI	V		
		VDDI=1.1~1.3V	Logic High level input voltage	0.88	1.35	V		
			Logic Low level input voltage	VSSI	0.55	V		
		Input Voltage Level (DSTB_SEL="1")	VDDI=1.65~3.3V		VDDIL=1.1~1.3V		Unit	
			Min.	Max.	Min.	Max.		
		VSEL =High	Logic High level input voltage	0.7xVDDI	VDDI	1.155	1.95	V
			Logic Low level input voltage	VSSI	0.3xVDDI	VSSI	0.585	V
VSEL =Low	Logic High level input voltage	0.88	1.35V	0.88	1.35V	V		
	Logic Low level input voltage	VSSI	0.55	VSSI	0.55	V		
TE (TE_L)	O	Tearing effect output pin to synchronize MCU to frame writing, activated by SW command. When this pin is not activated, this pin is output low. If not used, please open this pin.						
TE_R	O	Tearing effect output pin to synchronize MCU to frame writing, activated by SW command. The same output signal as TE (TE_L) pin. If not used, please open this pin.						
IM[2:0]	I	Interface type selection						
		IM[2:0]	Display Data	Command				
		000	80-series 8-bit MPU interface, D[7:0]	80-series 8-bit MPU interface, D[7:0]				
		001	80-series 16-bit MPU interface, D[15:0]	80-series 16-bit MPU interface, D[15:0]				
		010	80-series 24-bit MPU interface, D[23:0]	80-series 24-bit MPU interface, D[23:0]				
		011	RGB interface, D[23:0]	SPI, SDI/SDO serial data				
		100	RGB interface, D[23:0]	I2C interface, I2C_SDA serial data				
		101	MIPI DSI, HSSI_D0_P/N, HSSI_D1_P/N	MIPI DSI, HSSI_D0_P/N, HSSI_D1_P/N				
		110	MDDI, HSSI_D0_P/N, HSSI_D1_P/N	MDDI, HSSI_D0_P/N, HSSI_D1_P/N SPI, SDI/SDO serial data				
111	MDDI, HSSI_D0_P/N, HSSI_D1_P/N	MDDI, HSSI_D0_P/N, HSSI_D1_P/N I2C interface, I2C_SDA serial data						
IM3	I	Input pin to select the SCL rising/falling edge trigger for SPI I/F only. - IM3="0", SCL rising edge trigger - IM3="1", SCL falling edge trigger If not used, please connect to VSSI.						
I2C_SA[1:0]	I	I2C_SA[1:0]	Slave Address					
		00	10011 00					
		01	10011 01					
		10	10011 10					
		11	10011 11					

VSEL	I	Input pin to switch the I/O voltage. This VSEL function only apply for RESX, TE, LEDPWM, LEDON, KBBC pins. The VSEL dual IO function is valid when DSTB_SEL="1".					
		DSTB_SEL	VDDI	VSEL	DIOPWR	Output Voltage Level	
						TE	LEDON LEDPWM KBBC
		0	1.65~3.3V or 1.1~1.3V	X	Off	VOH=VDDI VOL=VSSI	VOH=VDDI or VDDA VOL=VSSI
		1	1.65~3.3V	Low	1.2V	VOH=1.2V VOL=VSSI	VOH=1.2V VOL=VSSI
				High	1.8V	VOH=VDDI or DIOPWR VOL=VSSI	VOH=VDDI or VDDA VOL=VSSI
		1	1.1~1.3V	Low	1.2V	VOH=1.2V VOL=VSSI	VOH=1.2V VOL=VSSI
				High	1.8V	VOH=1.8V VOL=VSSI	VOH=1.8V VOL=VSSI
		The input voltage range for VSEL pin:					
		Input Voltage Level		Min.	Max.	Unit	
Logic High level input voltage		0.88	VDDI	V			
Logic Low level input voltage		VSSI	0.55	V			
If not used, please connect to VDDI.							
GPO[3:0]	O	General purpose output pins. The output voltage swing is VDDI to VSSI. If not used, please open these pins.					
VGSW[3:0]	I	Input pin to select the different application.					
EXB1T	I	Input pin to select the external AVDD DC/DC voltage.					
		EXB1T	AVDD Voltage				
		0	Use internal DC/DC for AVDD				
		1	Use external DC/DC for AVDD				
If not used, please connect to VSSI.							
NBWSEL	I	Input pin to select the voltage sequence of V0 ~ V255.					
		NBWSEL	V0 ~ V255 voltage sequence				
		0	$V_{(00h)} > V_{(01h)} > \dots > V_{(FEh)} > V_{(FFh)}$ (Normally White)				
		1	$V_{(00h)} < V_{(01h)} < \dots < V_{(FEh)} < V_{(FFh)}$ (Normally Black)				
DSTB_SEL	I	Input pin to control DIOPWR regulator on/off.					
		DSTB_SEL	DIOPWR Regulator		VSEL Function		
		0	DIOPWR Off		Invalid		
		1	DIOPWR On		Valid		

NOTE: "1" = VDDI level, "0" = VSSI level.

4.7 Driver Output Pins

Symbol	I/O	Description
S1 ~ S1440	O	Pixel electrode driving output.
GOUT1 ~ GOUT32	O	Gate control signals for panel. The swing voltage level is VGHO to VGLO
SDUM0~3	O	Dummy Source, leave it Open if not used
VGHO	O	High voltage level for gate control signals and gate circuit of panel.
VGLO	O	Low voltage level for gate control signals and gate circuit of panel.
LVGL	O	Low voltage level for gate circuit of panel.
VCOM	O	Regulator output for common voltage of panel. Connect a capacitor for stabilization.

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4.8 DC/DC Converter Pins

Symbol	I/O	Description
AVDD	O	Output voltage from step-up circuit 1, generated from VDDB. Connect a capacitor for stabilization.
AVEE	O	Output voltage from step-up circuit 2, generated from VDDB. Connect a capacitor for stabilization.
VCL	O	Output voltage from step-up circuit 3, generated from VDDB. Connect a capacitor for stabilization.
VGH	O	Output voltage from step-up circuit 4. Connect a capacitor for stabilization.
VGLX	O	Output voltage from step-up circuit 5. Connect a capacitor for stabilization.
VGL	I	Substrate voltage for driver IC. Please connect VGL to VGLX.
C11P, C11N C12P, C12N C13P, C13N C14P, C14N	O	Capacitor connection pins for the step-up circuit which generate AVDD. Connect capacitor as requirement. When not in used, please open these pins.
C21P, C21N C22P, C22N C23P, C23N C24P, C24N	O	Capacitor connection pins for the step-up circuit which generate AVEE. Connect capacitor as requirement. When not in used, please open these pins.
C31P, C31N C32P, C32N	O	Capacitor connection pins for the step-up circuit which generate VCL. Connect capacitor as requirement.
C41P, C41N	O	Capacitor connection pins for the step-up circuit which generate VGH. Connect capacitor as requirement.
C51P, C51N	O	Capacitor connection pins for the step-up circuit which generate VGLX. Connect capacitor as requirement.
VRGH	O	Output voltage generated from AVDD. Connect a capacitor for stabilization. When not in use, please open this pin.
VGL_REG	O	Output voltage generated from VGLX. LDO output used for panel voltage. Connect a capacitor for stabilization. When not in use, please open this pin.
EXTP	O	PFM1 control output for DC/DC converter to generate AVDD. Connect to gate of external NMOS device. When not in use, please open this pin.
EXTN	O	PFM2 control output for DC/DC converter to generate AVEE. Connect to gate of external PMOS device. When not in use, please open this pin.
CSP	I	Current sensing input for PFM1 DC/DC converter (generate AVDD). When not in use, please connect to VSSB.
CSN	I	Current sensing input for PFM2 DC/DC converter (generate AVEE). When not in use, please connect to VSSB.
VREF_PWR	O	Regulator output for reference voltage. Connect a capacitor for stabilization.
VREFCP	O	Reference voltage for internal voltage generating circuit. Connect capacitor for stabilization.

Symbol	I/O	Description
VGMP	O	Output voltage generated from AVDD. LDO output for positive gamma high voltage generator.
VGSP	O	Output voltage generated from AVDD. LDO output for positive gamma low voltage generator.
VGMN	O	Output voltage generated from AVEE. LDO output for negative gamma high voltage generator.
VGSN	O	Output voltage generated from AVEE. LDO output for negative gamma low voltage generator.

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4.9 LABC and CABC Control Pins

Symbol	I/O	Description
LEDON	O	This pin is connect to the external LED driver. It is a LED driver control signal which is used for turning ON/OFF the LED backlight. If not used, please open this pin.
LEDPWM	O	This pin is connect to the external LED driver. It is a PWM type control signal for brightness of the LED backlight. The width of LEDPWM signal is set from 256 values between 0% (Low) and 100% (High) If not used, please open this pin.
KBBC	O	This pin is connect to the external keyboard LED driver. It is a PWM type control signal for brightness of the keyboard LED backlight. The width of KBBC signal is set from 256 values between 0% (Low) and 100% (High) If not used, please open this pin.

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4.10 Test Pins

Symbol	I/O	Description
PADA1 PADA2 PADA3 PADA4 PADB1 PADB2 PADB3 PADB4	I/O	- These test pins for chip attachment detection. PADA1 to PADA4 are output pins and PADB1 to PADB4 are input pins. - For normal operation: Connect PADA1 and PADB1 together by ITO trace. Connect PADA2 and PADB2 together by ITO trace. Connect PADA3 and PADB3 together by ITO trace. Connect PADA4 and PADB4 together by ITO trace.
CONTACT1A, CONTACT1B, CONTACT2A, CONTACT2B	I/O	- Test pin, for test bonding quality, IC internal will connect CONTACT1A with CONTACT1B, CONTACT2A with CONTACT2B
AVSS_AVDD	I	Test pin, must be connected to AVSS
AVEE_AVSS	I	Test pin, must be connected to AVEE
VCL_VDDB	I	Test pin, must be connected to VCL
VCL_AVSS	I	Test pin, must be connected to VCL
VGMN_VGMP	I	Test pin, must be connected to VGMN
VGSN_VGSP	I	Test pin, must be connected to VGSN
TEST0~7	I/O	Test pin, not accessible to user. Must be left open.
OSC_TEST	O	Test pin, not accessible to user. Must left open
VDDI_OPT1~2	O	Use them to fix the electrical potentials of unused interface pins and fixed pins. When not in use, leave it open.
VSSI_OPT1	O	Use them to fix the electrical potentials of unused interface pins and fixed pins. When not in use, leave it open.
VSSIDUM0~106	O	-These pins are dummy with VSSI potential (not have any function inside). -Signal traces can't pass through on glass under these pads.

5 FUNCTIONAL DESCRIPTION

5.1 MPU Interface

NT35510 can interface with MPU at high speed. However, if the interface cycle time is faster than the limit, MPU needs to have dummy wait(s) to meet the cycle time limit.

5.1.1 Interface Type Selection

The selection of a given interfaces are done by setting IM2, IM1 and IM0 pins as show in **Table 5.1.1**

Table 5.1.1 Interface Type Selection

IM3	IM2	IM1	IM0	SRAM	Register
X	0	0	0	80-series 8-bit MPU interface, D[7:0]	80-series 8-bit MPU interface, D[7:0]
X	0	0	1	80-series 16-bit MPU interface, D[15:0]	80-series 16-bit MPU interface, D[15:0]
X	0	1	0	80-series 24-bit MPU interface, D[23:0]	80-series 24-bit MPU interface, D[23:0]
0	0	1	1	RGB interface, D[23:0]	SPI, SDI/SDO serial data, SCL rising trigger
1	0	1	1	RGB interface, D[23:0]	SPI, SDI/SDO serial data, SCL falling trigger
X	1	0	0	RGB interface, D[23:0]	I2C interface, I2C_SDA serial data
X	1	0	1	MIPI DSI, HSSI_D0_P/N, HSSI_D1_P/N	MIPI DSI, HSSI_D0_P/N, HSSI_D1_P/N
0	1	1	0	MDDI, HSSI_D0_P/N, HSSI_D1_P/N	MDDI, HSSI_D0_P/N, HSSI_D1_P/N SPI, SDI/SDO serial data, SCL rising trigger
1	1	1	0	MDDI, HSSI_D0_P/N, HSSI_D1_P/N	MDDI, HSSI_D0_P/N, HSSI_D1_P/N SPI, SDI/SDO serial data, SCL falling trigger
X	1	1	1	MDDI, HSSI_D0_P/N, HSSI_D1_P/N	MDDI, HSSI_D0_P/N, HSSI_D1_P/N I2C interface, I2C_SDA serial data

NOTES:

1. X = Don't Care.

5.1.2 80-series MPU Interface

The MCU uses an 11-wires 8-data or 19-wires 16-data or 27-wires 24-data parallel interface.

The chip-select CSX (active low) enables and disables the parallel interface. WRX is the parallel data write, RDX is the parallel data read and D[23:0] is parallel data.

The Graphics Controller Chip reads the data at the rising edge of WRX signal. The D/CX is the data/command flag. When D/CX='1', D[23:0] bits are display RAM data or command parameters. When D/C='0', D[23:0] bits are commands.

The 8080-series bi-directional interface can be used for communication between the micro controller and LCD driver chip. Interface bus width can be selected with IM2, IM1 and IM0.

The interface functions of 80-series parallel interface are given in **Table 5.1.2**.

Table 5.1.2 Parallel interface function (80-Series)

IM2	IM1	IM0	Interface	D/CX	RDX	WRX	Function
0	0	0	8-bit Parallel	0	1	↑	Write 16-bit command, D[7:0]
				1	1	↑	Write 16/18/24-bit display data or 16-bit parameter, D[7:0]
				1	↑	1	Read 16/18/24-bit display data, D[7:0]
				1	↑	1	Read 16-bit parameter or status, D[7:0]
0	0	1	16-bit Parallel	0	1	↑	Write 16-bit command, D[7:0]
				1	1	↑	Write 16/18/24-bit display data or 16-bit parameter, D[15:0]
				1	↑	1	Read 16/18/24-bit display data, D[15:0]
				1	↑	1	Read 16-bit parameter or status, D[15:0]
0	1	0	24-bit Parallel	0	1	↑	Write 16-bit command, D[23:0]
				1	1	↑	Write 16/18/24-bit display data or 16-bit parameter, D[23:0]
				1	↑	1	Read 16/18/24-bit display data, D[23:0]
				1	↑	1	Read 16-bit parameter or status, D[23:0]

5.1.2.1 WRITE CYCLE SEQUENCE

The write cycle means that the host writes information (command or/and data) to the display via the interface. Each write cycle (WRX high-low-high sequence) consists of 3 control (D/CX, RDX, WRX) and data signals (D[23:0]). D/CX bit is a control signal, which tells if the data is a command or a data. The data signals are the command if the control signal is low (= '0') and vice versa it is data (= '1').

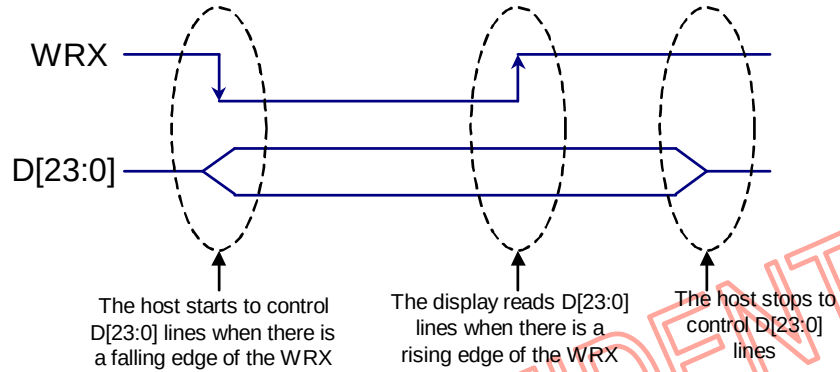


Fig. 5.1.1 80-Series WRX protocol

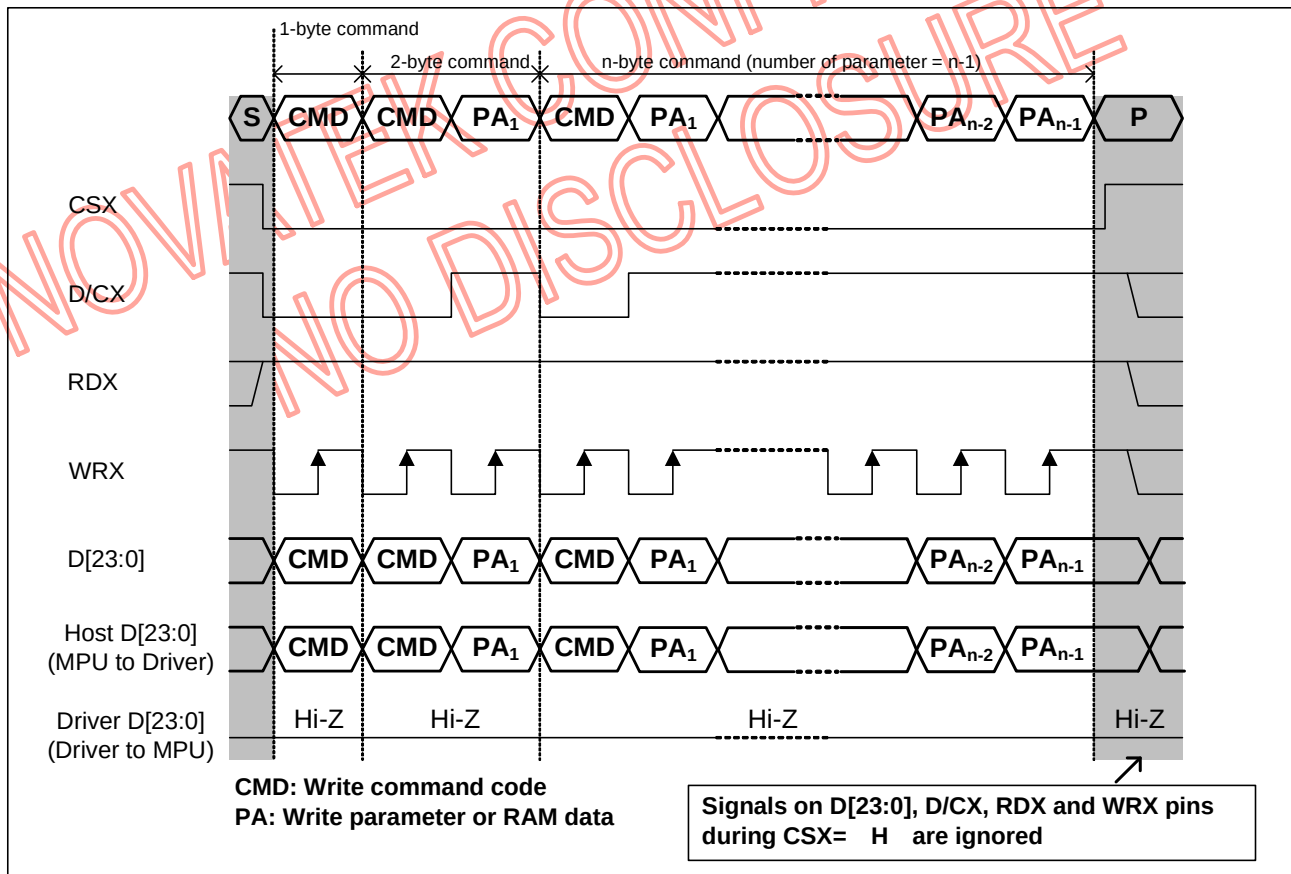


Fig. 5.1.2 80-Series parallel bus protocol, write to register or display RAM

5.1.2.2 READ CYCLE SEQUENCE

The read cycle (RDX high-low-high sequence) means that the host reads information from display via interface. The display sends data (D[17:0]) to the host when there is a falling edge of RDX and the host reads data when there is a rising edge of RDX.

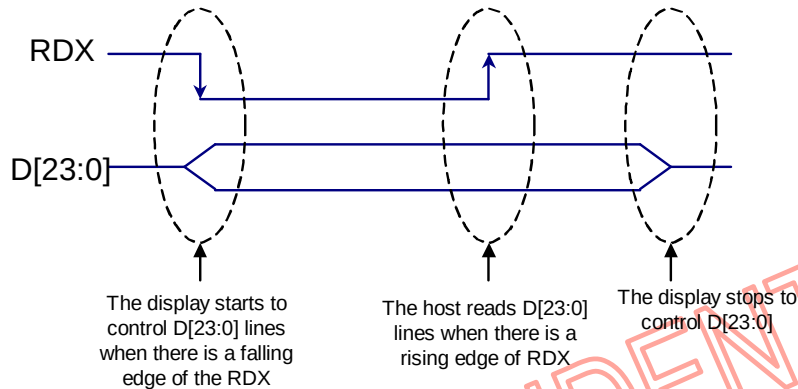


Fig. 5.1.3 80-Series RDX protocol

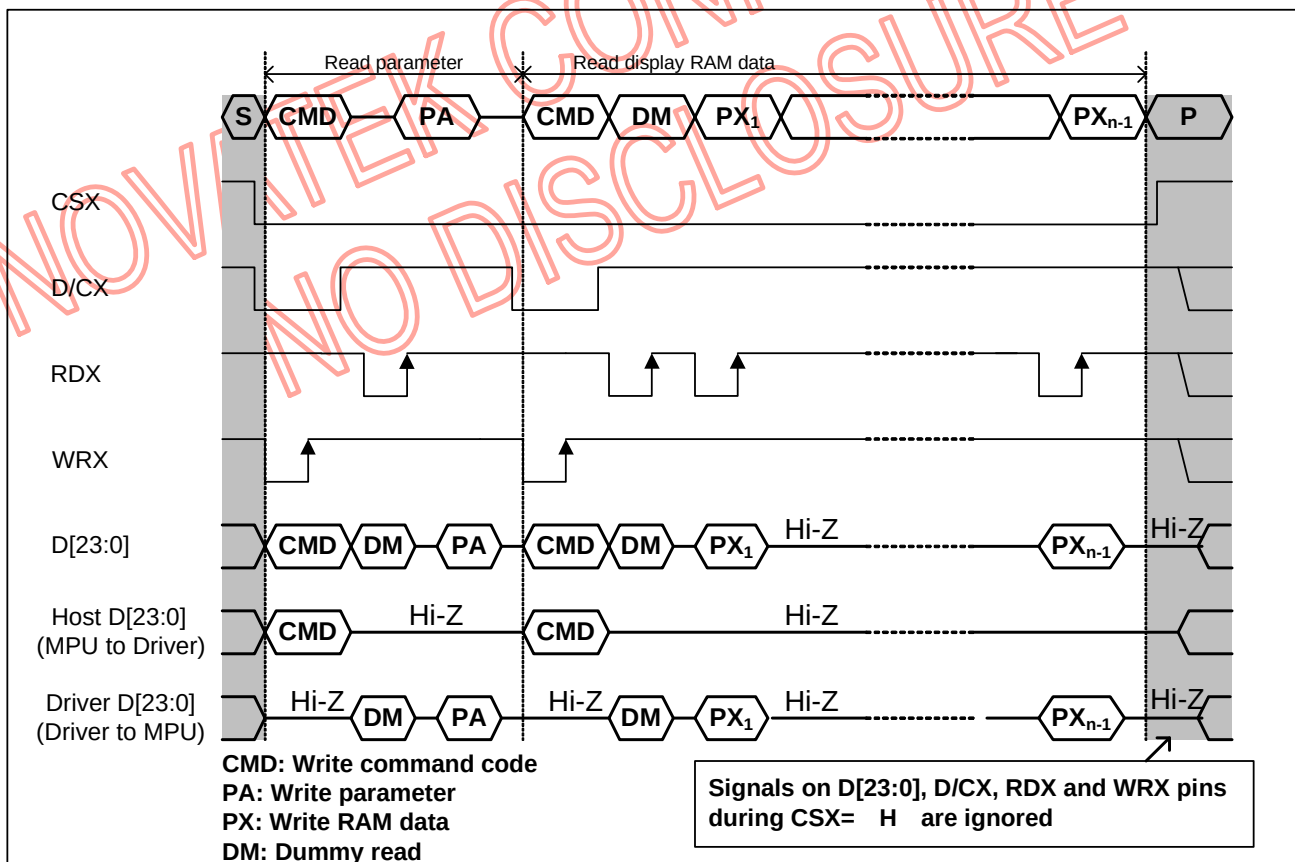


Fig. 5.1.4 80-Series parallel bus protocol, read from register or display RAM

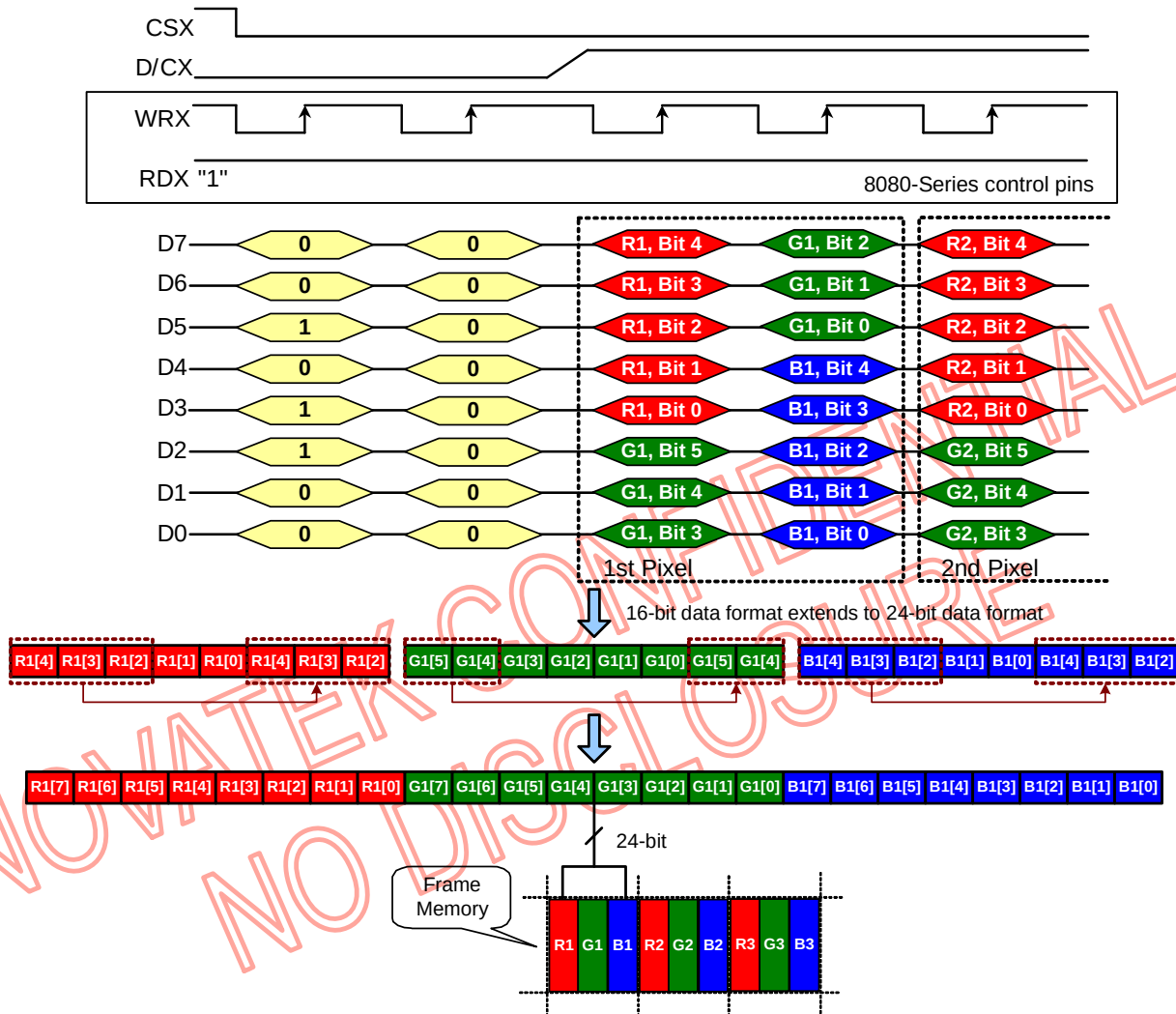
5.1.2.3 8-BIT PARALLEL INTERFACE FOR DATA RAM WRITE

Different display data formats are available for three color depths supported by the LCM listed below.

Register Command	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Register
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	1	0	1	1	0	0	2Ch
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	00h
3A00h	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Color
0005h	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	R4	R3	R2	R1	R0	G5	G4	G3	65K-Color
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	G2	G1	G0	B4	B3	B2	B1	B0	
0006h	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	R5	R4	R3	R2	R1	R0	x	x	262K-Color
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	G5	G4	G3	G2	G1	G0	x	x	
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	B5	B4	B3	B2	B1	B0	x	x	
0007h	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	R7	R6	R5	R4	R3	R2	R1	R0	16.7M-Color
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	G7	G6	G5	G4	G3	G2	G1	G0	
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	B7	B6	B5	B4	B3	B2	B1	B0	

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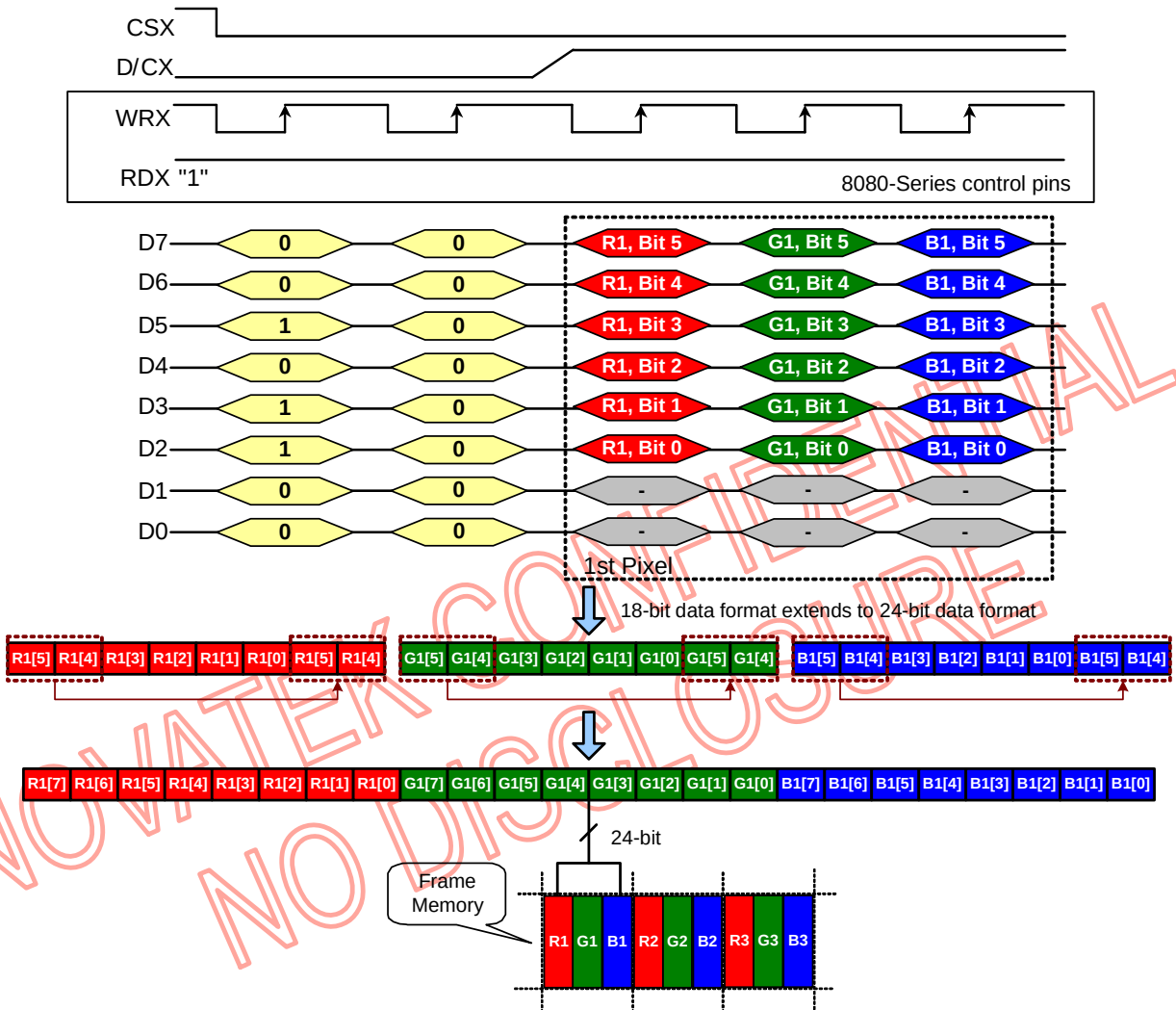
- 65K colors, RGB is 5-6-5-bit pixel data input



NOTES:

1. 2 times transfer is used to transmit 1 pixel data with the 16-bit color depth information.
2. The most significant bits are Rx4, Gx5 and Bx4.
3. The least significant bits are Rx0, Gx0 and Bx0.

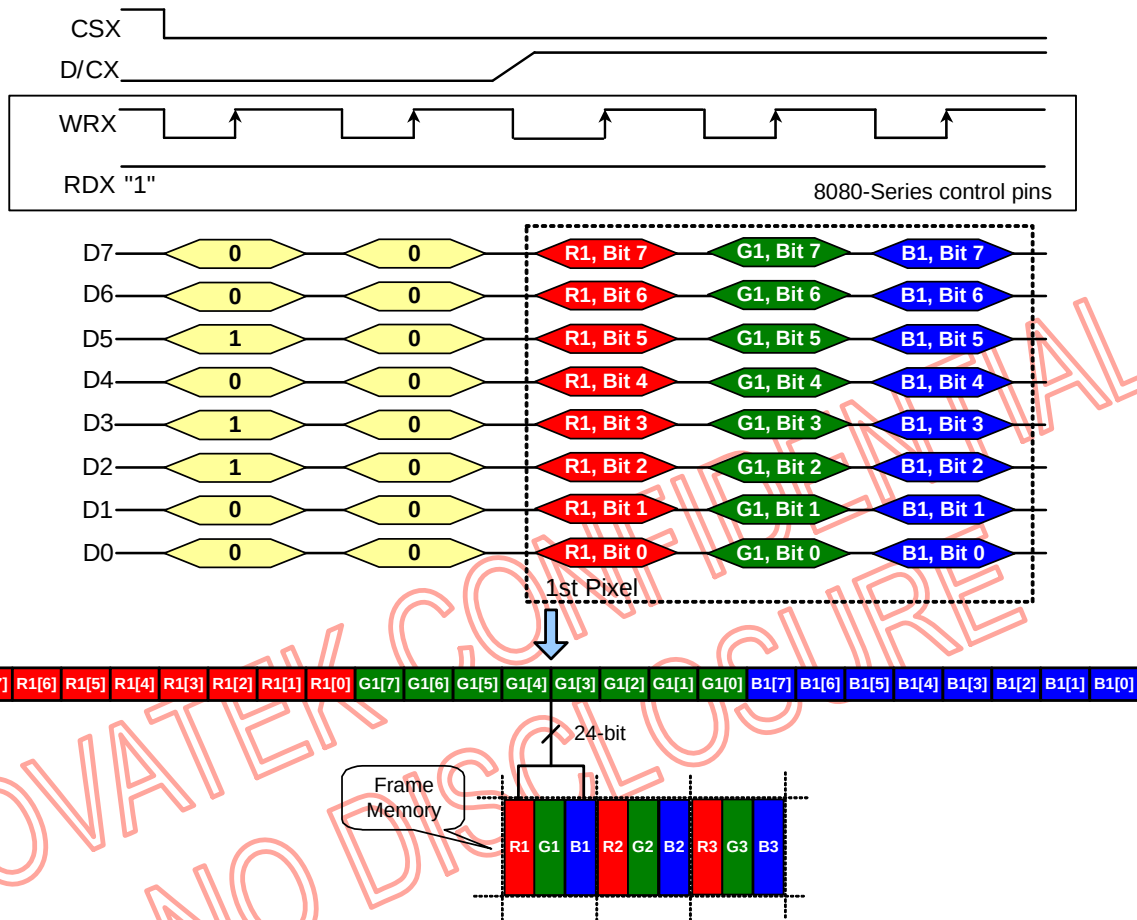
- 262K colors, RGB is 6-6-6-bit pixel data input



NOTES:

1. 3 times transfer is used to transmit 1 pixel data with the 18-bit color depth information.
2. The most significant bits are Rx5, Gx5 and Bx5.
3. The least significant bits are Rx0, Gx0 and Bx0.

- 16M colors, RGB is 8-8-8-bit pixel data input



NOTES:

1. 3 times transfer is used to transmit 1 pixel data with the 24-bit color depth information.
2. The most significant bits are Rx7, Gx7 and Bx7.
3. The least significant bits are Rx0, Gx0 and Bx0.

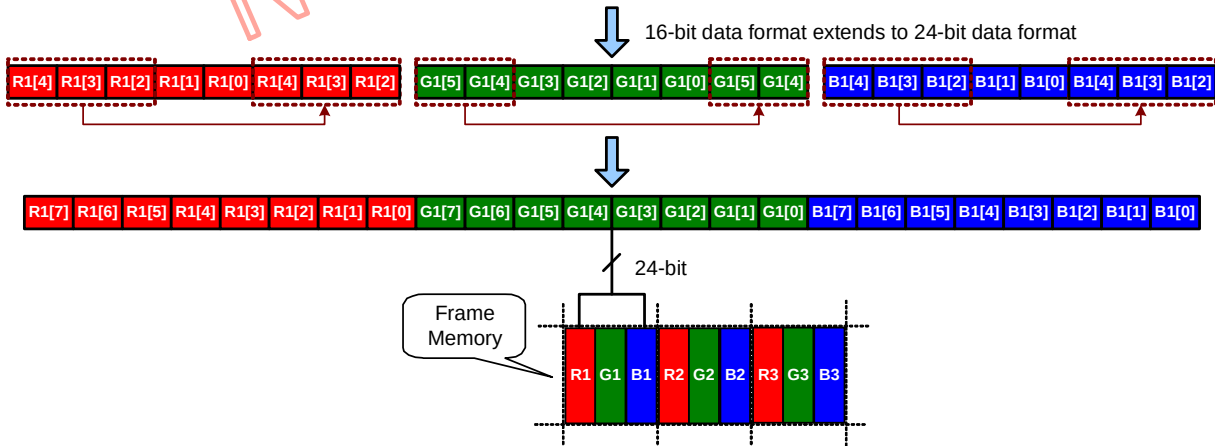
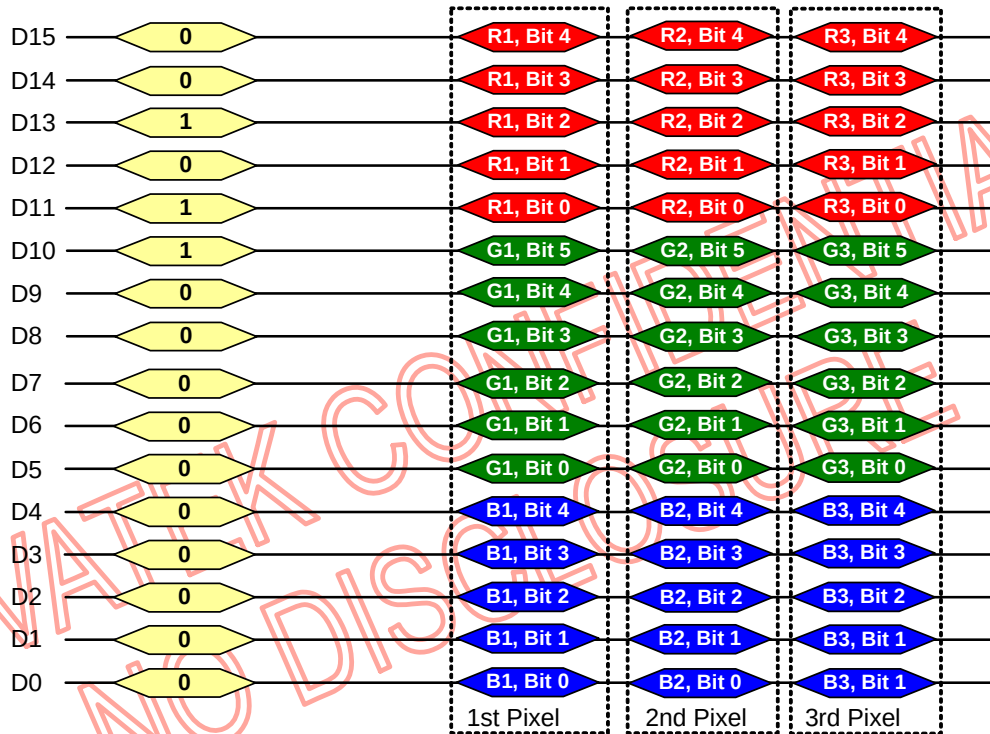
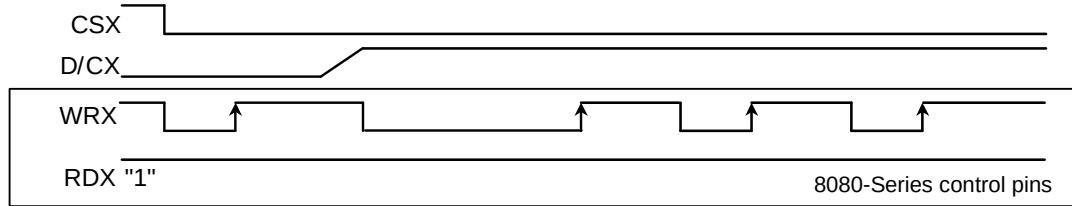
5.1.2.4 16-BIT PARALLEL INTERFACE FOR DATA RAM WRITE

Different display data formats are available for three color depths supported by the LCM listed below.

Register Command	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Register
3A00h	x	x	x	x	x	x	x	x	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	2C00h
0005h	x	x	x	x	x	x	x	x	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B4	B3	B2	B1	B0	65K-Color
0006h	x	x	x	x	x	x	x	x	R5	R4	R3	R2	R1	R0	x	x	G5	G4	G3	G2	G1	G0	x	x	262K-Color
	x	x	x	x	x	x	x	x	B5	B4	B3	B2	B1	B0	x	x	R5	R4	R3	R2	R1	R0	x	x	
	x	x	x	x	x	x	x	x	G5	G4	G3	G2	G1	G0	x	x	B5	B4	B3	B2	B1	B0	x	x	
0007h	x	x	x	x	x	x	x	x	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	16.7M-Color
	x	x	x	x	x	x	x	x	B7	B6	B5	B4	B3	B2	B1	B0	R7	R6	R5	R4	R3	R2	R1	R0	
	x	x	x	x	x	x	x	x	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0	

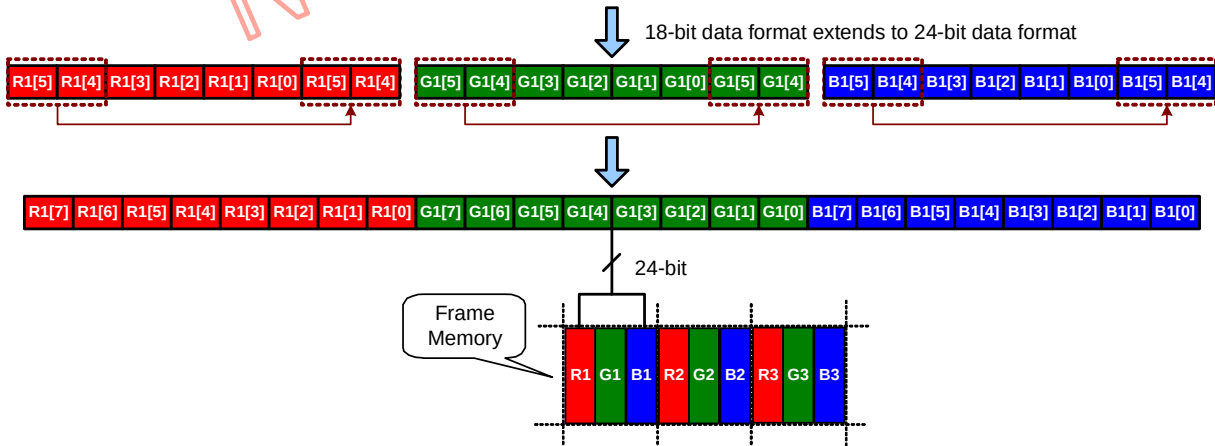
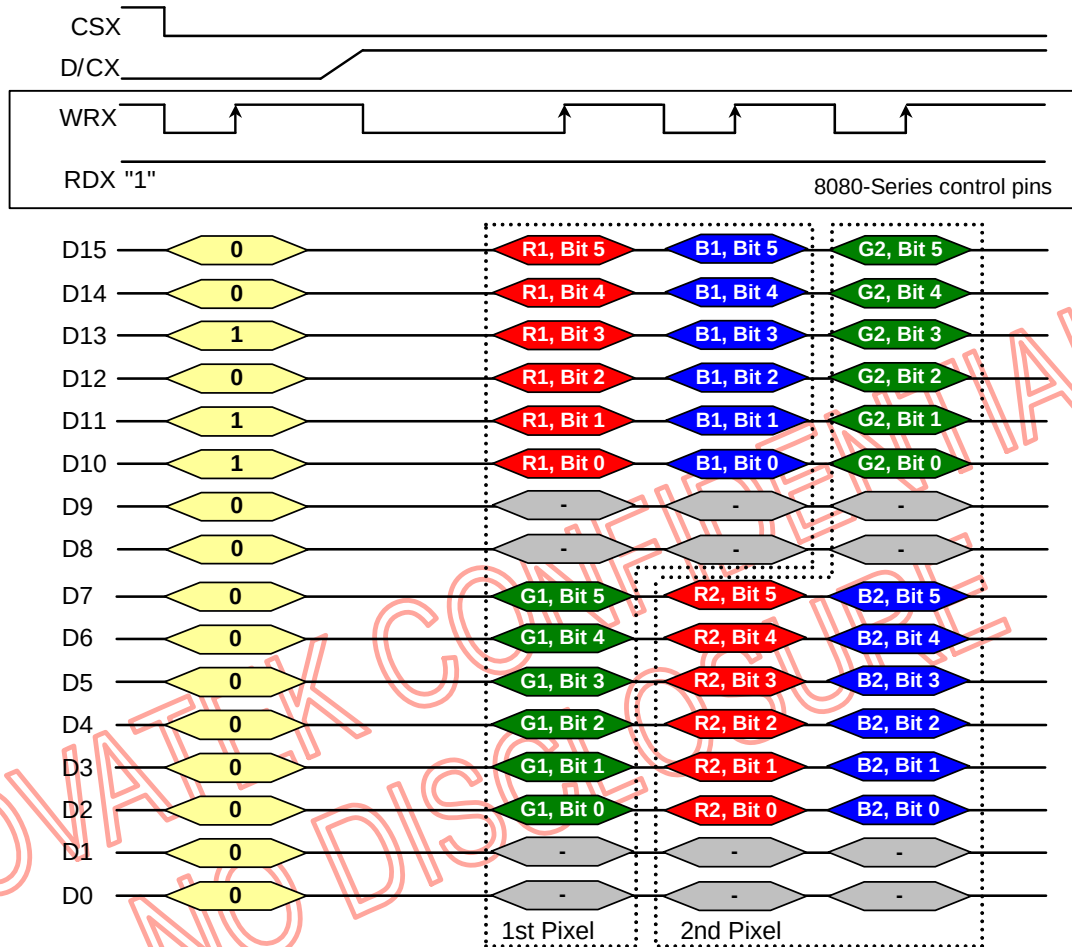
NOVATEK CONFIDENTIAL
NO DISCLOSURE

- 65K colors, RGB is 5-6-5-bit pixel data input



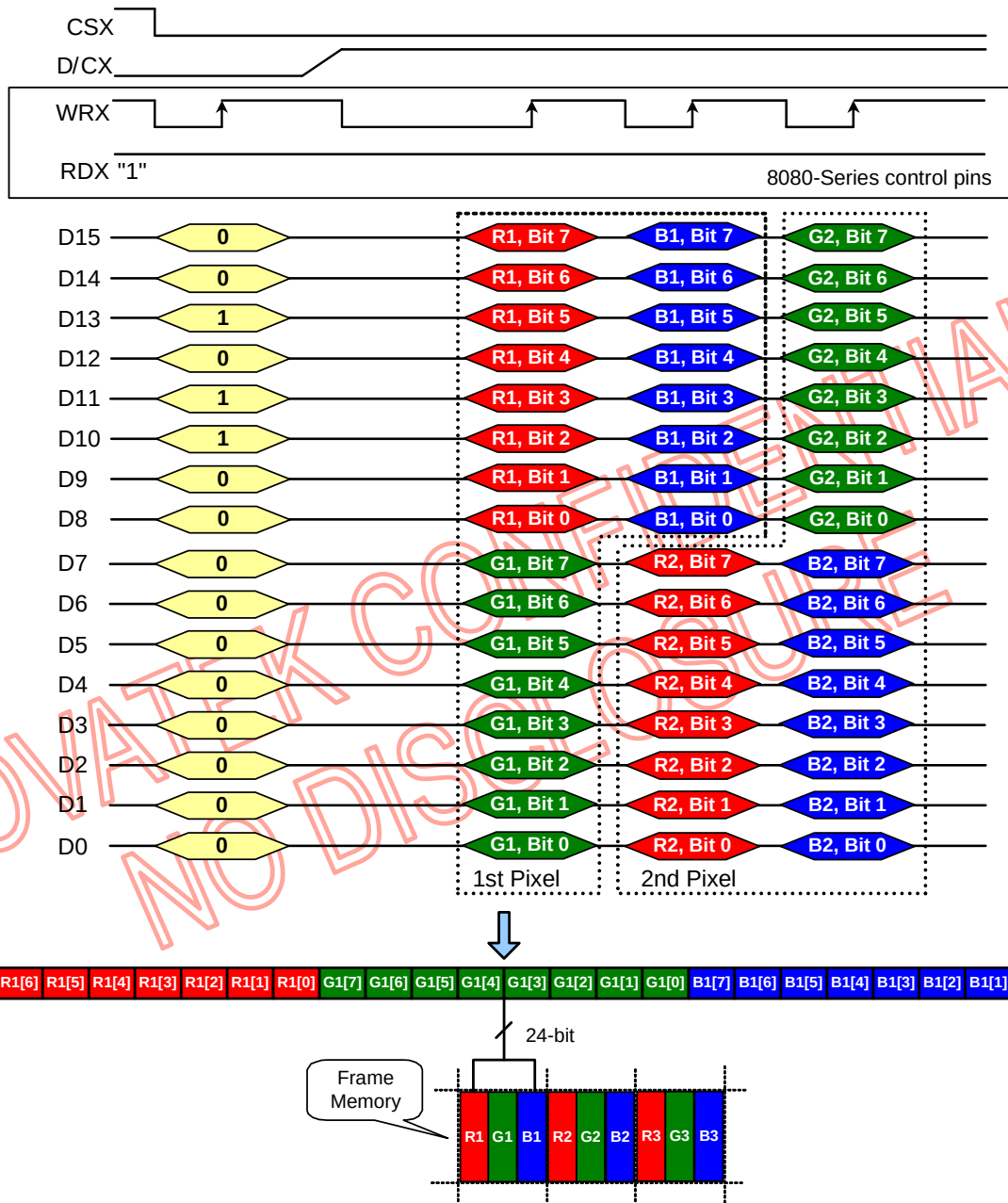
1. In one transfer (D15 to D0), 1 pixel data transmitted with the 16-bit color depth information.
2. The most significant bits are Rx4, Gx5 and Bx4.
3. The least significant bits are Rx0, Gx0 and Bx0.

- 262K colors, RGB is 6-6-6-bit pixel data input



1. 3 times transfer is used to transmit 2 pixel data or 2 times transfer is used to transmit 1 pixel data with the 18-bit color depth information.
2. The most significant bits are Rx5, Gx5 and Bx5.
3. The least significant bits are Rx0, Gx0 and Bx0.

- 16M colors, RGB is 8-8-8-bit pixel data input



1. 3 times transfer is used to transmit 2 pixel data or 2 times transfer is used to transmit 1 pixel data with the 24-bit color depth information.
2. The most significant bits are Rx7, Gx7 and Bx7.
3. The least significant bits are Rx0, Gx0 and Bx0.

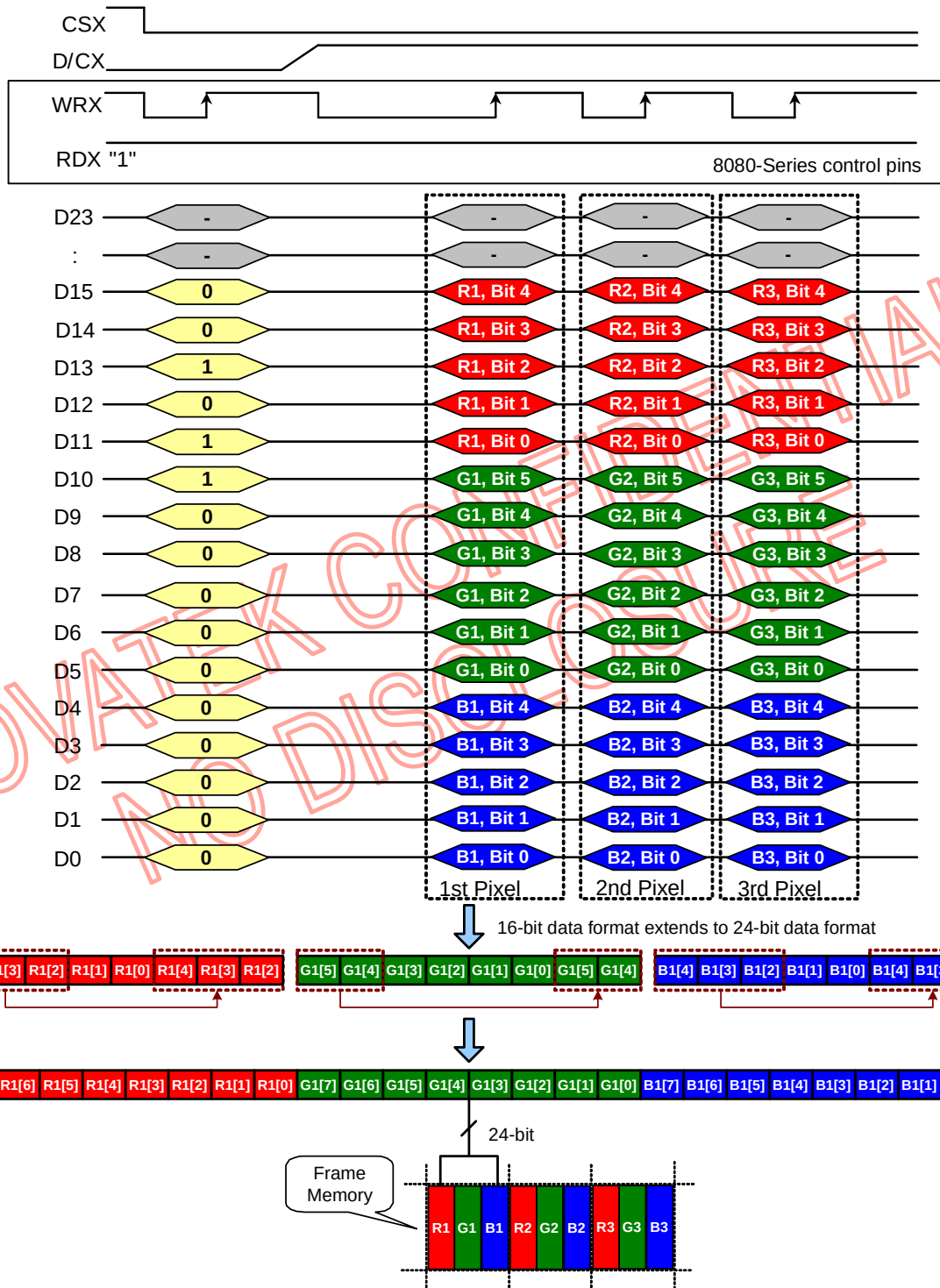
5.1.2.5 24-BIT PARALLEL INTERFACE FOR DATA RAM WRITE

Different display data formats are available for three color depths supported by the LCM listed below.

Register Command	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Register
3A00h	x	x	x	x	x	x	x	x	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	2C00h
0005h	x	x	x	x	x	x	x	x	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B4	B3	B2	B1	B0	65K-Color
0006h	x	x	x	x	x	x	R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0	262K-Color
0007h	R7	R6	R5	R4	R3	R2	R1	G7	G6	G5	G4	G3	G2	G1	G0	R0	B7	B6	B5	B4	B3	B2	B1	B0	16.7M-Color

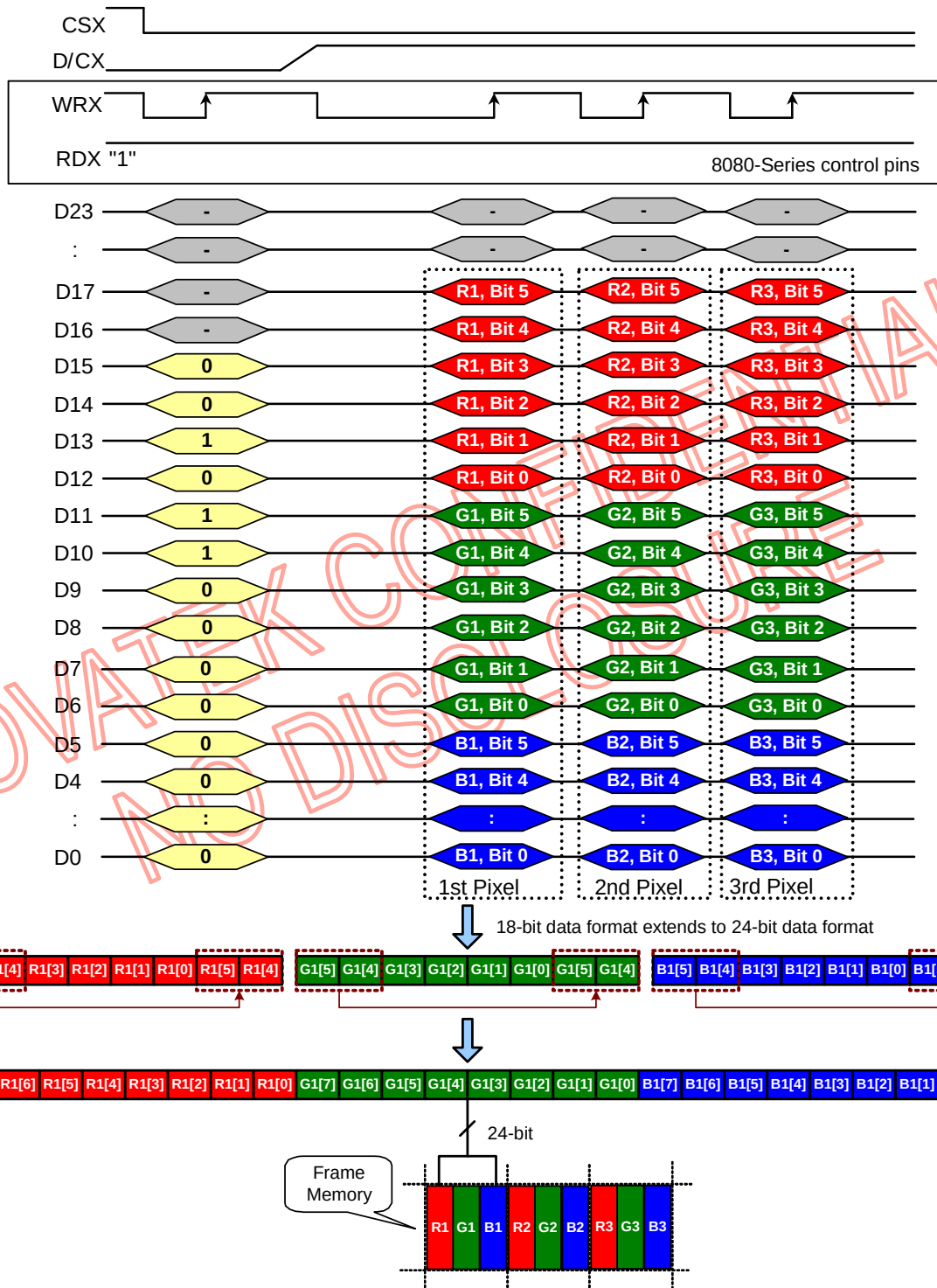
NOVATEK CONFIDENTIAL
NO DISCLOSURE

- 65K colors, RGB is 5-6-5-bit pixel data input



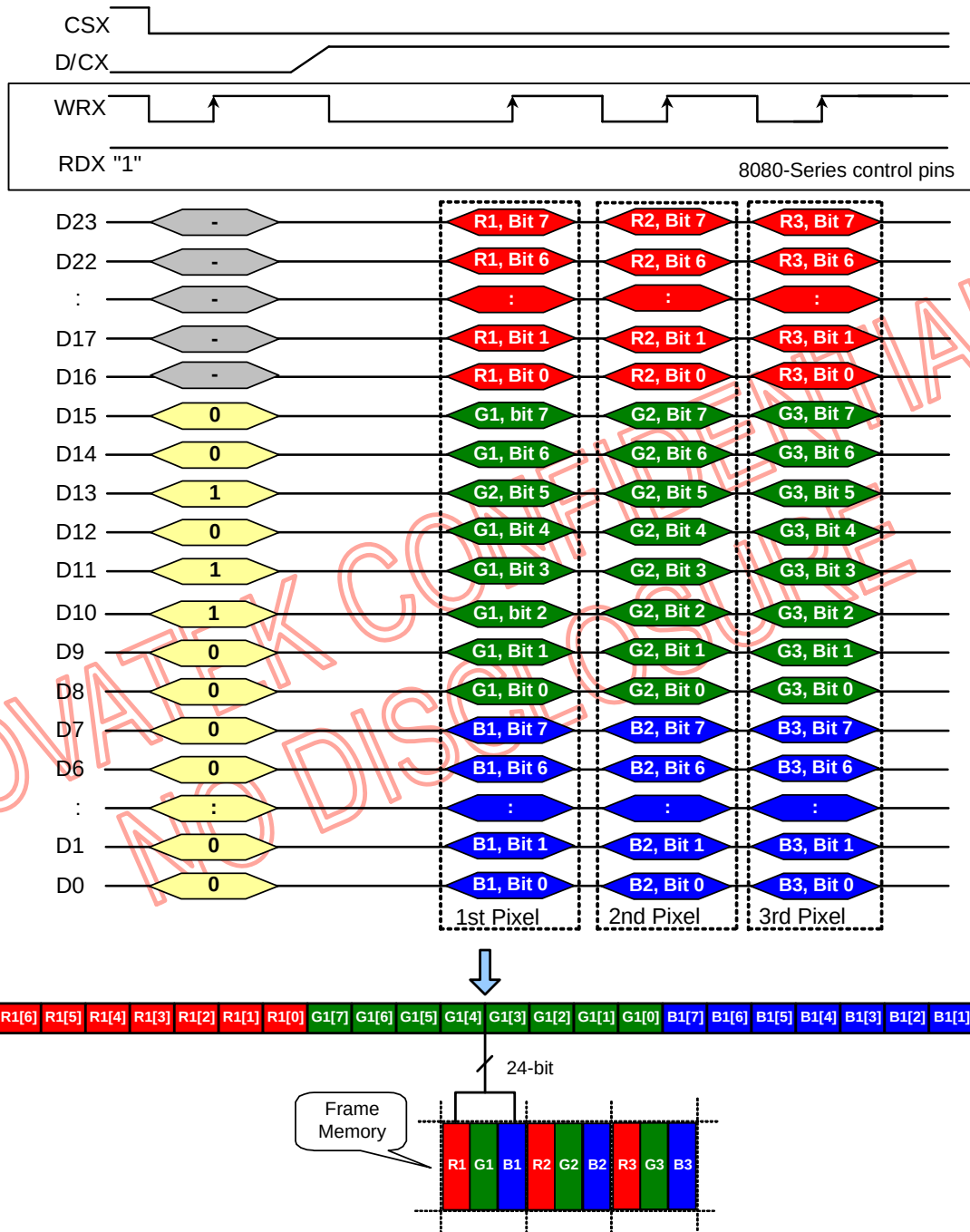
1. In one transfer (D15 to D0), 1 pixel data transmitted with the 16-bit color depth information.
2. The most significant bits are Rx4, Gx5 and Bx4.
3. The least significant bits are Rx0, Gx0 and Bx0.

- 262K colors, RGB is 6-6-6-bi pixel data t input



1. In one transfer (D17 to D0), 1 pixel data transmitted with the 18-bit color depth information.
2. The most significant bits are Rx5, Gx5 and Bx5.
3. The least significant bits are Rx0, Gx0 and Bx0.

- 16M colors, RGB is 8-8-8-bit pixel data input



1. In one transfer (D23 to D0), 1 pixel data transmitted with the 24-bit color depth information.
2. The most significant bits are Rx7, Gx7 and Bx7.
3. The least significant bits are Rx0, Gx0 and Bx0.

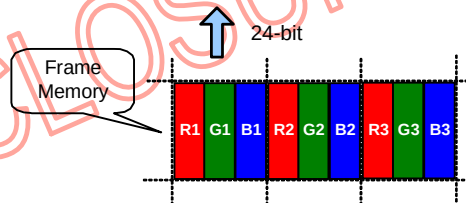
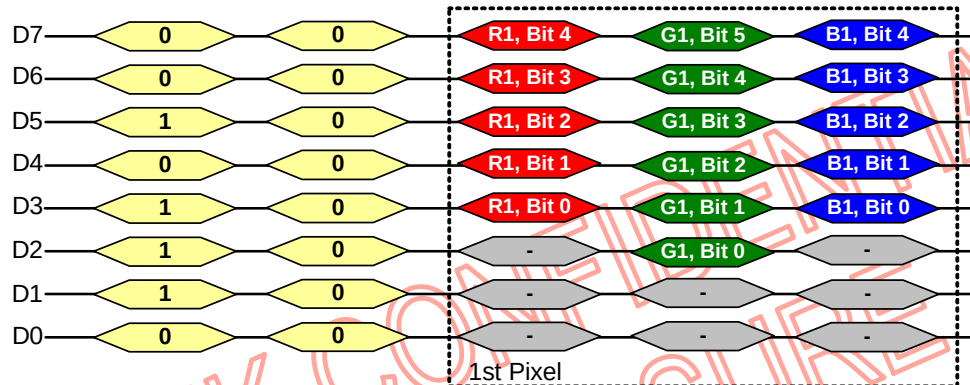
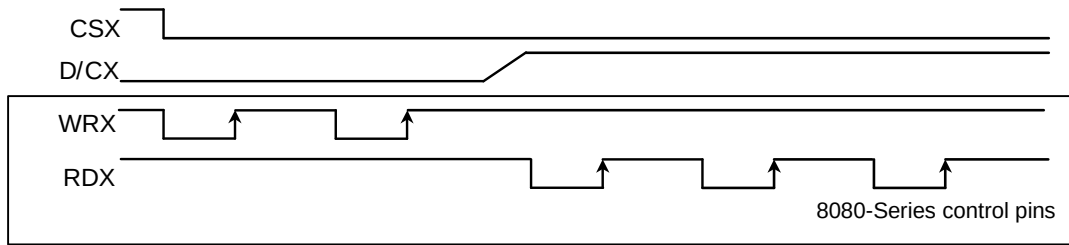
5.1.2.6 8-BIT PARALLEL INTERFACE FOR DATA RAM READ

Different display data formats are available for three color depths supported by the LCM listed below.

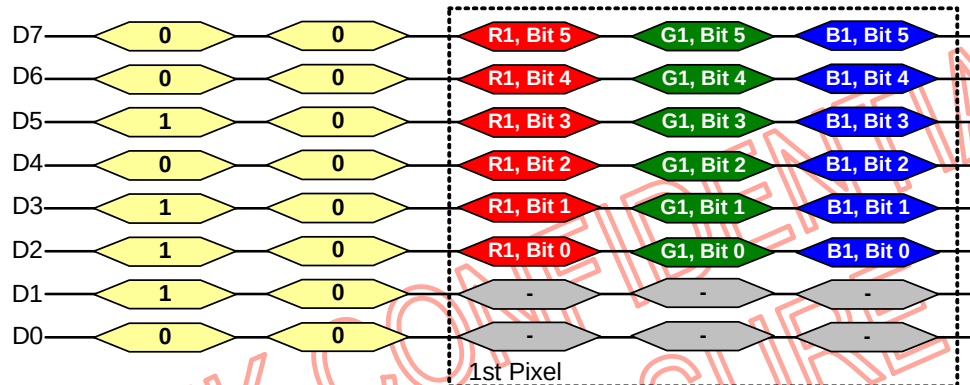
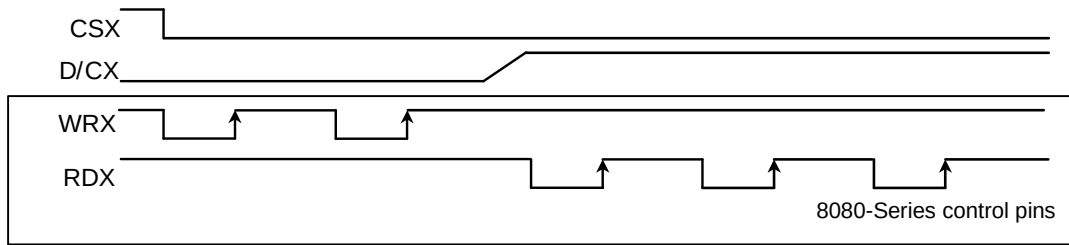
Register Command	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Register
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	1	0	1	1	1	0	2Eh
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	00h
3A00h	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Color
0005h	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	R4	R3	R2	R1	R0	x	x	x	65K-Color
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	G5	G4	G3	G2	G1	G0	x	x	
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	B4	B3	B2	B1	B0	x	x	x	
0006h	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	R5	R4	R3	R2	R1	R0	x	x	262K-Color
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	G5	G4	G3	G2	G1	G0	x	x	
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	B5	B4	B3	B2	B1	B0	x	x	
0007h	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	R7	R6	R5	R4	R3	R2	R1	R0	16.7M-Color
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	G7	G6	G5	G4	G3	G2	G1	G0	
	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	B7	B6	B5	B4	B3	B2	B1	B0	

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NO DISCLOSURE

- 65K colors, RGB is 5-6-5-bit pixel data output



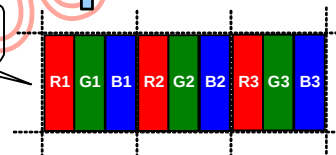
- 262K colors, RGB is 6-6-6-bit pixel data output



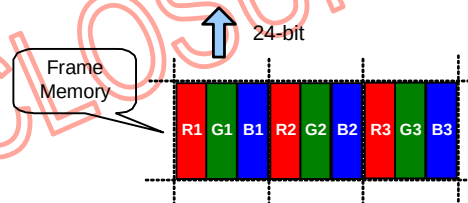
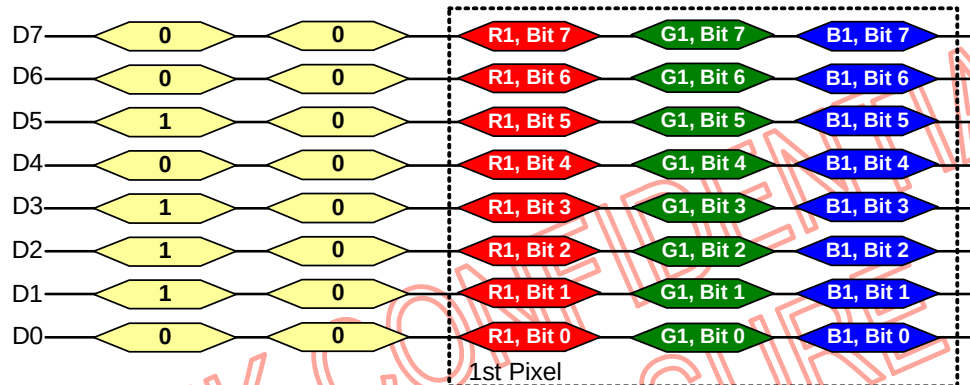
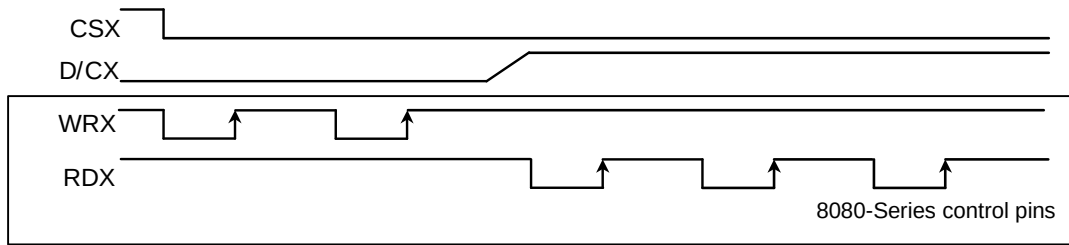
1st Pixel

Frame Memory

24-bit



- 16M colors, RGB is 8-8-8-bit pixel data output



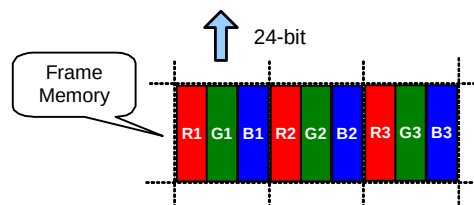
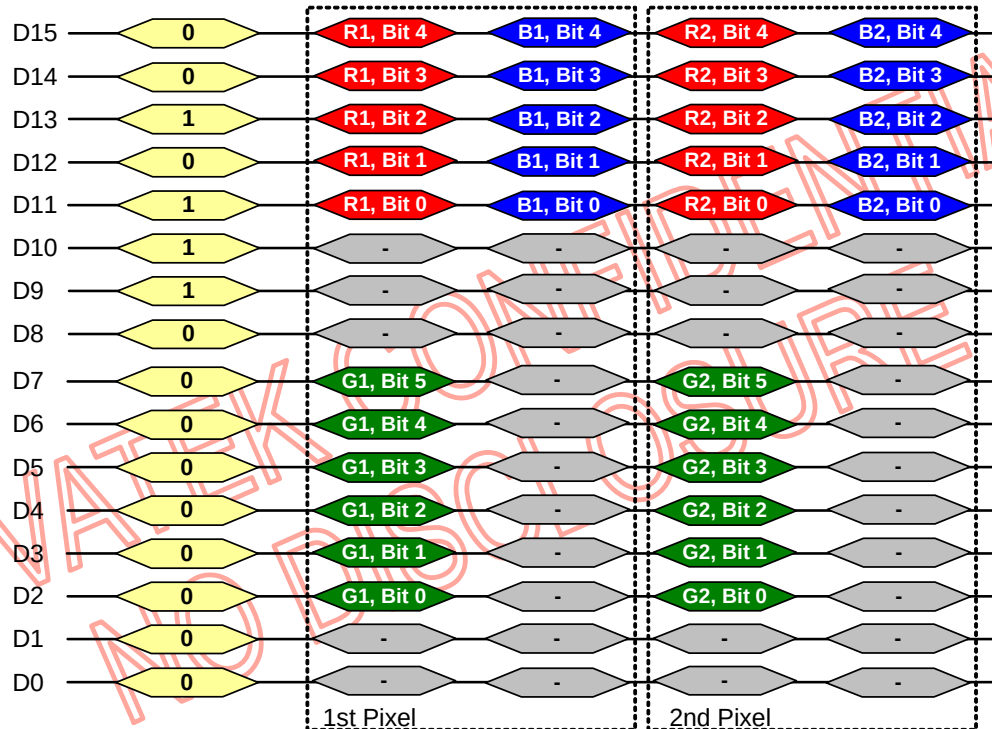
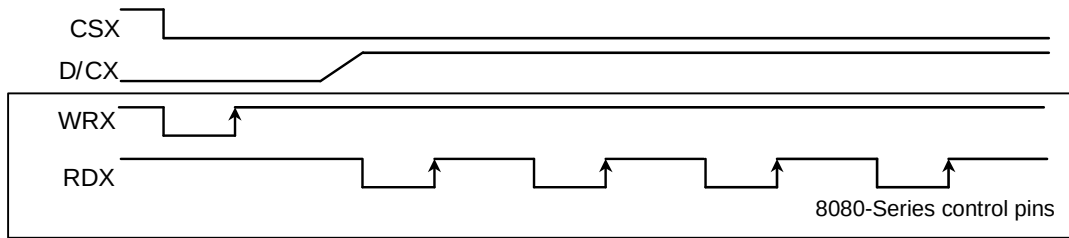
5.1.2.7 16-BIT PARALLEL INTERFACE FOR DATA RAM READ

Different display data formats are available for three color depths supported by the LCM listed below.

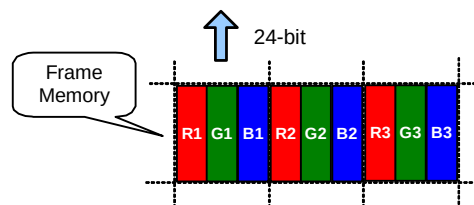
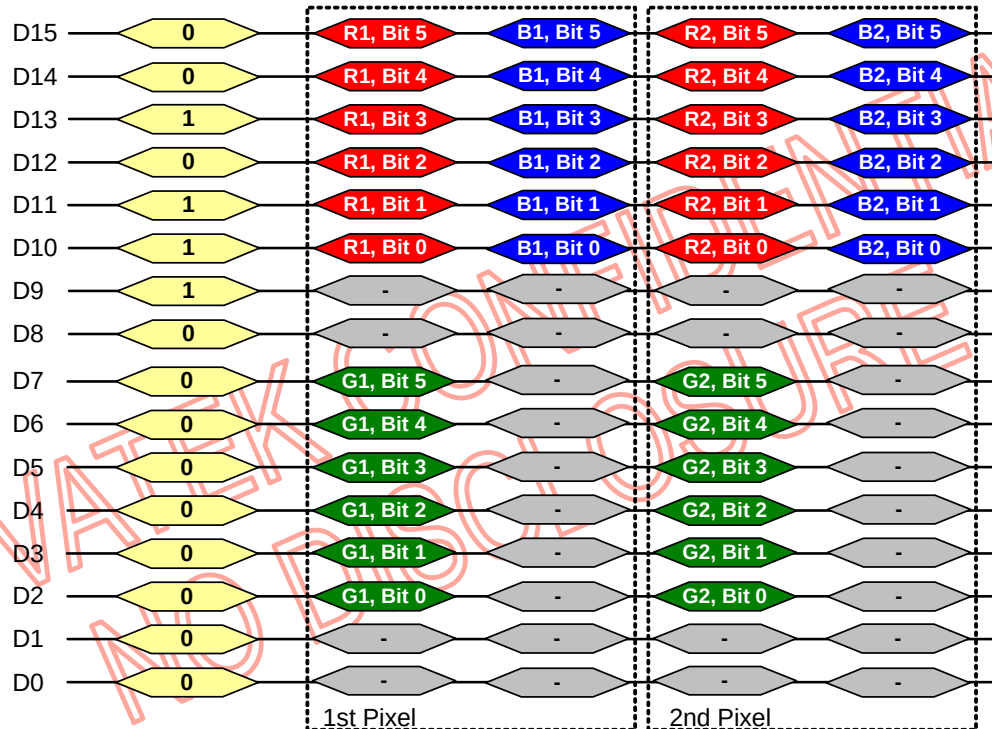
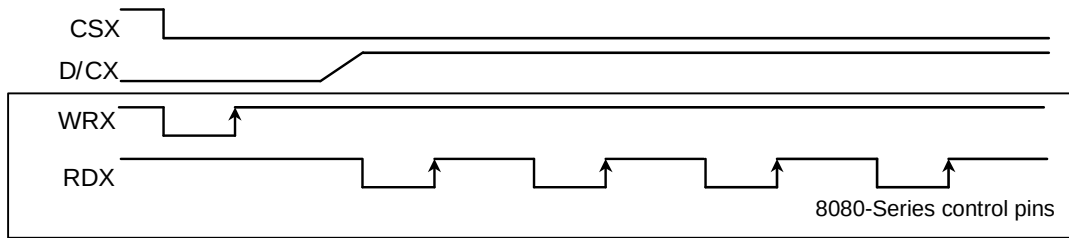
Register	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Register
Command	x	x	x	x	x	x	x	x	0	0	1	0	1	1	1	0	0	0	0	0	0	0	0	0	2E00h
3A00h	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Color
0005h	x	x	x	x	x	x	x	x	R4	R3	R2	R1	R0	x	x	x	G5	G4	G3	G2	G1	G0	x	x	65K-Color
	x	x	x	x	x	x	x	x	B4	B3	B2	B1	B0	x	x	x	x	x	x	x	x	x	x	x	
0006h	x	x	x	x	x	x	x	x	R5	R4	R3	R2	R1	R0	x	x	G5	G4	G3	G2	G1	G0	x	x	262K-Color
	x	x	x	x	x	x	x	x	B5	B4	B3	B2	B1	B0	x	x	x	x	x	x	x	x	x	x	
0007h	x	x	x	x	x	x	x	x	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	16.7M-Color
	x	x	x	x	x	x	x	x	B7	B6	B5	B4	B3	B2	B1	B0	x	x	x	x	x	x	x	x	

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NO DISCLOSURE

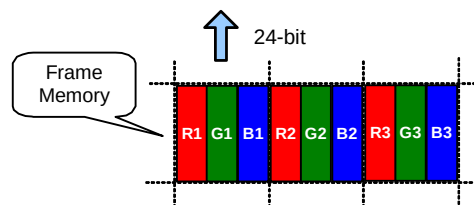
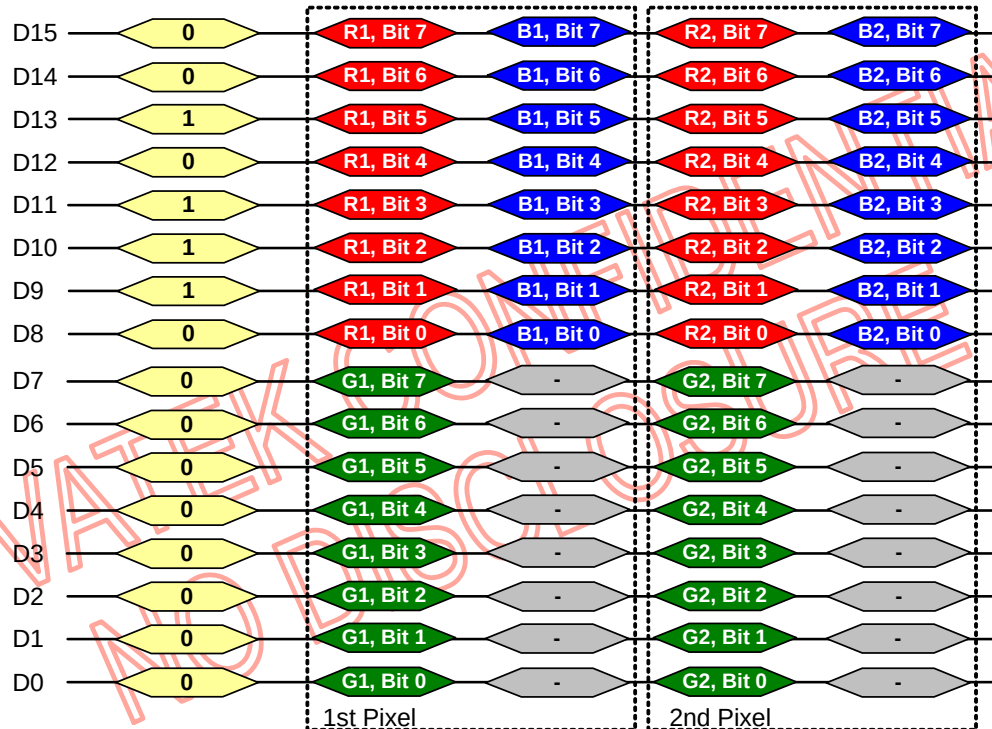
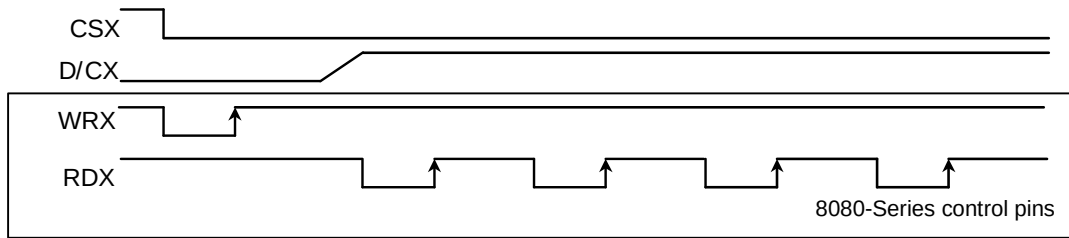
- 65K colors, RGB is 5-6-5-bit pixel data output



- 262K colors, RGB is 6-6-6-bit pixel data output



- 16M colors, RGB is 8-8-8-bit pixel data output



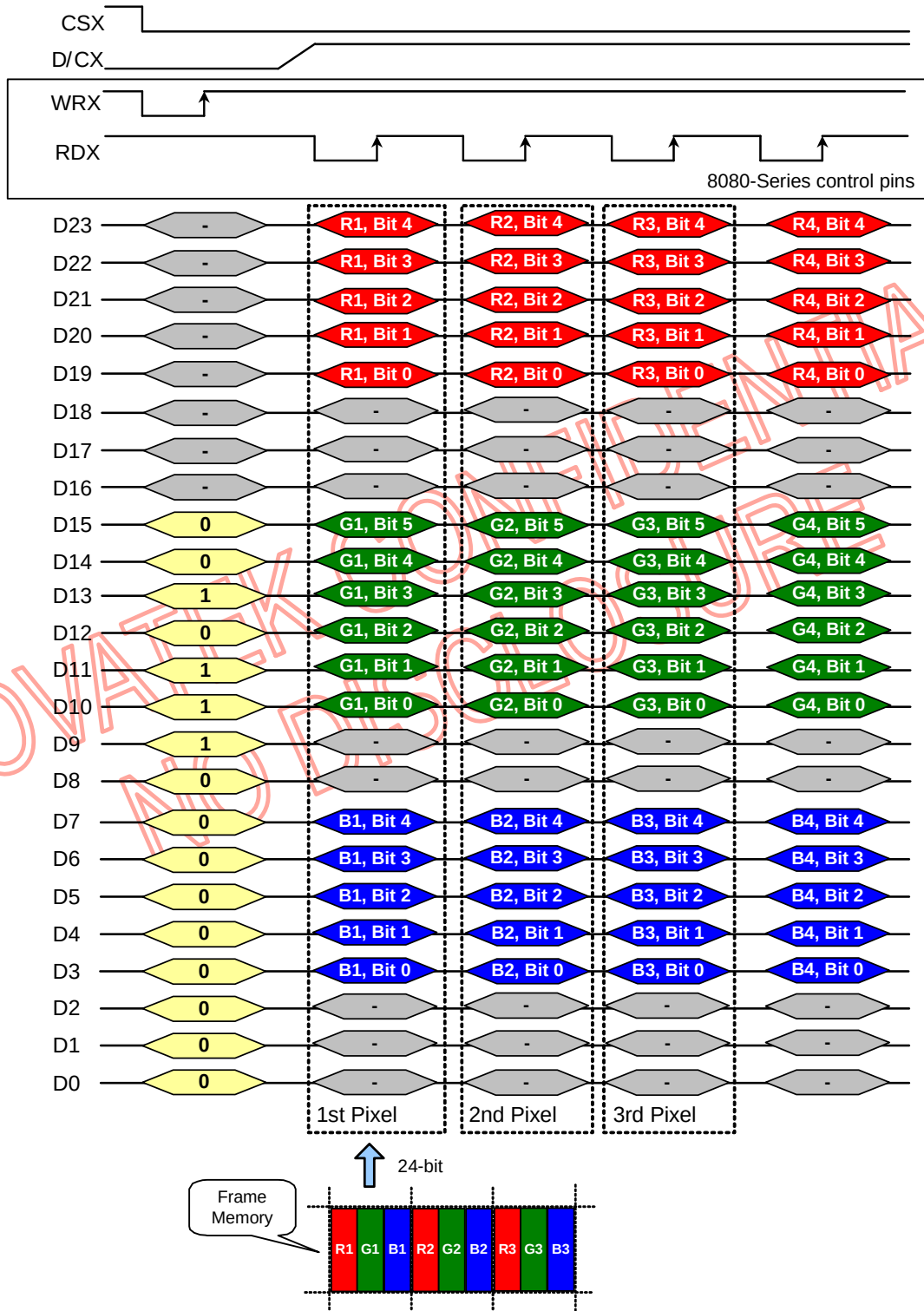
5.1.2.8 24-BIT PARALLEL INTERFACE FOR DATA RAM READ

Different display data formats are available for three color depths supported by the LCM listed below.

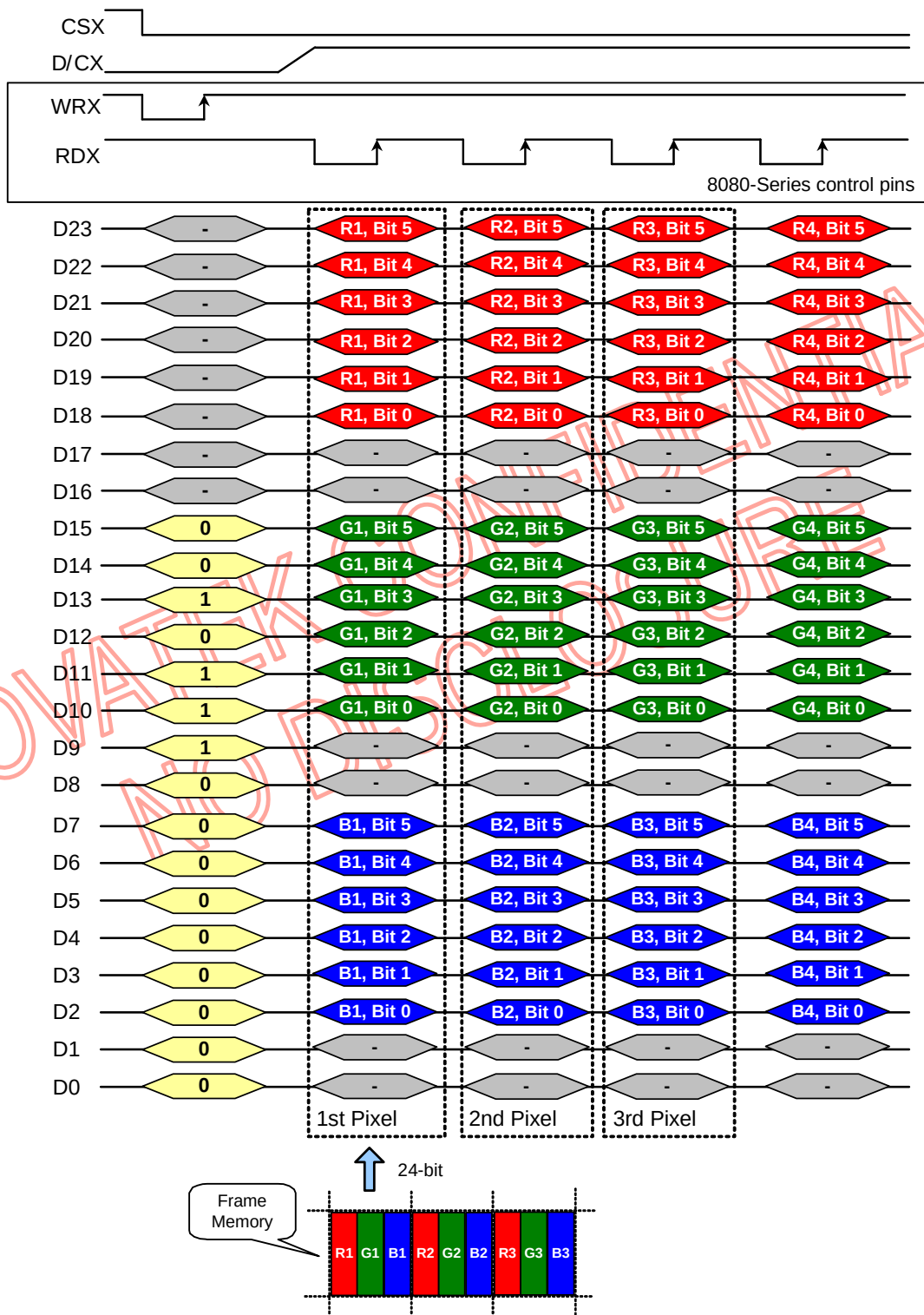
Register Command	D23	D22	D21	D20	D19	D18	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Register
3A00h	x	x	x	x	x	x	x	x	0	0	1	0	1	1	1	0	0	0	0	0	0	0	0	0	2E00h
0005h	R4	R3	R2	R1	R0	x	x	x	G5	G4	G3	G2	G1	G0	x	x	B4	B3	B2	B1	B0	x	x	x	65K-Color
0006h	R5	R4	R3	R2	R1	R0	x	x	G5	G4	G3	G2	G1	G0	x	x	B5	B4	B3	B2	B1	B0	x	x	262K-Color
0007h	R7	R6	R5	R4	R3	R2	R1	G7	G6	G5	G4	G3	G2	G1	G0	R0	B7	B6	B5	B4	B3	B2	B1	B0	16.7M-Color

NOVATEK CONFIDENTIAL
NO DISCLOSURE

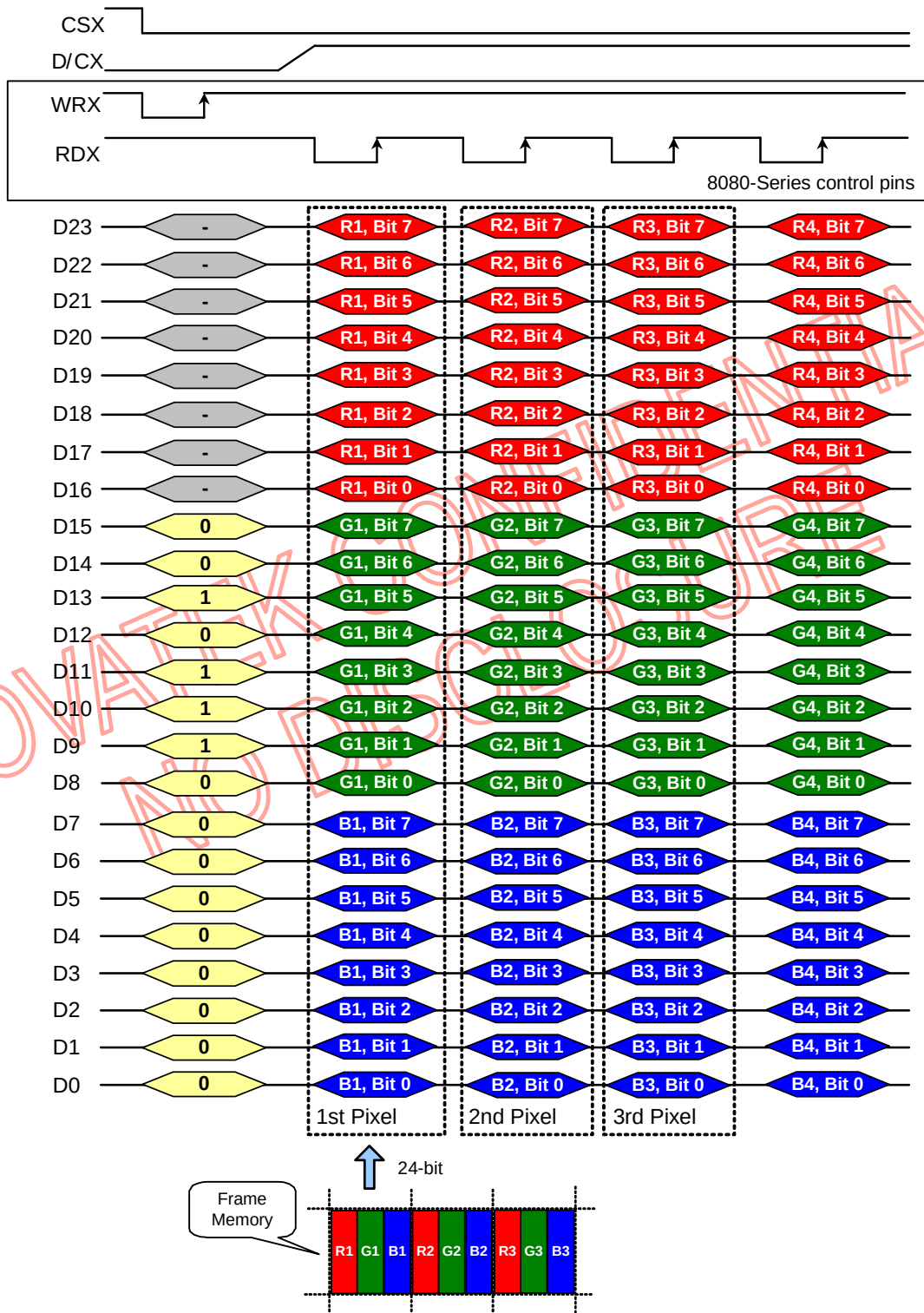
- 65K colors, RGB is 5-6-5-bit pixel data output



- 262K colors, RGB is 6-6-6-bit pixel data output



- 16M colors, RGB is 8-8-8-bit pixel data output



5.1.3 Serial Interface

The selection of this interface is done by IM3, IM2, IM1 and IM0.

The serial interface can select IM3 = 0 or 1 to decide the trigger edge of serial clock (SCL) is rising edge or falling edge. The serial interface is used to communication between the micro controller and the LCD driver chip. It contains CSX (chip select), SCL (serial clock), SDI (serial data input) and SDO (serial data output). Serial clock (SCL) is used for interface with MPU only, so it can be stopped when no communication is necessary. If the host places the SDI line into high-impedance state during the read intervals, then the SDI and SDO can be tied together.

5.1.3.1 WRITE MODE

The write mode of the interface means the micro controller writes commands and data to the NT35510. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or SDI data have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

When CSX is high, SCL clock is ignored. During the high time of CSX the serial interface is initialized. At the falling CSX edge, SCL can be high or low (see **Fig. 5.1.5**). SDI/SDO are sampled at the rising edge of SCL. R/W indicates, whether the byte is read command (R/W = '1') or write command (R/W = '0'). It is sampled when first rising SCL edge. If CSX stays low after the last bit of command/data byte, the serial interface expects the R/W bit of the next byte at the next rising edge of SCL.

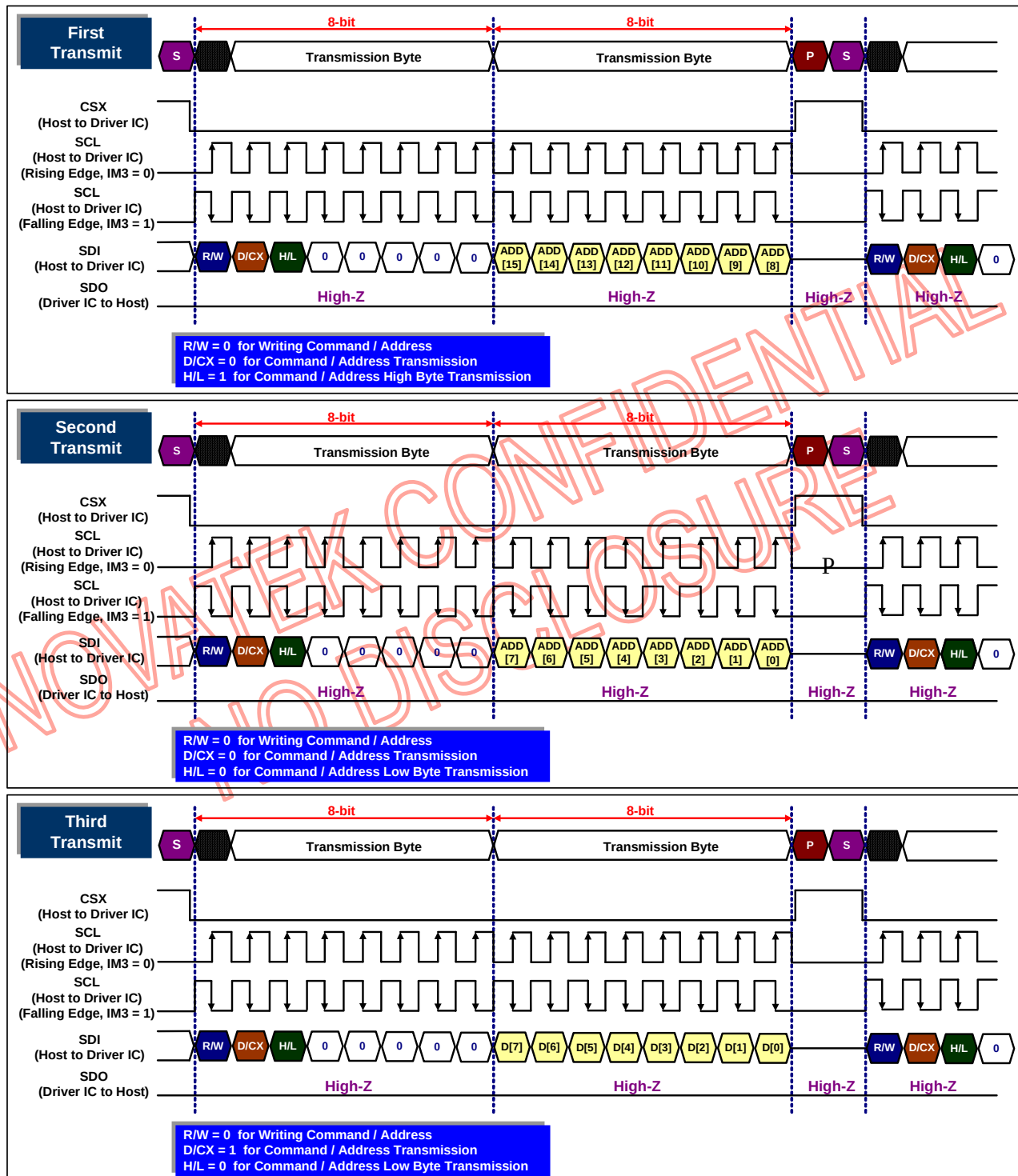


Fig. 5.1.5 Serial bus protocol for register write mode

5.1.3.2 READ MODE

The read mode of the interface means that the micro controller reads register value from the NT35510. To do so the micro controller first has to send a command and then the following byte is transmitted in the opposite direction. After that CSX is required to go high before a new command is send (see **Fig. 5.1.6**). The NT35510 samples the SDI (input data) at the rising edges, but shifts SDO (output data) at the falling SCL edges. Thus the micro controller is supported to read data at the rising SCL edges. After the read status command has been sent, the SDI line must be set to tri-state no later than at the falling SCL edge of the last bit. For the memory data read, a dummy clock cycle is needed (16 SCL clocks) to wait the memory data send out in SPI interface. But it doesn't need any dummy clock when execute the command data read.

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NO DISCLOSURE

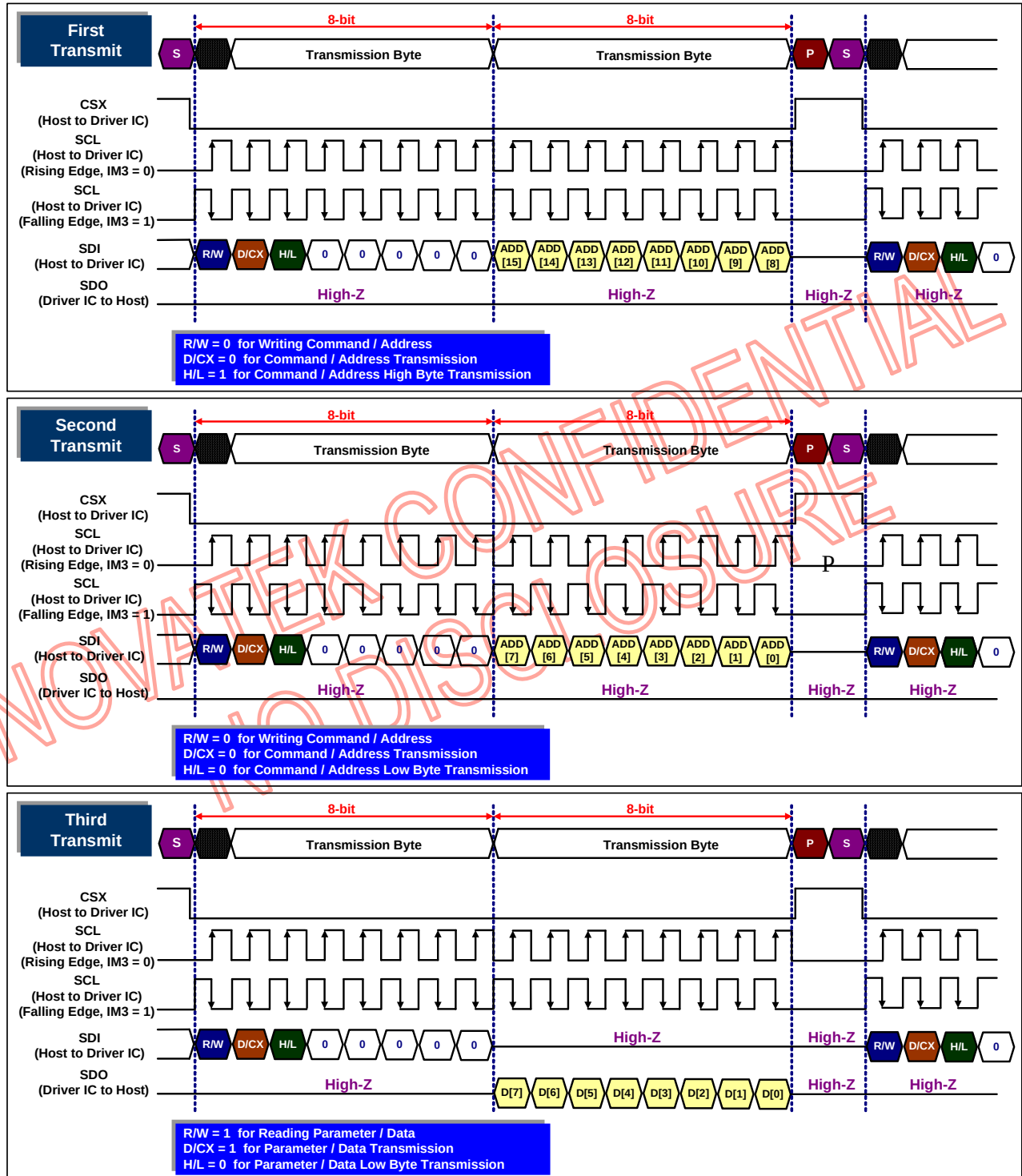
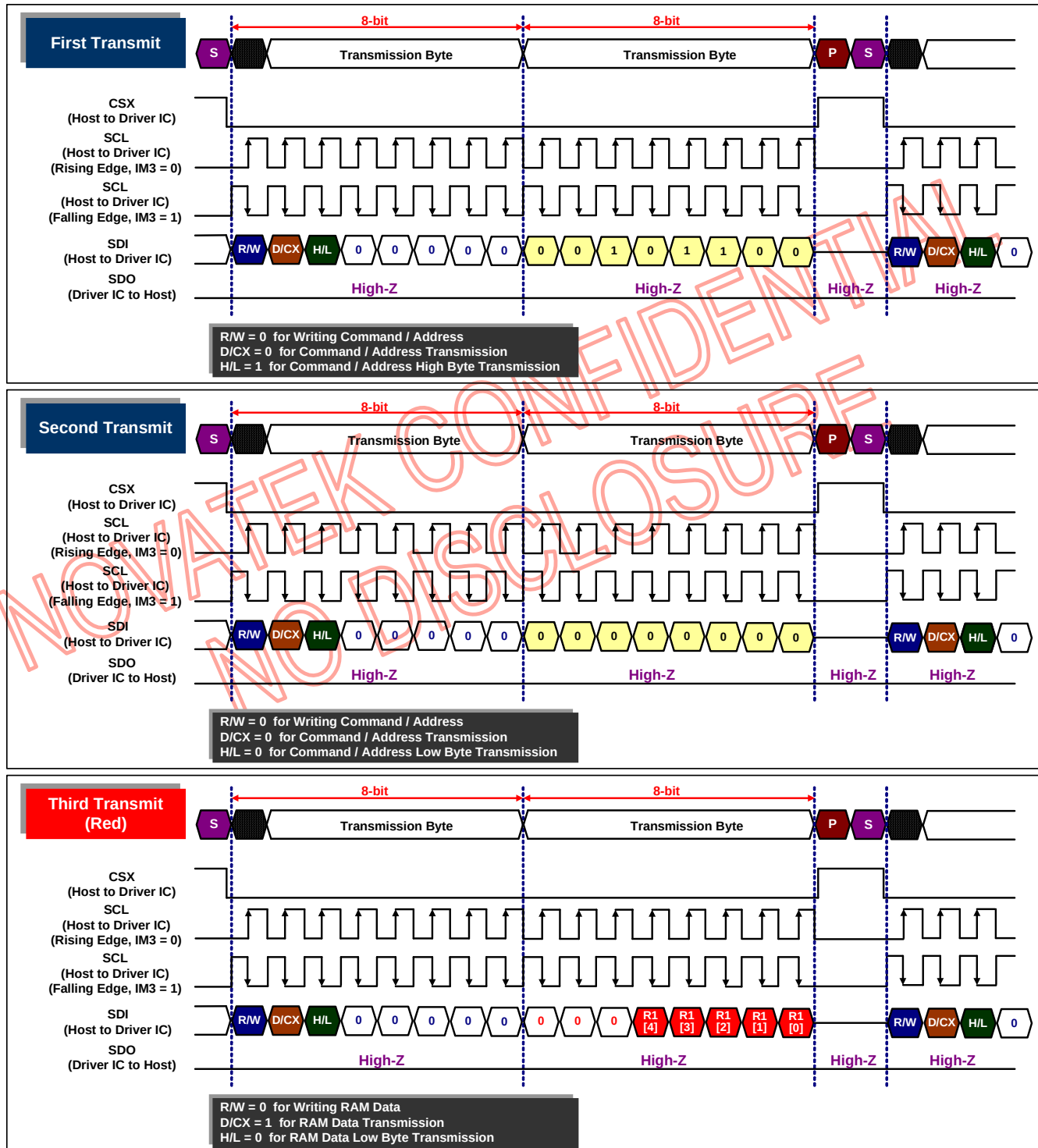


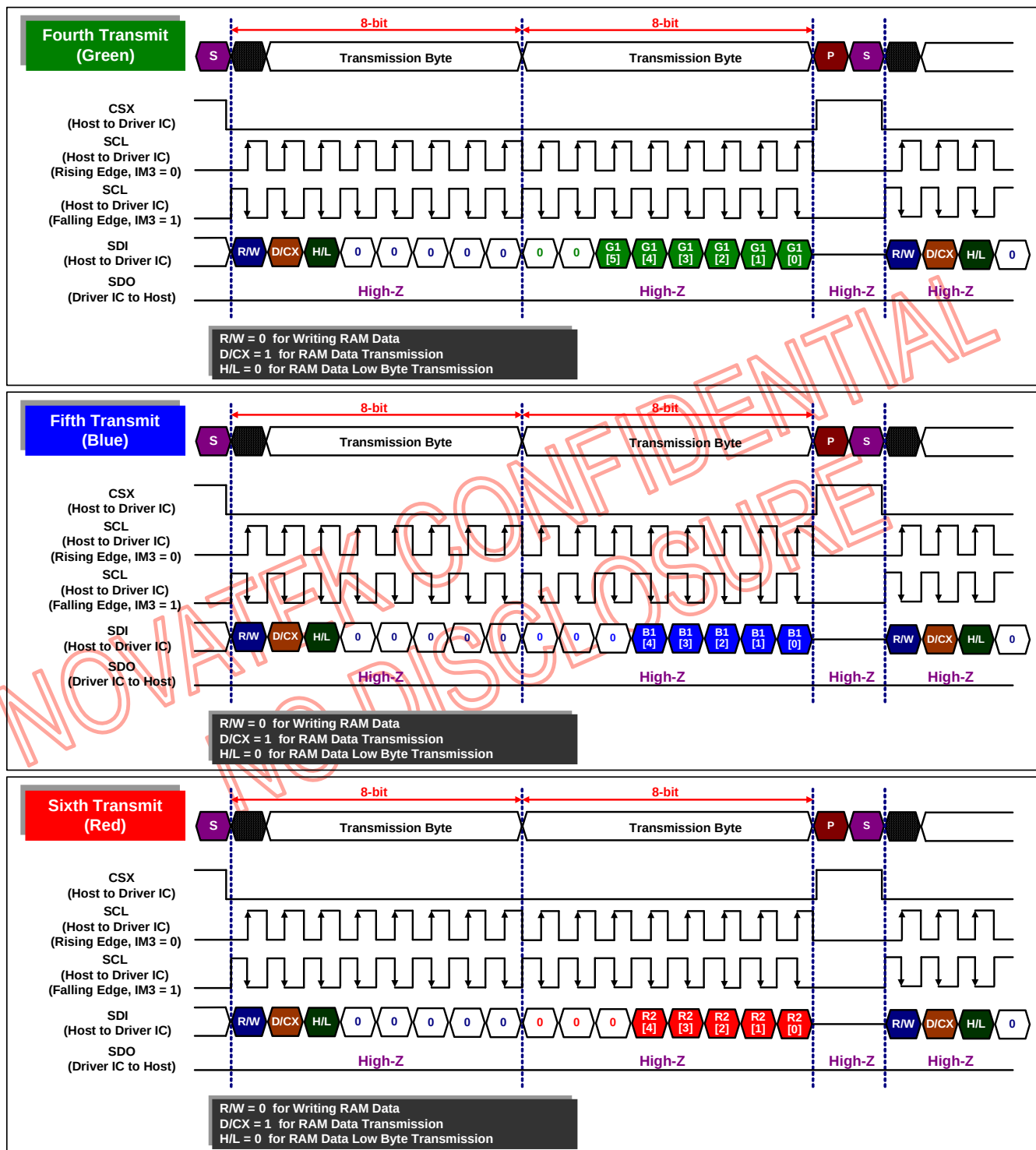
Fig. 5.1.6 Serial bus protocol for register read mode

5.1.3.3 SERIAL INTERFACE FOR DATA RAM WRITE

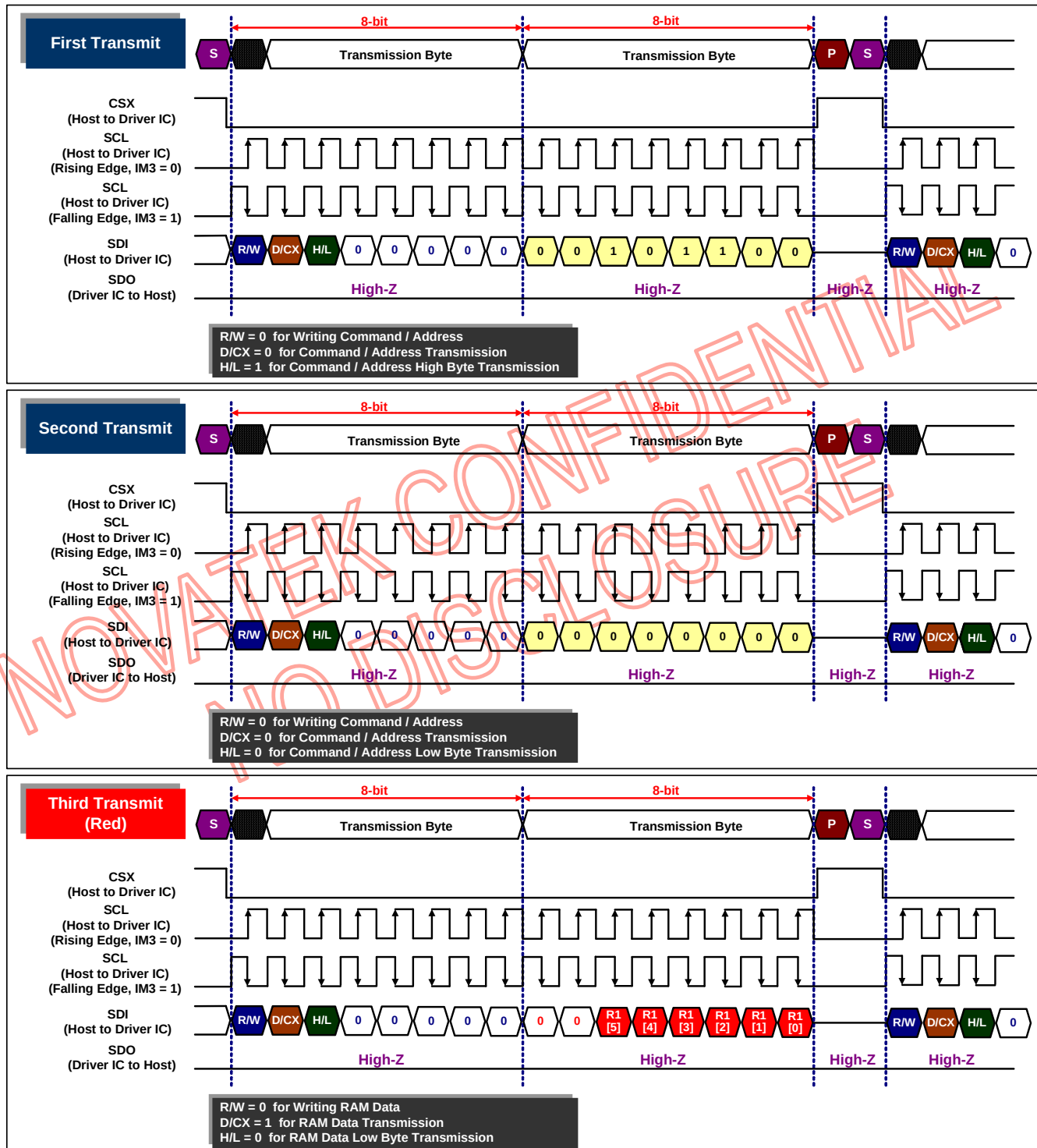
Different display data formats are available for three color depths supported by the LCM listed below:

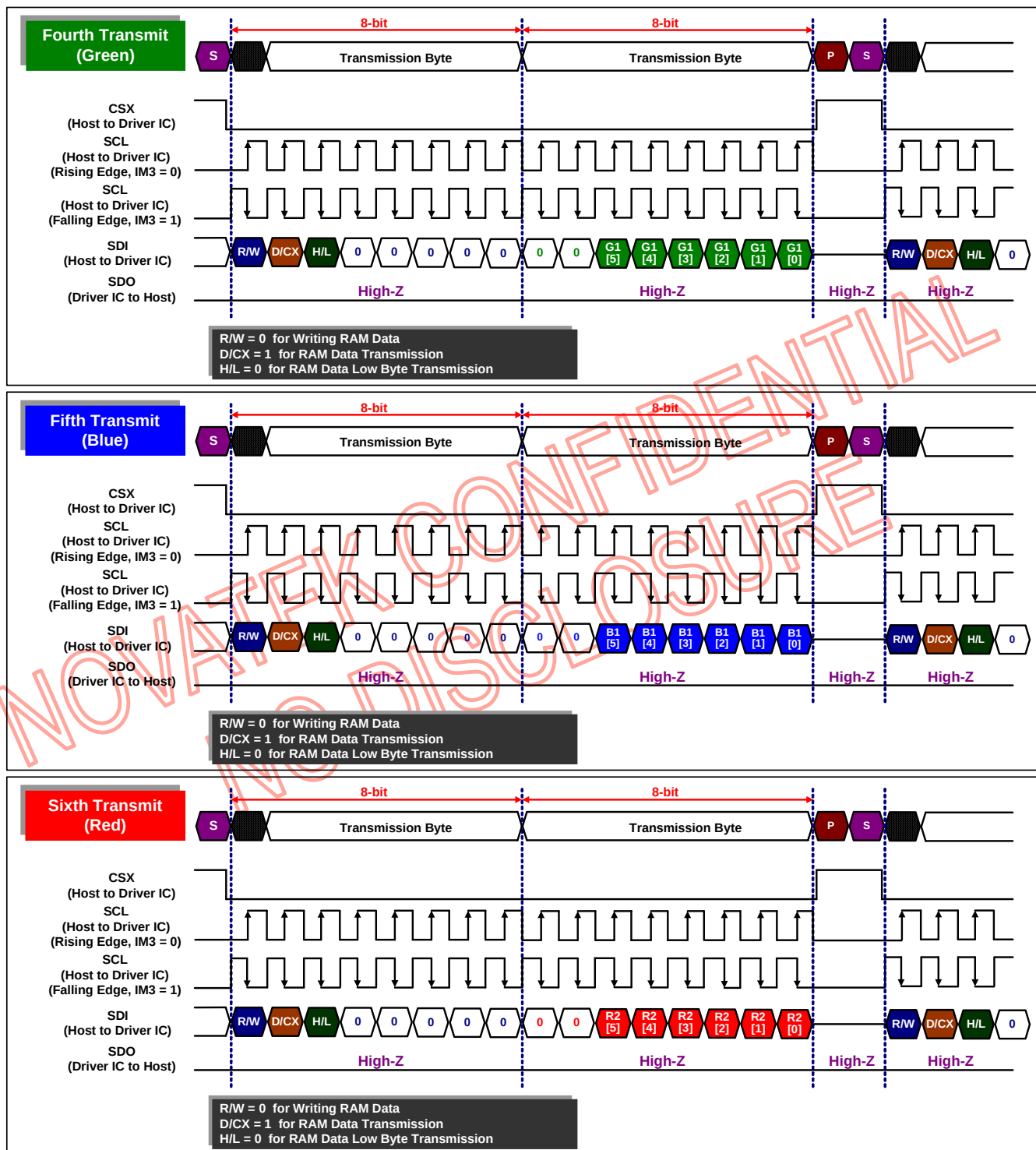
- 65K colors, RGB is 5-6-5-bit pixel data input (parameter of command 3A00h is 0x0005)





- 262K colors, RGB is 6-6-6-bit pixel data input (parameter of command 3A00h is 0x0006)





First Transmit

CSX (Host to Driver IC)
SCL (Host to Driver IC) (Rising Edge, IM3 = 0)
SCL (Host to Driver IC) (Falling Edge, IM3 = 1)
SDI (Host to Driver IC)
SDO (Driver IC to Host)

8-bit Transmission Byte
8-bit Transmission Byte

R/W D/CX H/L 0 0 0 0 0 0 0 0 0 0 1 0 1 1 0 0 0 0

High-Z High-Z High-Z High-Z

R/W = 0 for Writing Command / Address
D/CX = 0 for Command / Address Transmission
H/L = 1 for Command / Address High Byte Transmission

Second Transmit

CSX (Host to Driver IC)
SCL (Host to Driver IC) (Rising Edge, IM3 = 0)
SCL (Host to Driver IC) (Falling Edge, IM3 = 1)
SDI (Host to Driver IC)
SDO (Driver IC to Host)

8-bit Transmission Byte
8-bit Transmission Byte

R/W D/CX H/L 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

High-Z High-Z High-Z High-Z

R/W = 0 for Writing Command / Address
D/CX = 0 for Command / Address Transmission
H/L = 0 for Command / Address Low Byte Transmission

Third Transmit (Red)

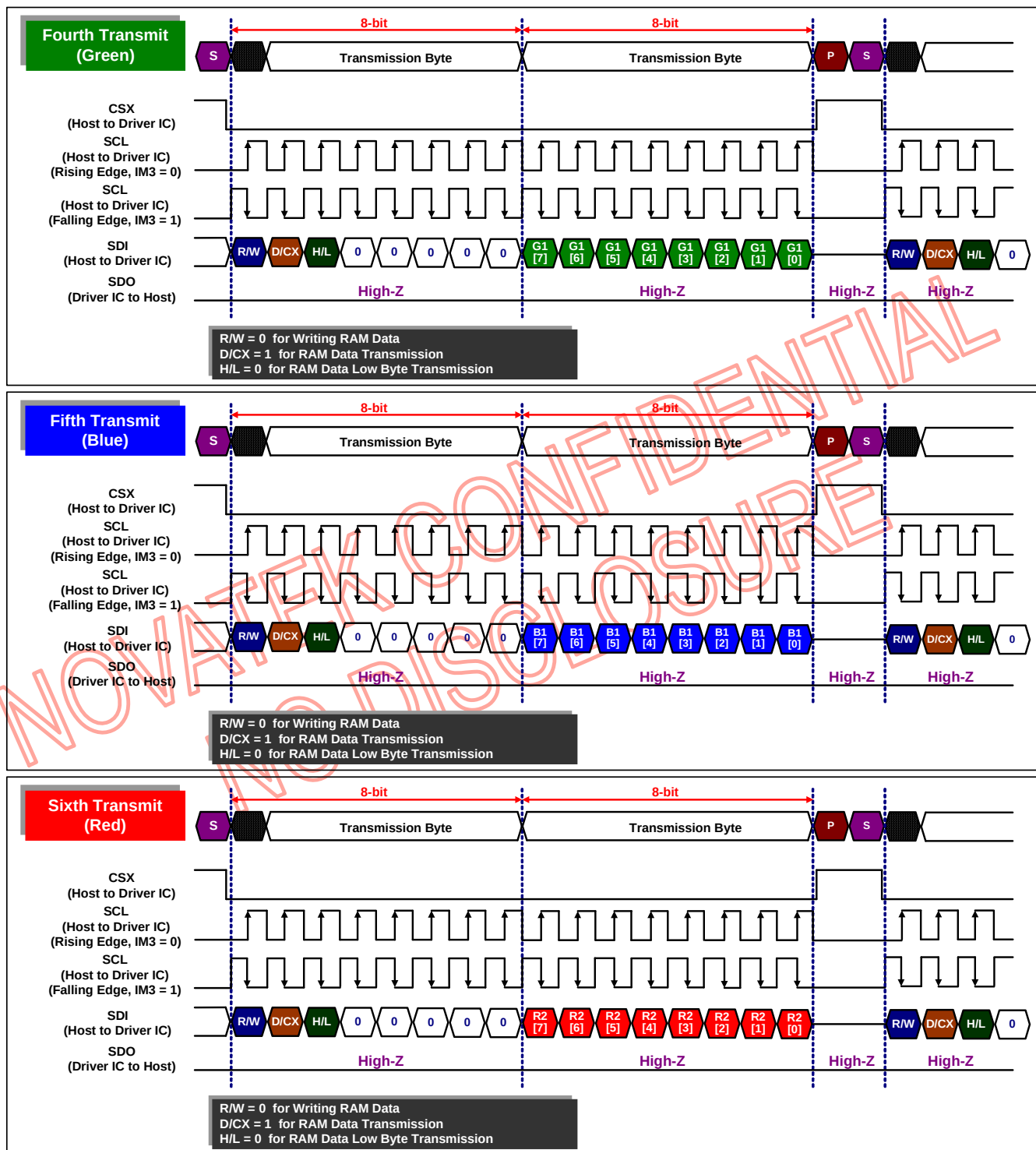
CSX (Host to Driver IC)
SCL (Host to Driver IC) (Rising Edge, IM3 = 0)
SCL (Host to Driver IC) (Falling Edge, IM3 = 1)
SDI (Host to Driver IC)
SDO (Driver IC to Host)

8-bit Transmission Byte
8-bit Transmission Byte

R/W D/CX H/L 0 0 0 0 0 0 R1 [7] R1 [6] R1 [5] R1 [4] R1 [3] R1 [2] R1 [1] R1 [0]

High-Z High-Z High-Z High-Z

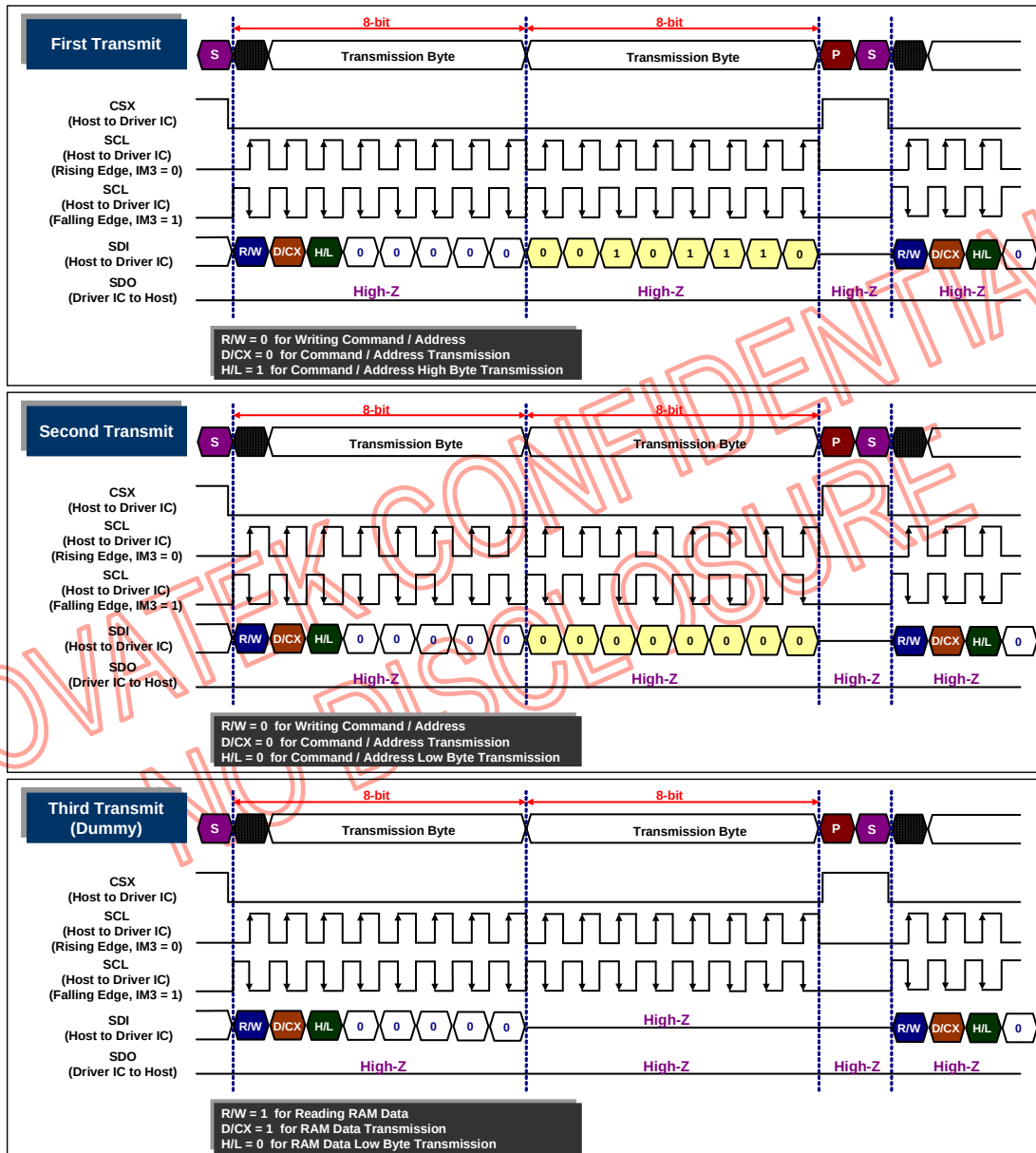
R/W = 0 for Writing RAM Data
D/CX = 1 for RAM Data Transmission
H/L = 0 for RAM Data Low Byte Transmission

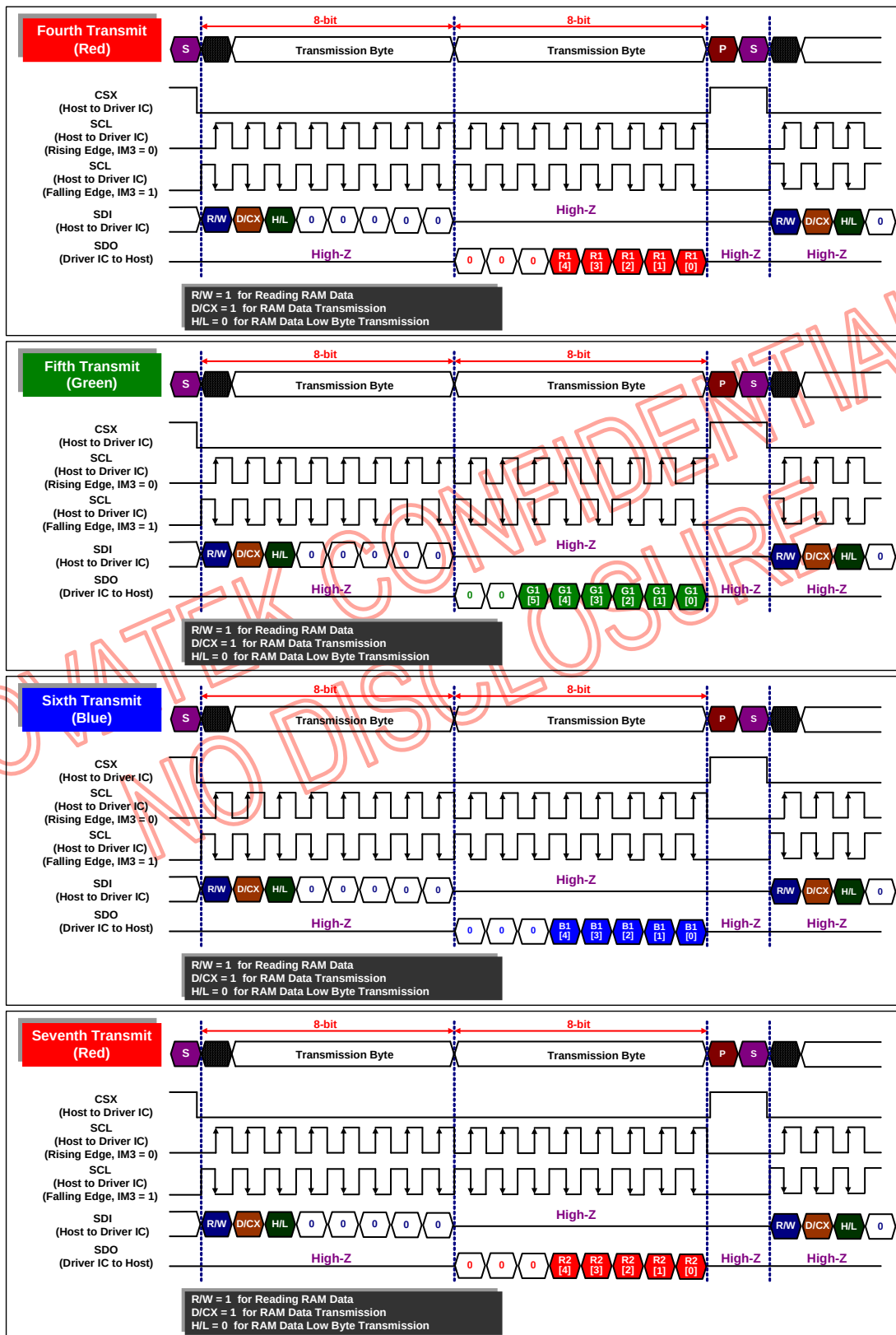


5.1.3.4 SERIAL INTERFACE FOR DATA RAM READ

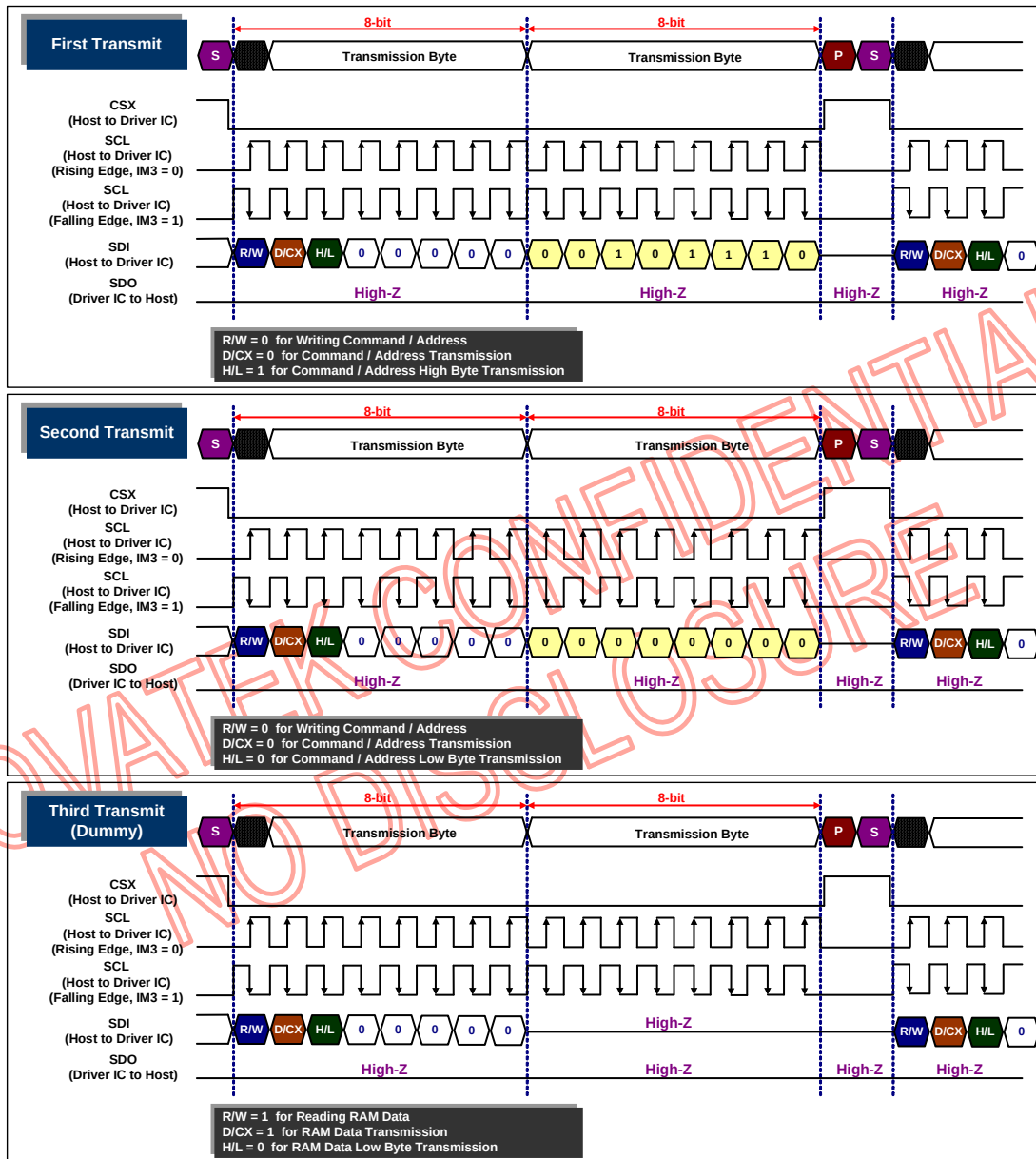
Different display data formats are available for three color depths supported by the LCM listed below:

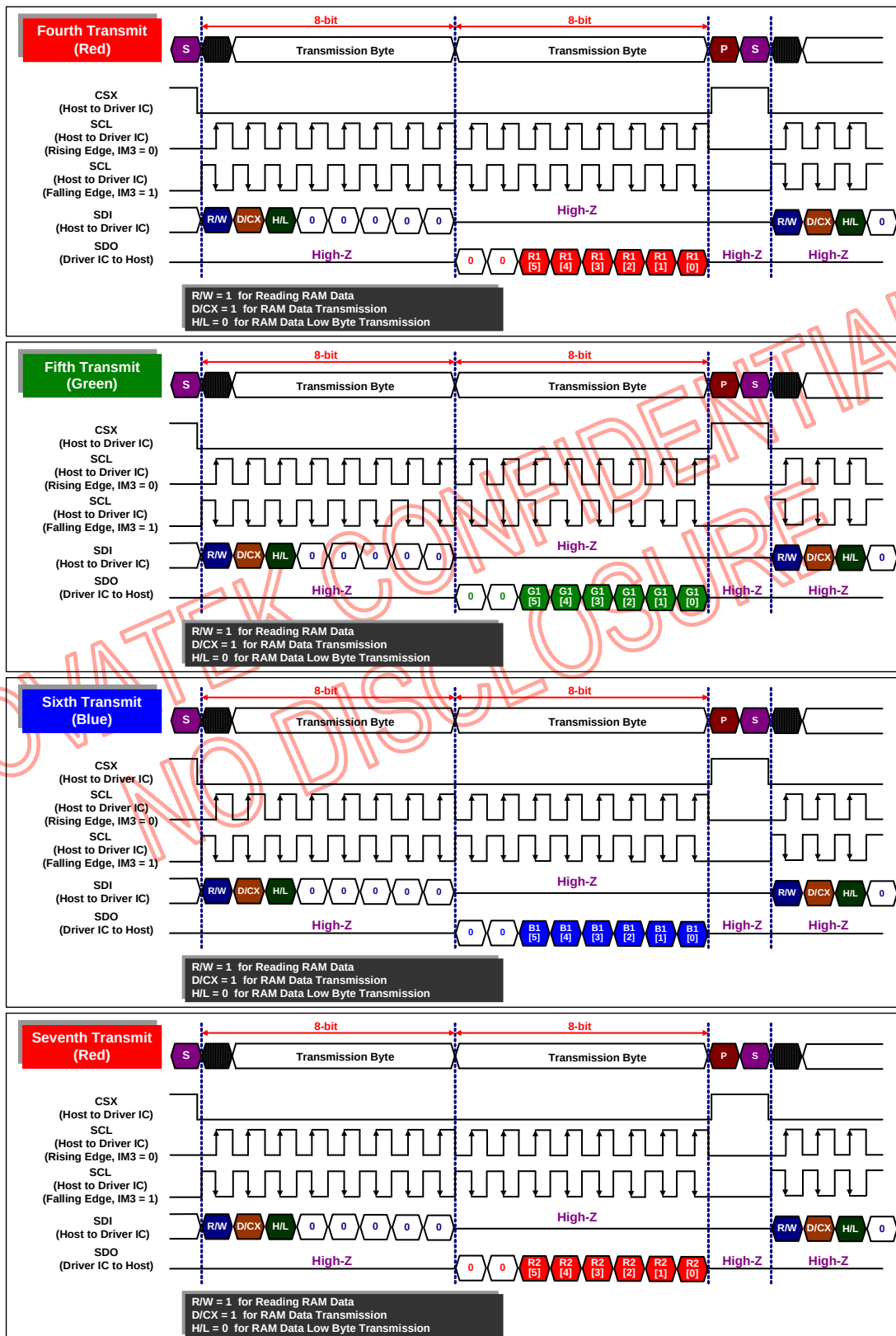
- 65K colors, RGB is 5-6-5-bit pixel data output (parameter of command 3A00h is 0x0005)



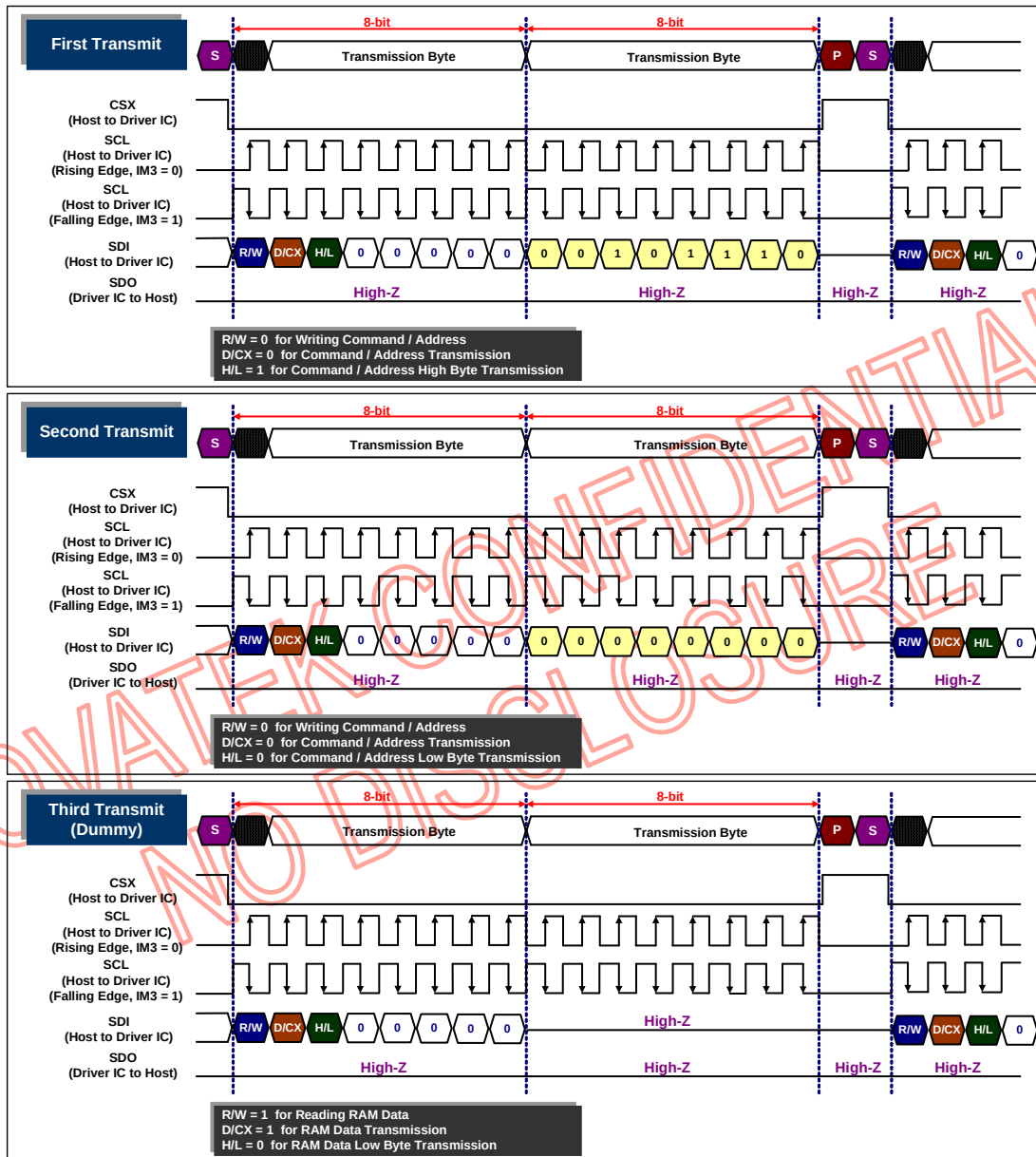


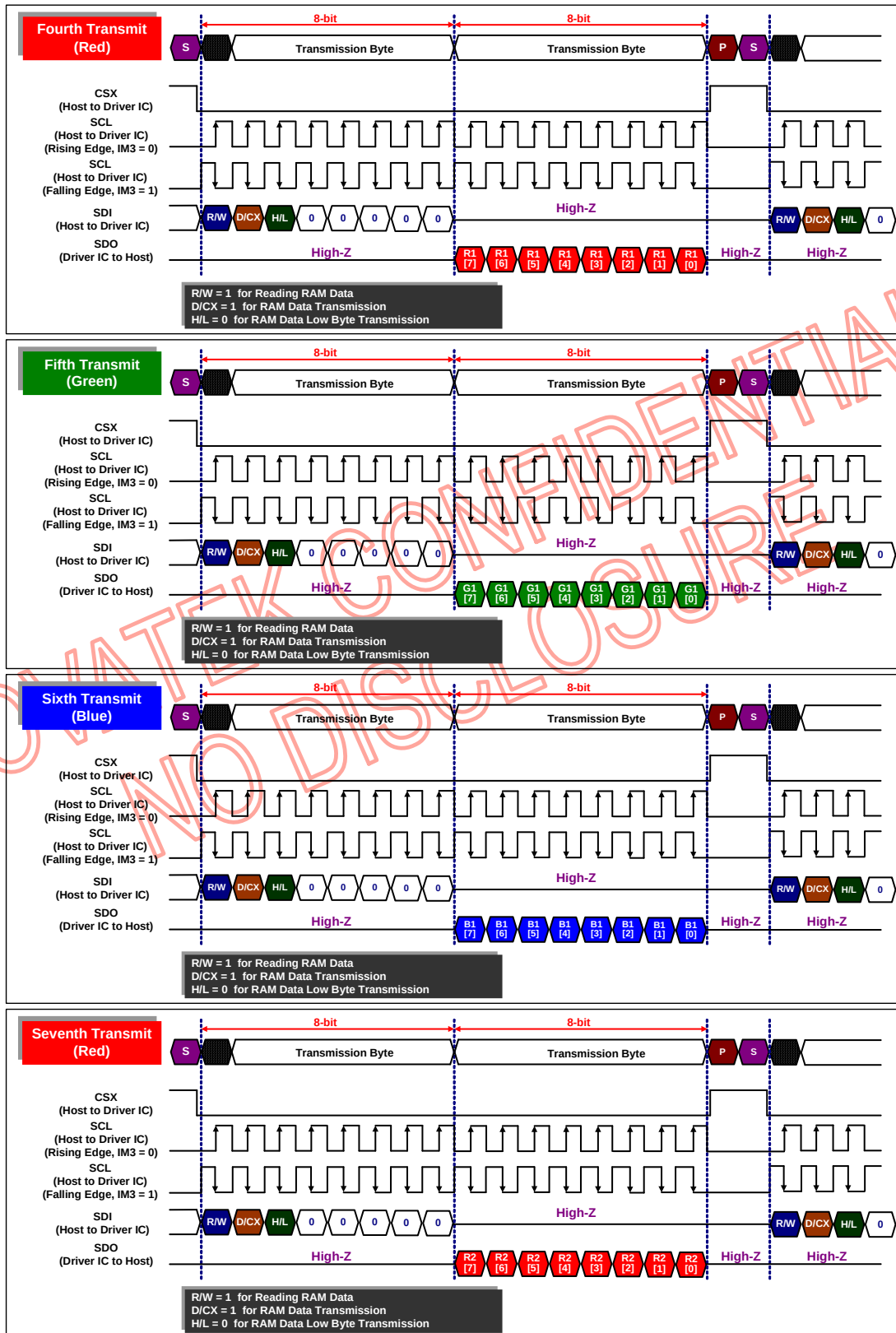
- 262K colors, RGB is 6-6-6-bit pixel data output (parameter of command 3A00h is 0x0006)





- 16.7M colors, RGB is 8-8-8-bit pixel data output (parameter of command 3A00h is 0x0007)





5.2 I2C Interface

The I2C-bus is for bi-directional, two-line communication between different ICs or modules. The two lines are the Serial Data line (I2C_SDA) and the Serial Clock Line (I2C_SCL). Both lines must be connected to a positive supply via pull-up resistors. Data transfer can be initiated only when the bus is not busy. Each byte of eight bits is followed by an acknowledge bit. The acknowledge bit is a HIGH level signal put on the bus by the transmitter during which time the master generates an extra acknowledgement related clock pulse. A slave receiver which is addressed must generate an acknowledgement after the reception of each byte. Also a master receiver must generate an acknowledgement after the reception of each byte that has been clocked out of the slave transmitter.

(a) I2C-Bus Protocol:

Before any data is transmitted on the I2C-bus, the device, which should respond is addressed first. There are four slave address can be selected by MCU. The slave addressing is always carried out with the first byte transmitted after the START procedure.

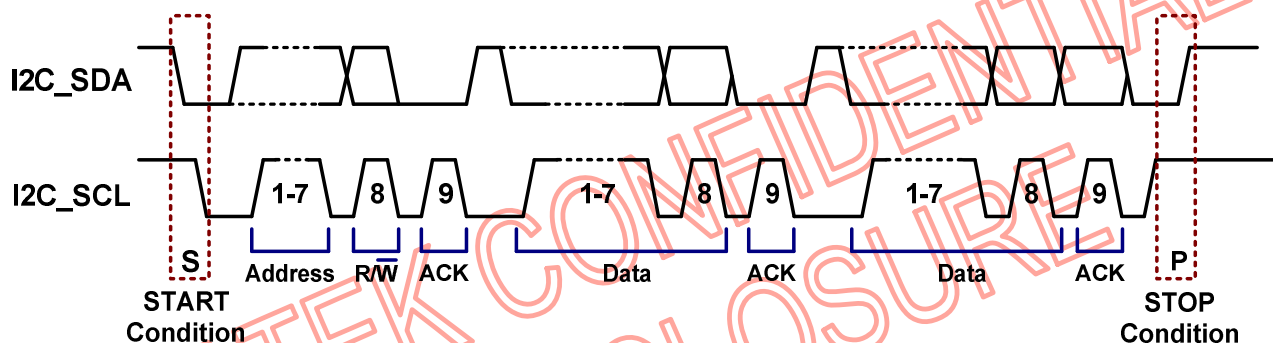


Fig. 5.2.1 Definition of I2C-Bus Protocol

(b) Definitions:

- Transmitter: The device which sends the data to the bus.
- Receiver: The device which receives the data from the bus.
- Master: The device which initiates a transfer, generates clock signals and terminates a transfer.
- Slave: The device addressed by a master.
- Multi-master: More than one master can attempt to control the bus at the same time without corrupting the message.
- Arbitration: Procedure to ensure that, if more than one master simultaneously tries to control the bus, only one is allowed to do so and the message is not corrupted.
- Synchronization: Procedure to synchronize the clock signals of two or more devices.

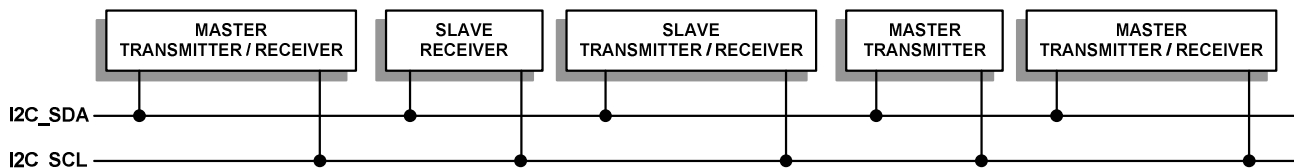


Fig. 5.2.2 System Configuration

5.2.1 Slave Address of I2C

NT35510 supports four slave addresses, 1001100, 1001101, 1001110 and 1001111 after the START procedure via I2C bus for MCU usage. There are 2 hard pins, I2C_SA1 and I2C_SA0 to determine the difference slave address. The slave address selection is described as the following table. The I2C interface address is selected from the external MPU.

Table 5.1.1 Selection Table of Slave Address

I2C_SA1	I2C_SA0	Slave Address	Notes
0	0	1001100	0000xxx and 1111xxx: Reversed
0	1	1001101	
1	0	1001110	
1	1	1001111	

5.2.2 Register Write Sequence of I2C Interface

NT35510 supports register write sequence via I2C-bus transfer. The detail transference sequences are illustrated and described as below.

- (1) Data transfers for register writing follow the format is shown in Fig.5.2.2.
- (2) After the START condition (S), a slave address is sent. R/W bit is setting to "zero" for WRITE.
- (3) The slave issues an ACK to master.
- (4) 16 bits register high byte address transfer first. Then transfer the register low byte address.
- (5) 16 bits register high byte data of parameter transfer first. Then transfer the register low byte data parameter.
- (6) A data transfer is always terminated by a STOP condition.

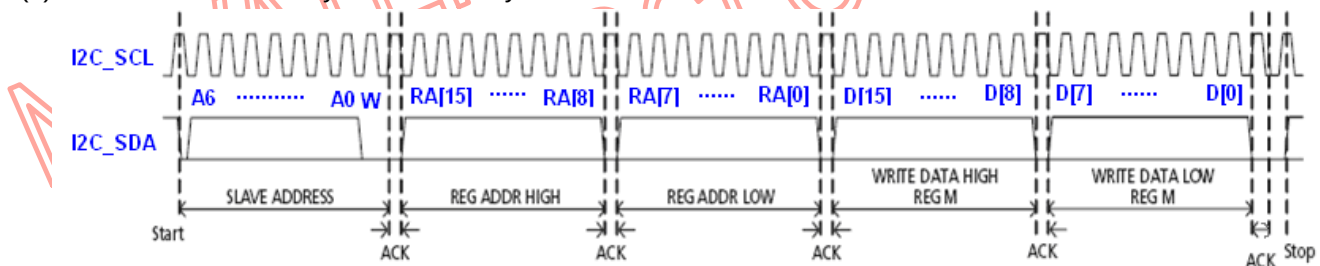


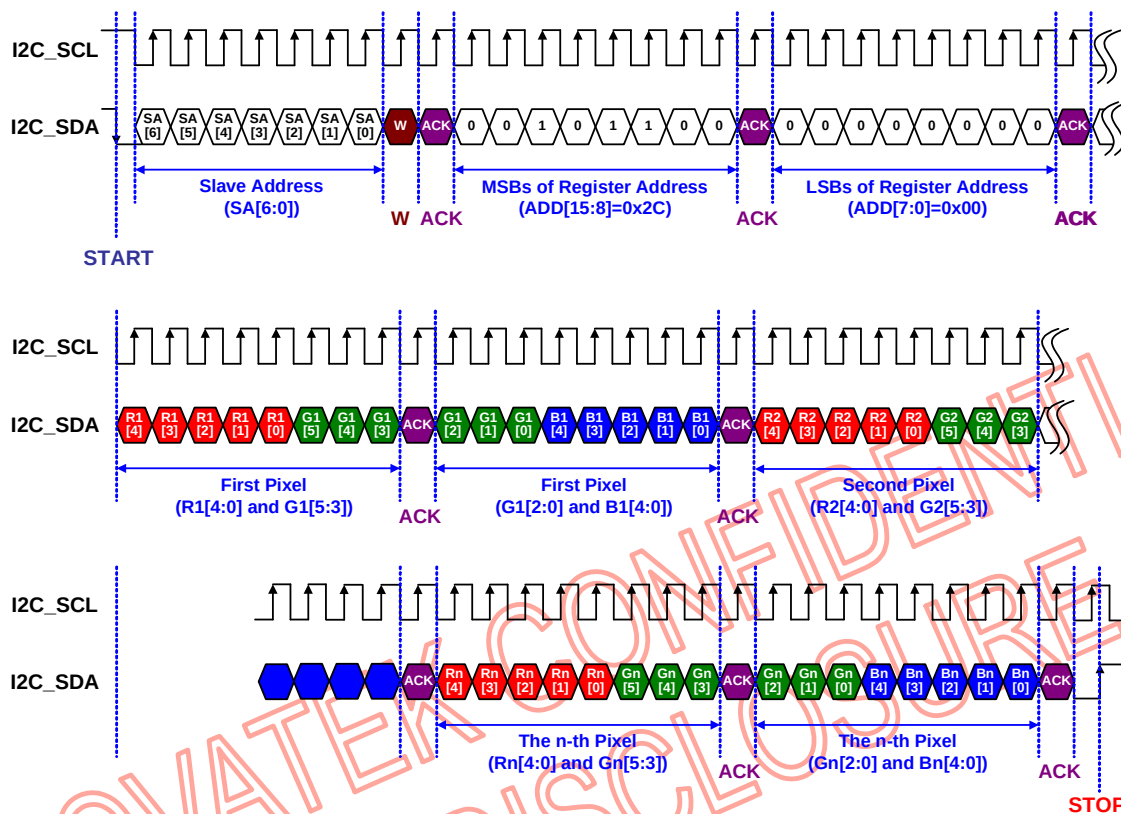
Fig. 5.2.3 Register Writing Timing of I2C Interface

5.2.3 RAM Data Write Sequence of I2C Interface

NT35510 supports sequential RAM data writing via I2C-Bus. NT35510 will increase the RAM address automatic by window address when the Host MCU write the RAM data via this way. The transfer protocol of window address setting can refer to the 5.2.3 Register Write Sequence. Different display data formats are available for three color depths supported by the LCM.

The sequential RAM writing timing is shown in below.

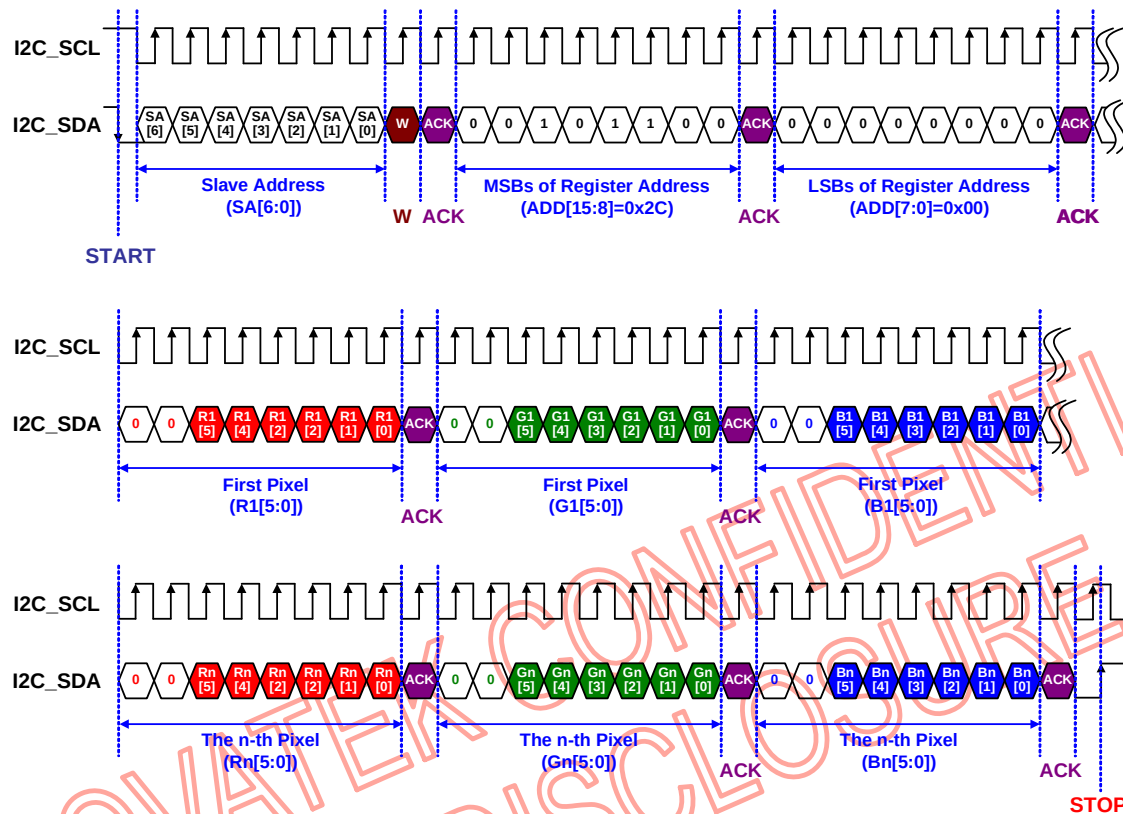
- 65K colors, RGB is 5-6-5-bit pixel data input (parameter of command 3A00h is 0x0005)



W: Write Bit, where W="0"
ACK: Acknowledge Bit, where ACK="0"

SA[6:0]: Slave Address
ADD[15:0]: Register Address, where ADD[15:0]="0x2C00"
R1[4:0], R2[4:0], ..., Rn[4:0]: The red color data of each pixel
G1[5:0], G2[5:0], ..., Gn[5:0]: The green color data of each pixel
B1[4:0], B2[4:0], ..., Bn[4:0]: The blue color data of each pixel

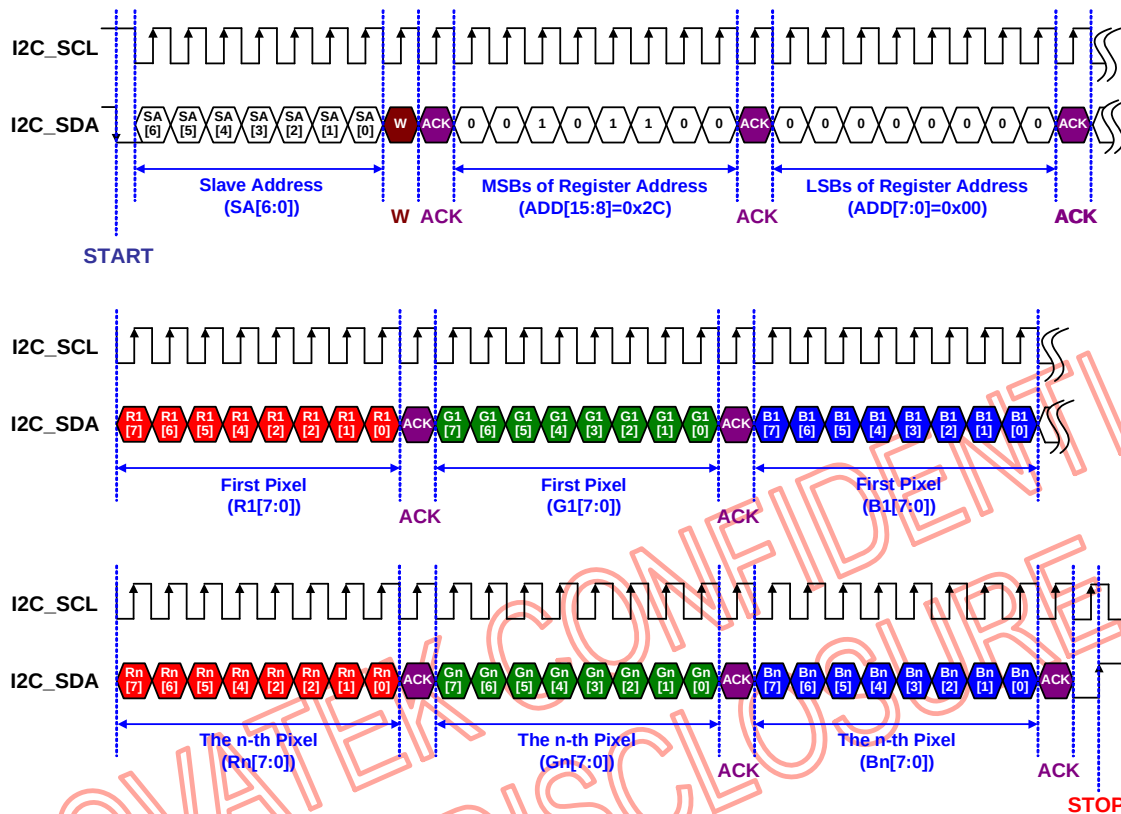
- 262K colors, RGB is 6-6-6-bit pixel data input (parameter of command 3A00h is 0x0006)



W: Write Bit, where W="0"
ACK: Acknowledge Bit, where ACK="0"

SA[6:0]: Slave Address
ADD[15:0]: Register Address, where ADD[15:0]="0x2C00"
R1[5:0], R2[5:0], ..., Rn[5:0]: The red color data of each pixel
G1[5:0], G2[5:0], ..., Gn[5:0]: The green color data of each pixel
B1[5:0], B2[5:0], ..., Bn[5:0]: The blue color data of each pixel

- 16.7M colors, RGB is 8-8-8-bit pixel data input (parameter of command 3A00h is 0x0007)



W: Write Bit, where W="0"
ACK: Acknowledge Bit, where ACK="0"

SA[6:0]: Slave Address
ADD[15:0]: Register Address, where ADD[15:0]="0x2C00"
R1[7:0], R2[7:0], ..., Rn[7:0]: The red color data of each pixel
G1[7:0], G2[7:0], ..., Gn[7:0]: The green color data of each pixel
B1[7:0], B2[7:0], ..., Bn[7:0]: The blue color data of each pixel

5.2.4 Register Read Sequence of I2C Interface

NT35510 supports register read sequence via I2C-bus transfer. Register data reading transfers follow the format and is shown in Fig.5.2.4.

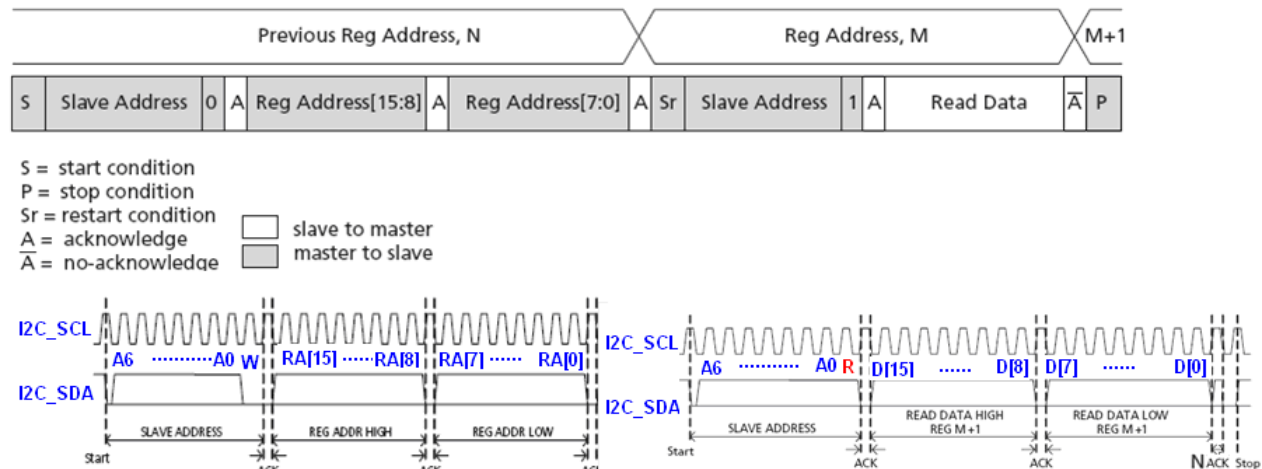


Fig. 5.2.4 Register Reading Timing of I2C Interface

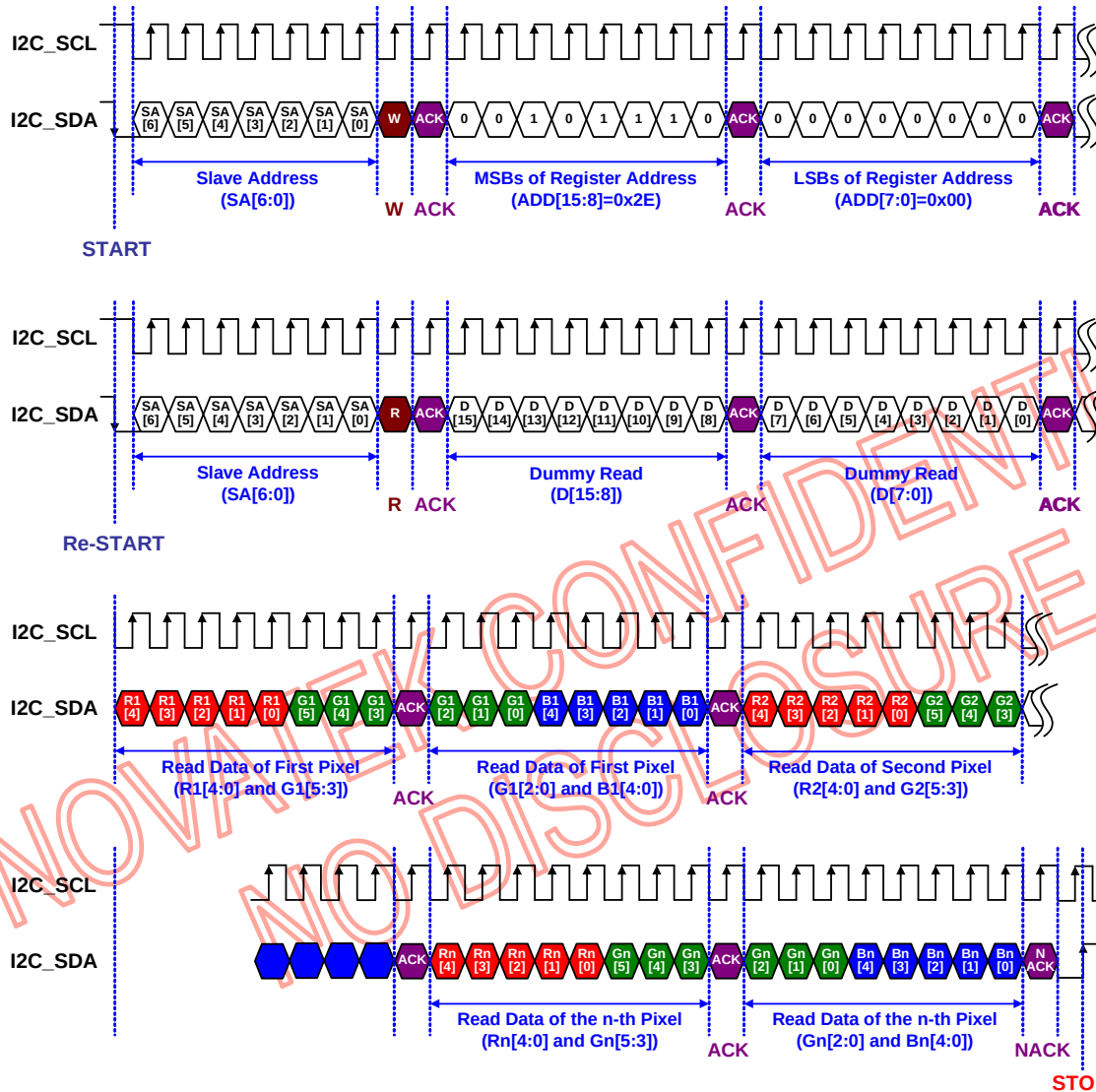
5.2.5 RAM Data Read Sequence of I2C Interface

NT35510 supports RAM data read function for I2C interface.

The master MCU need to send the RAM address of reading first and transfer protocol can refer to the 5.2.3 Register Write Sequence. Then the master MCU need to send the RAM data read register "2E00h" to NT35510. And finally, the MCU can send the following RAM data reading timing to feedback single RAM data value by one complete I2C packet.

The RAM data reading timing is shown in below.

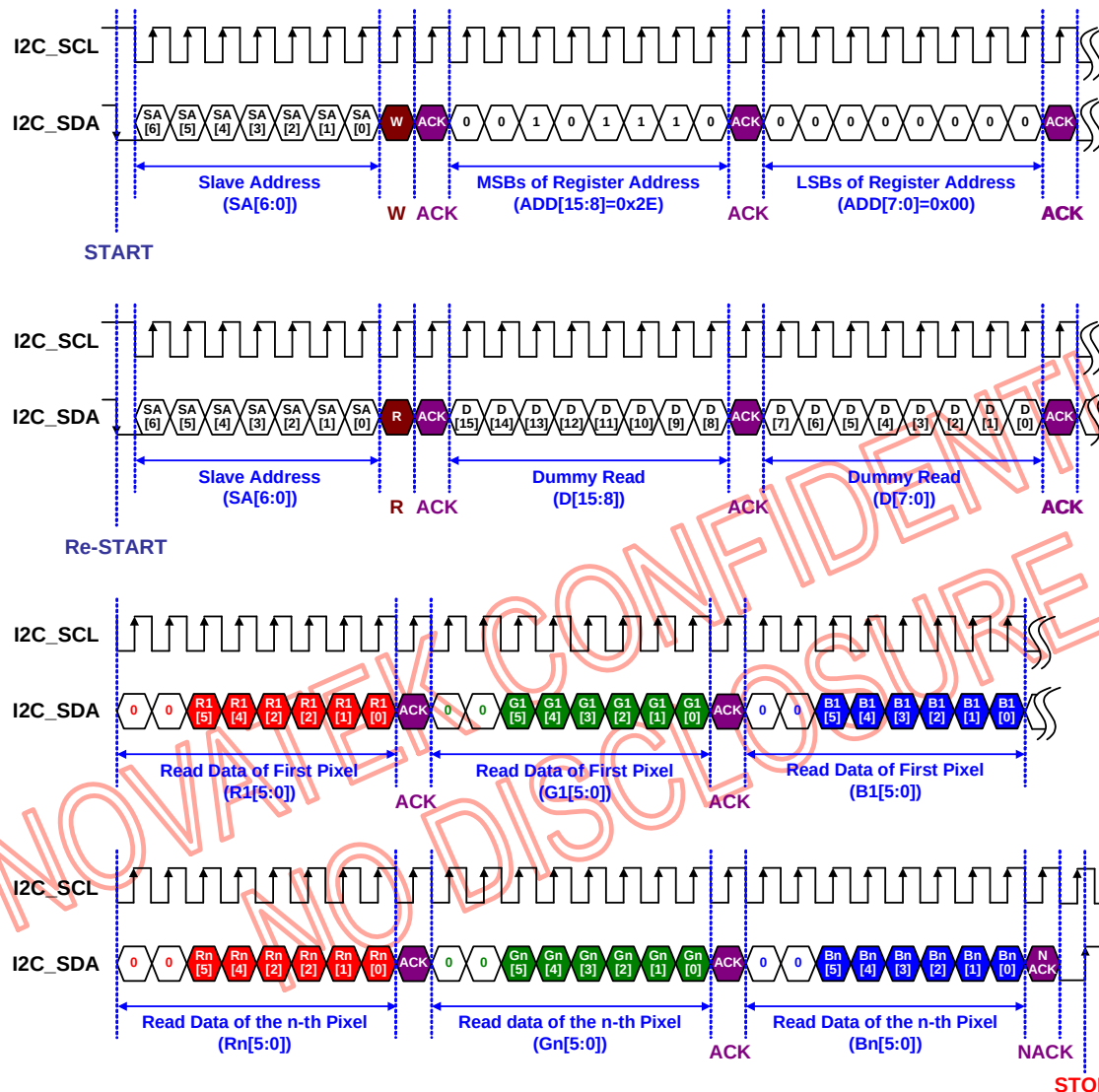
- 65K colors, RGB is 5-6-5-bit pixel data output (parameter of command 3A00h is 0x0005)



W: Write Bit, where W="0"
R: Read Bit, where R="1"
ACK: Acknowledge Bit, where ACK="0"
NACK: Non-acknowledge Bit, where NACK="1"

SA[6:0]: Slave Address
ADD[15:0]: Register Address, where ADD[15:0]="0x2E00"
R1[4:0], R2[4:0], ..., Rn[4:0]: The red color data of each pixel
G1[5:0], G2[5:0], ..., Gn[5:0]: The green color data of each pixel
B1[4:0], B2[4:0], ..., Bn[4:0]: The blue color data of each pixel

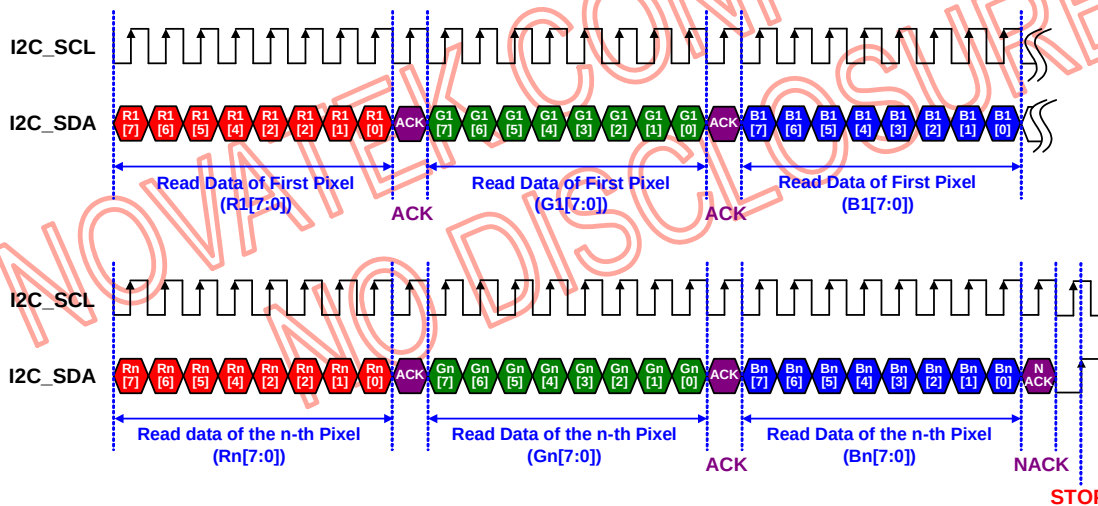
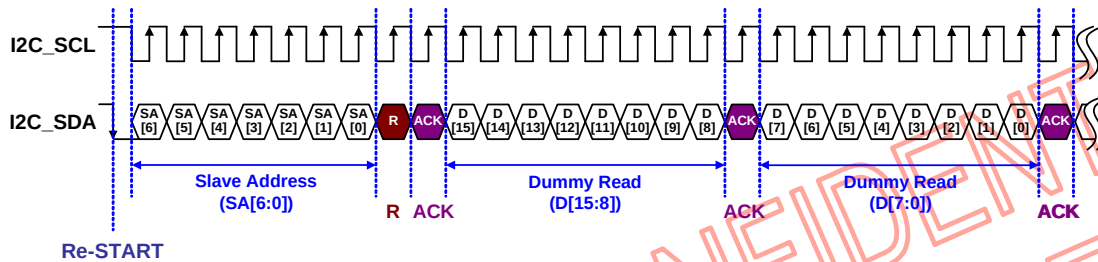
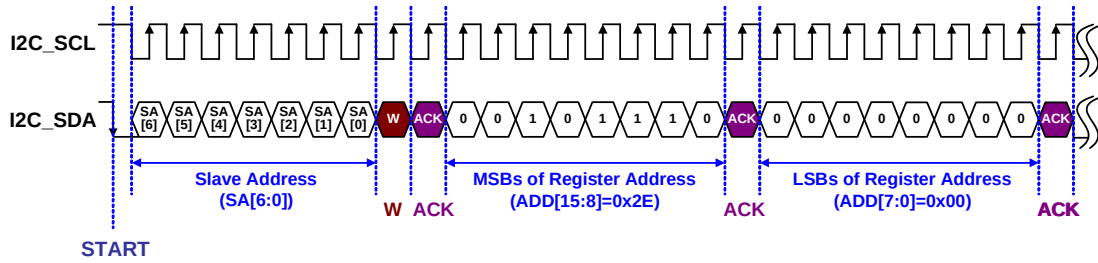
- 262K colors, RGB is 6-6-6-bit pixel data output (parameter of command 3A00h is 0x0006)



W: Write Bit, where W="0"
R: Read Bit, where R="1"
ACK: Acknowledge Bit, where ACK="0"
NACK: Non-acknowledge Bit, where NACK="1"

SA[6:0]: Slave Address
ADD[15:0]: Register Address, where ADD[15:0]="0x2E00"
R1[5:0], R2[5:0], ..., Rn[5:0]: The red color data of each pixel
G1[5:0], G2[5:0], ..., Gn[5:0]: The green color data of each pixel
B1[5:0], B2[5:0], ..., Bn[5:0]: The blue color data of each pixel

- 16.7M colors, RGB is 8-8-8-bit pixel data output (parameter of command 3A00h is 0x0007)



W: Write Bit, where W="0"
 R: Read Bit, where R="1"
 ACK: Acknowledge Bit, where ACK="0"
 NACK: Non-acknowledge Bit, where NACK="1"

SA[6:0]: Slave Address
 ADD[15:0]: Register Address, where ADD[15:0]="0x2E00"
 R1[7:0], R2[7:0], ..., Rn[7:0]: The red color data of each pixel
 G1[7:0], G2[7:0], ..., Gn[7:0]: The green color data of each pixel
 B1[7:0], B2[7:0], ..., Bn[7:0]: The blue color data of each pixel

5.3 MIPI Interface

The Display Serial Interface standard defines protocols between a host processor and peripheral devices that adhere to MIPI Alliance standards for mobile device interfaces. The DSI standard builds on existing standards by adopting pixel formats and command set defined in MIPI Alliance standards.

DSI-compliant peripherals support either of two basic modes of operation: Command Mode and Video Mode. Which mode is used depends on the architecture and capabilities of the peripheral. The mode definitions reflect the primary intended use of DSI for display interconnect, but are not intended to restrict DSI from operating in other applications.

Typically, a peripheral is capable of Command Mode operation or Video Mode operation. Some Video Mode display modules also include a simplified form of Command Mode operation in which the display module may refresh its screen from a reduced-size, or partial, frame buffer, and the interface (DSI) to the host processor may be shut down to reduce power consumption.

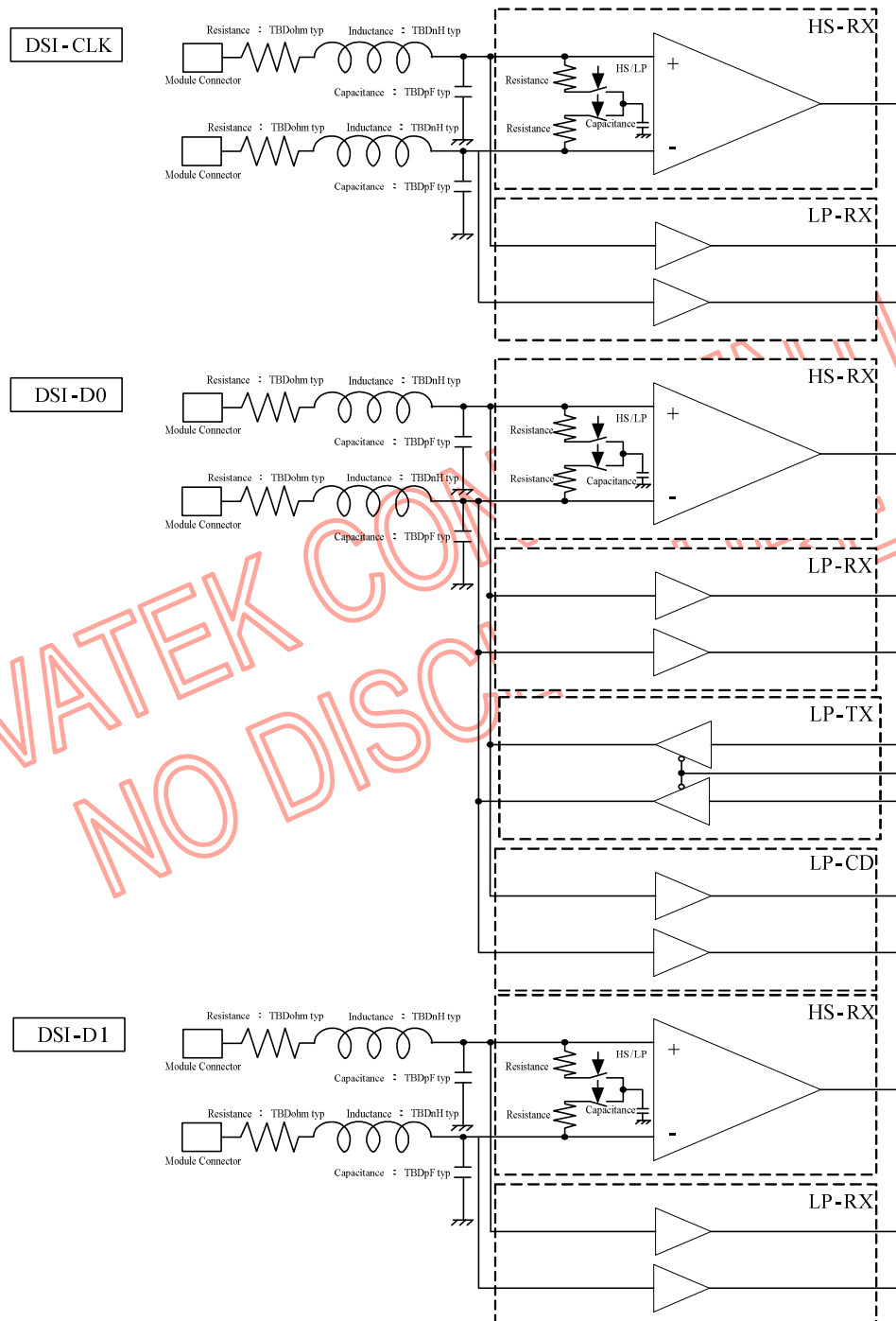
Command Mode refers to operation in which transactions primarily take the form of sending commands and data to a peripheral, such as a display module, that incorporates a display controller. The display controller may include local registers and a frame buffer. Systems using Command Mode write to, and read from, the registers and frame buffer memory. The host processor indirectly controls activity at the peripheral by sending commands, parameters and data to the display controller. The host processor can also read display module status information or the contents of the frame memory. Command Mode operation requires a bidirectional interface.

Video Mode refers to operation in which transfers from the host processor to the peripheral take the form of a real-time pixel stream. In normal operation, the display module relies on the host processor to provide image data at sufficient bandwidth to avoid flicker or other visible artifacts in the displayed image. Video information should only be transmitted using High Speed Mode. Some Video Mode architectures may include a simple timing controller and partial frame buffer, used to maintain a partial-screen or lower-resolution image in standby or Low Power Mode. This permits the interface to be shut down to reduce power consumption. To reduce complexity and cost, systems that only operate in Video Mode may use a unidirectional data path.

Configuration:

Lane Pair	MCU (Master) Display Module (Slave)
Clock Lane	Unidirectional Lane ■ Clock Only ■ Escape Mode(ULPS Only)
Data Lane 0	Bi-directional Lane ■ Forward High-Speed ■ Bi-directional Escape Mode ■ Bi-directional LPDT
Data Lane 1	Unidirectional Lane ■ Forward High-Speed ■ Escape Mode (ULPM only) ■ No LPDT

5.3.1 Display Module Pin Configuration for DSI



5.3.2 Display Serial Interface (DSI)

5.3.2.1 GENERAL DESCRIPTION

Communication sequences between the MCU and the display module are described on chapter “5.3.2.3.3 Communication Sequences”.

The communication can be separated 2 different levels between the MCU and the display module:

- Low level communication what is done on the interface level
- High level communication what is done on the packet level

5.3.2.2 INTERFACE LEVEL COMMUNICATION

5.3.2.2.1 GENERAL

The display module uses data and clock lane differential pairs for DSI (DSI-1M). Both differential lane pairs can be driven Low Power (LP) or High Speed (HS) mode.

Low Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disable (A termination resistor of the receiver is disable) and it can be driven into a low power mode.

High Speed mode means that differential pairs (The termination resistor of the receiver is enable) are not used in the single end mode. There are used different modes and protocol in each mode when there is wanted to transfer information from the MCU to the display module and vice versa.

The State Codes of the High Speed (HS) and Low Power (LP) lane pair are defined below.

Lane Pair State Code	Line DC Voltage Levels		High Speed(HS)	Low-Power(LP)	
	Dn+ -line	Dn- -line	Burst Mode	Control Mode	Escape Mode
HS-0	Low (HS)	High (HS)	Differential-0	Note 1	Note 1
HS-1	High (HS)	Low (HS)	Differential-1	Note 1	Note 1
LP-00	Low (LP)	Low (LP)	Not Defined	Bridge	Space
LP-01	Low (LP)	High (LP)	Not Defined	HS-Request	Mark-0
LP-10	High (LP)	Low (LP)	Not Defined	LP-Request	Mark-1
LP-11	High (LP)	High (LP)	Not Defined	Stop	Note 2

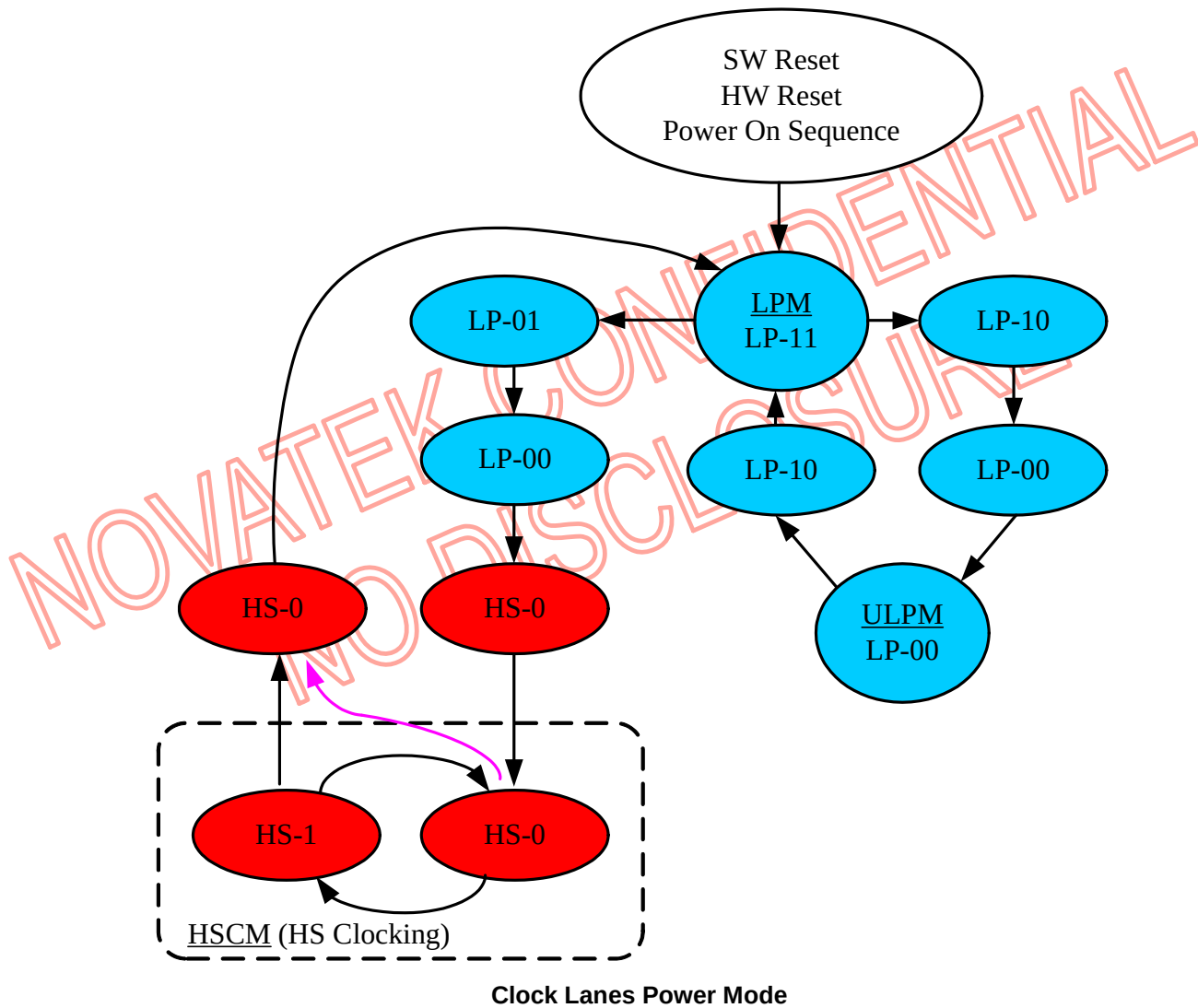
NOTES:

1. Low-Power Receivers (LP-Rx) of the lane pair are checking the LP-00 state code, when the Lane Pair is in the High Speed (HS) mode.
2. If Low-Power Receivers (LP-Rx) of the lane pair recognizes LP-11 state code, the lane pair returns to LP-11 of the Control Mode.

5.3.2.2.2 DSI-CLK LANES

DSI-CLK+/- lanes can be driven into three different power modes: Low Power Mode (LPM), Ultra Low Power Mode (ULPM) or High Speed Clock Mode (HSCM). Clock lanes are in a single end mode (LP = Low Power) when there is entering or leaving Low Power Mode (LPM) or Ultra Low Power Mode (ULPM). Clock lanes are in the single end mode (LP = Low Power) when there is entering in or leaving out High Speed Clock Mode (HSCM). These entering and leaving protocols are using clock lanes in the single end mode to generate an entering or leaving sequences.

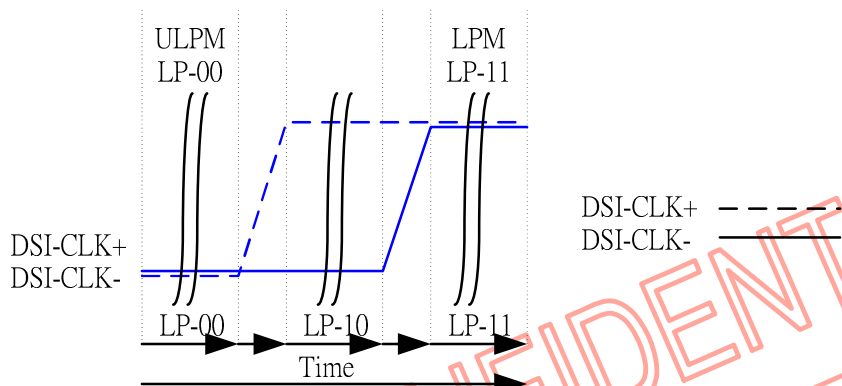
The principle flow chart of the different clock lanes power modes is illustrated below.



5.3.2.2.1 LOW POWER MODE (LPM)

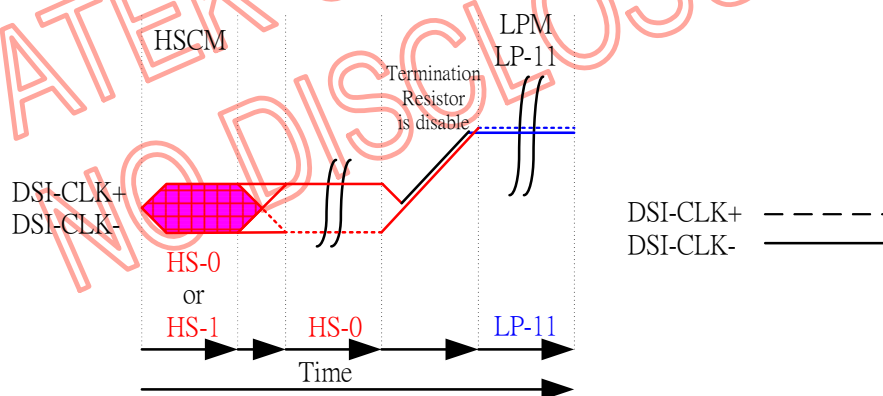
DSI-CLK+/- lanes can be driven to the Low Power Mode (LPM), when DSI-CLK lanes are entering LP-11 State Code, in three different ways:

- 1) After SW Reset, HW Reset or Power On Sequence =>LP-11
- 2) After DSI-CLK+/- lanes are leaving Ultra Low Power Mode (ULPM, LP-00 State Code) =>LP-10 =>LP-11 (LPM). This sequence is illustrated below.



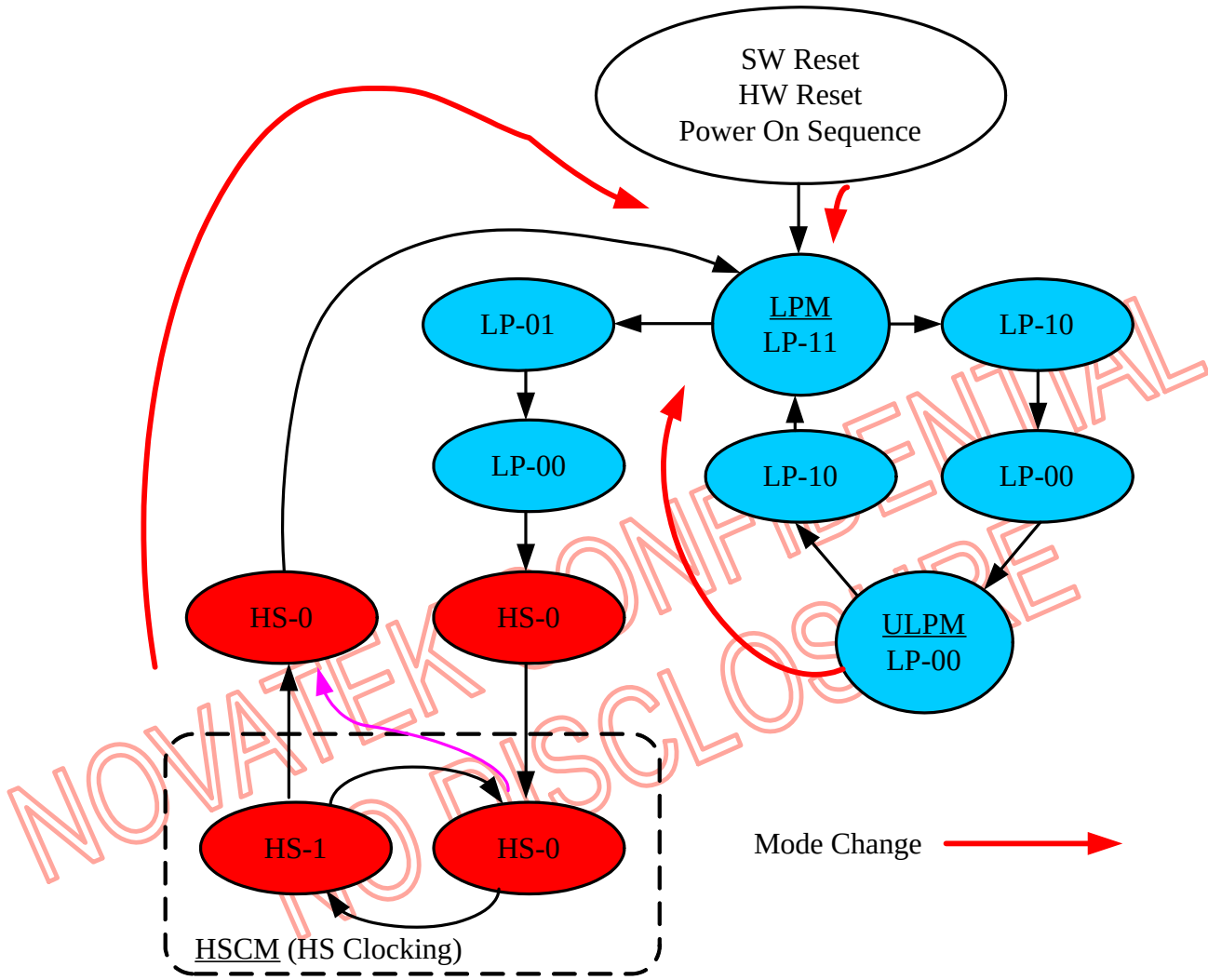
From ULPM to LPM

- 3) After DSI-CLK+/- lanes are leaving High Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) =>HS-0 =>LP-11 (LPM). This sequence and all three mode changes are illustrated below.



From High Speed Clock Mode (HSCM) to LPM

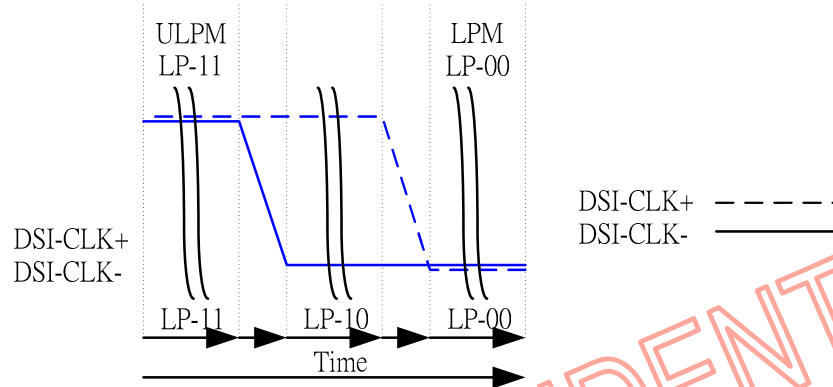
All three mode changes are illustrated a flow chart below.



All Three Mode Change to LPM on the Flow Chart

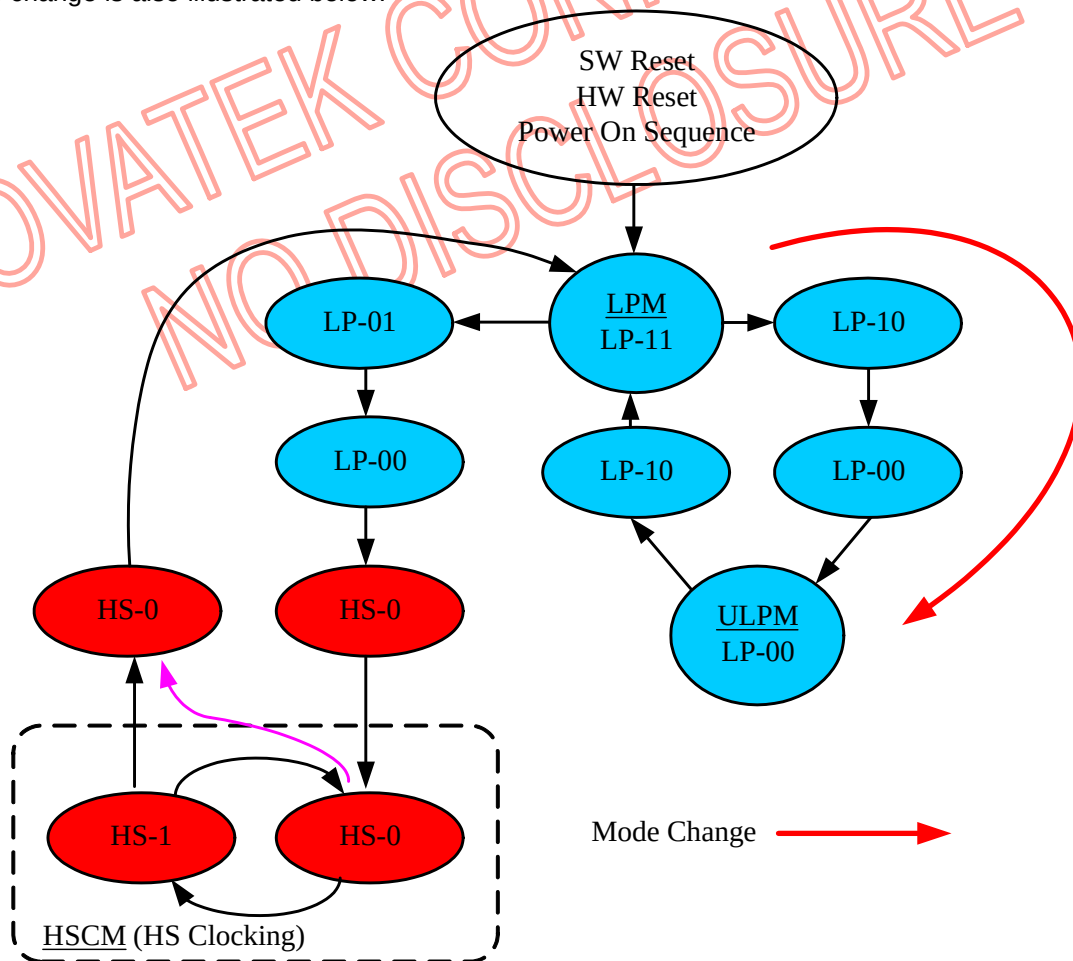
5.3.2.2.2 ULTRA LOW POWER MODE (ULPM)

DSI-CLK+/- lanes can be driven to the Ultra Low power Mode (ULPM), when DSI-CLK lanes are entering LP-00 State Code. The only entering possibility is from the Low Power Mode (LPM, LP-11 State Code) =>LP-10 =>LP-00 (ULPM). This sequence is illustrated below.



From LPM to ULPM

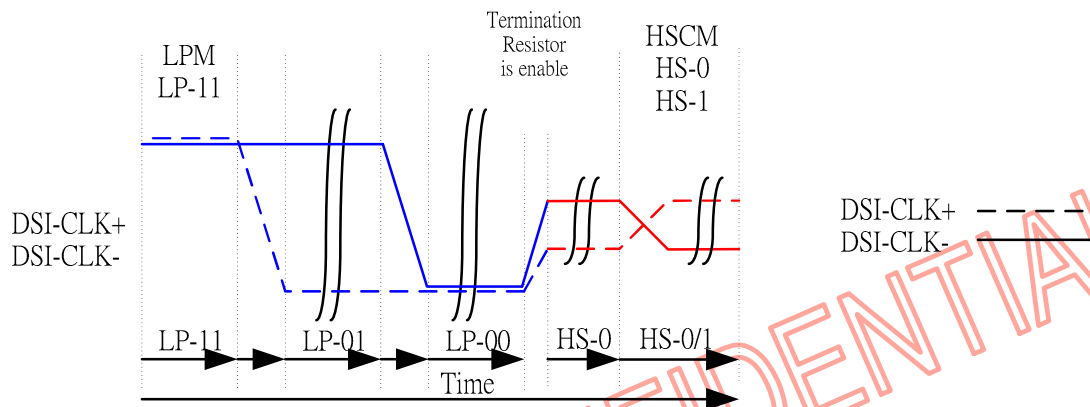
The mode change is also illustrated below.



Mode Change from LPM to ULPM on the Flow Chart

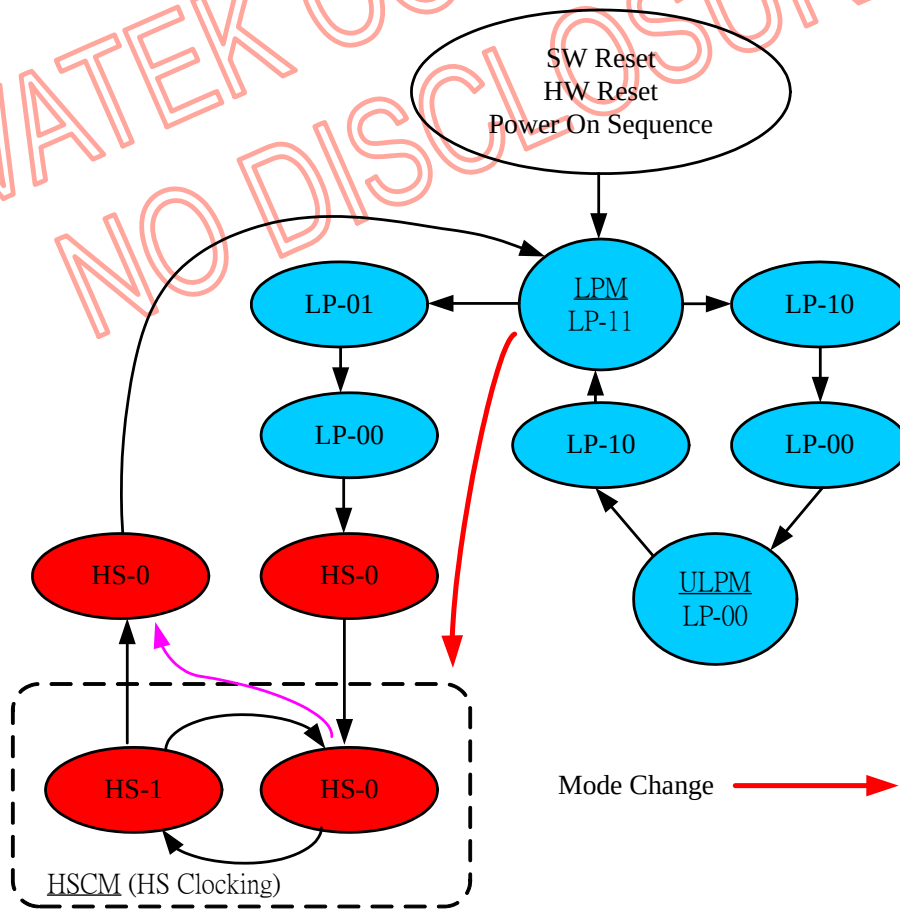
5.3.2.2.2.3 HIGH SPEED CLOCK MODE (HSCM)

DSI-CLK+/- lanes can be driven to the High Speed Clock Mode (HSCM), when DSI-CLK lanes are starting to work between HS-0 and HS-1 State Codes. The only entering possibility is from the Low Power Mode (LPM, LP-11 State Code) =>LP-01 =>LP-00 =>HS-0 =>HS-0/1 (HSCM). This sequence is illustrated below.



From LPM to HSCM

The mode change is also illustrated below.

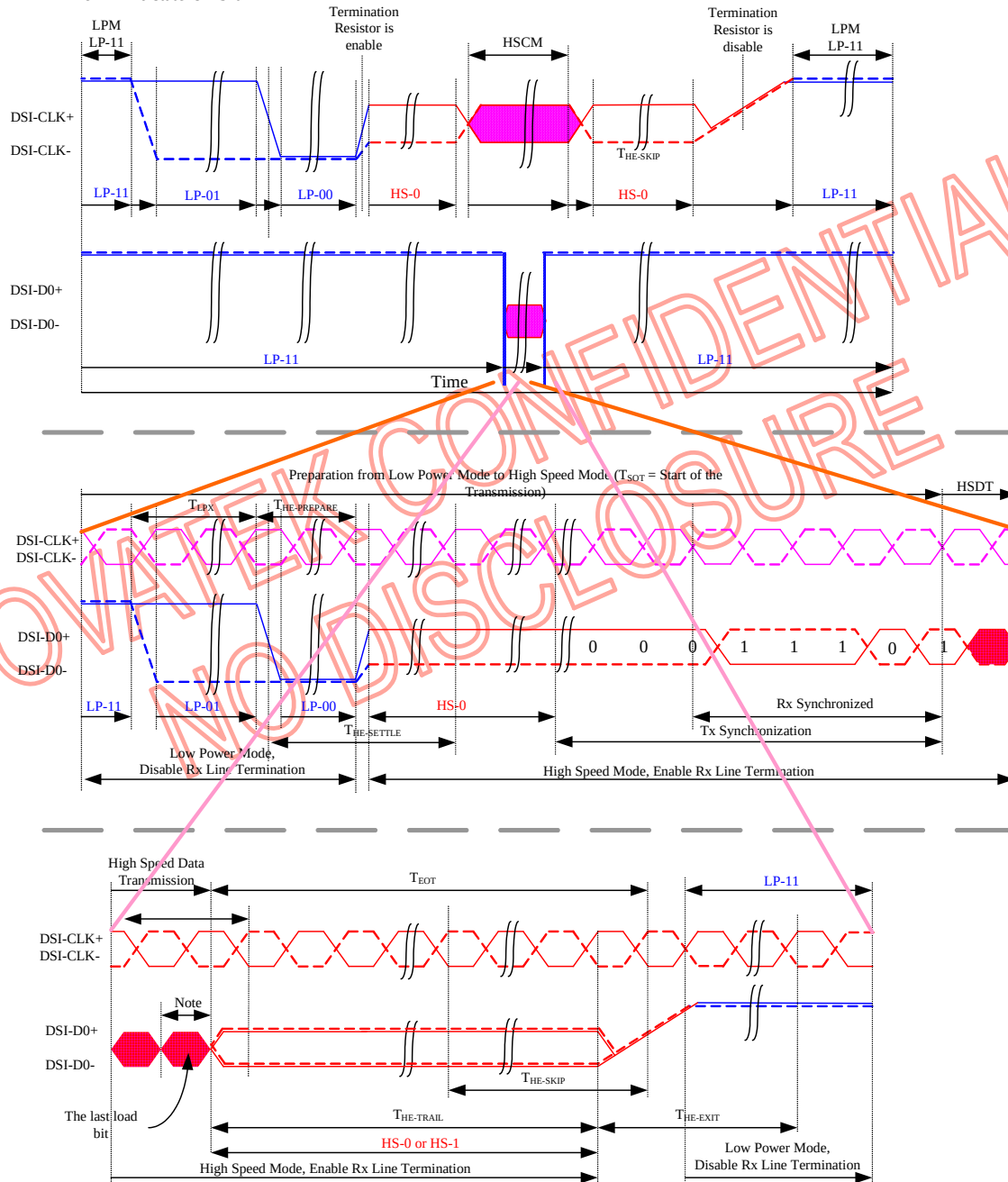


Mode Change from LPM to HSCM on the Flow Chart

The high speed clock (DSI-CLK+/-) is started before high speed data is sent via DSI-Dn+/- lanes. The high speed clock continues clocking after the high speed data sending has been stopped.

The burst of the high speed clock consists of :

- o Even number of transitions
- o Start state is HS-0
- o End state is HS-0



Note :

If the last load bit is HS-0, the transmitter changes from HS-0 to HS-1.
If the last load bit is HS-1, the transmitter changes from HS-1 to HS-0.

DSI-CLK+, DSI-D0+ - - - - -
DSI-CLK-, DSI-D0- ————

High Speed Clock Burst

5.3.2.2.3 DSI-DATA LANES

5.3.2.2.3.1 GENERAL

DSI-D0+/- Data Lanes can be driven in different modes which are:

- Escape Mode
- High-Speed Data Transmission
- Bus Turnaround Request

These modes and their entering codes are defined on the following table.

Mode	Entering Mode Sequence	Leaving Mode Sequence
Escape Mode	LP-11=>LP-10=>LP-00=>LP-01=>LP-00	LP-00=>LP-10=>LP-11(Mark-1)
High-Speed Data Transmission	LP-11=>LP-01=>LP-00=>HS-0	(HS-0 or HS-1) =>LP-11
Bus Turnaround Request	LP-11=>LP-10=>LP-00=>LP-10=>LP-00	High-Z

Notes:

1. DSI-D0+/- data lanes are used.
2. More information on section "Bus Turnaround (BTA)"

5.3.2.2.3.2 ESCAPE MODES

Data lanes (DSI-D0+/-) can be used in different Escape Modes when data lanes are in Low Power (LP) mode.

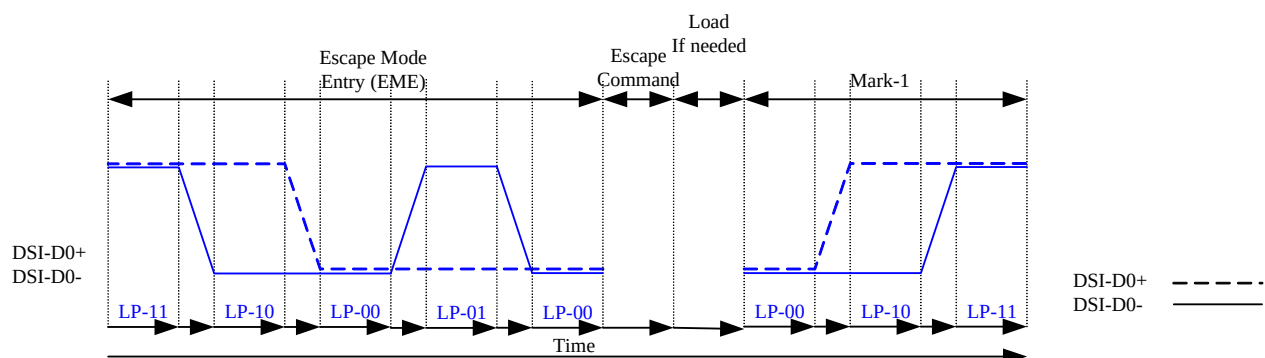
These Escape Modes are used to:

- Send "Low-Power Data Transmission" (LPDT) e.g. from the MCU to the display module
- Drive data lanes to "Ultra-Low Power State" (ULPS)
- Indicate "Remote Application Reset" (RAR), which is reset the display module
- Indicate "Tearing Effect" (TEE), which is used for a TE trigger event from the display module to the MCU
- Indicate "Acknowledge" (ACK), which is used for a non-error event from the display module to the MCU

The basic sequence of the Escape Mode is as follow

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Escape Command (EC), which is coded, when one of the data lanes is changing from low-to-high-to-low then this changed data lane is presenting a value of the current data bit (DSI-D0+ = 1, DSI-D0- = 0) e.g. when DSI-D0- is changing from low-to-high-to-low, the receiver is latching a data bit, which value is logical 0. The receiver is using this low-to-high-to-low transition for its internal clock.
- A load if it is needed
- Exit Escape (Mark-1) LP-00 =>LP-10 =>LP-11
- End: LP-11

This basic construction is illustrated below:



General Escape Mode Sequence

The number of the different Escape Commands (EC) is eight. These eight different escape commands (EC) can be divided 2 different groups: Mode or Trigger. The MCU is informing to the display module that it is controlling data lanes (DSI-D0+/-) with the mode e.g. The MCU can inform to the display module that it can put data lanes in the low power mode. The MCU is waiting from the display module event information, which has been set by the MCU, with the trigger e.g. when the display module reaches a new V-synch, the display module sent to the MCU a TE trigger (TEE), if the MCU has been requested it.

Escape commands are defined on the next table.

This basic construction is illustrated below:

Escape Command	Command Type Mode/Trigger	Entry Command Pattern (First Bit => Last Bit Transmitted)	Dn	D0
Low-Power Data Transmission	Mode	1110 0001 _{bin}	-	X
Ultra-Low Power Mode	Mode	0001 1110 _{bin}	X	X
Underfined-1, Note 1	Mode	1001 1111 _{bin}	-	-
Underfined-2, Note 1	Mode	1101 1110 _{bin}	-	-
Remote Application Reset	Trigger	0110 0010 _{bin}	-	X
Tearing Effect	Trigger	0101 1101 _{bin}	-	X
Acknowledge	Trigger	0010 0001 _{bin}	-	X
Unknow-5, Note 1	Trigger	1010 0000 _{bin}	-	-

Notes:

1. This Escape command support has not been implemented on the display module.
2. n=1.
3. "X"=Supported
4. "-"=Not Supported

Low-Power Data Transmission (LPDT)

The MCU can send data to the display module in Low-Power Data Transmission (LPDT) mode when data lanes are entering in Escape Mode and Low-Power Data Transmission (LPDT) command has been sent to the display module. The display module is also using the same sequence when it is sending data to the MCU.

The Low Power Data Transmission (LPDT) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Low-Power Data Transmission (LPDT) command in Escape Mode: 1110 0001 (First to Last bit)
- Load (Data):
 - One or more bytes (8 bit)
 - Data lanes are in pause mode when data lanes are stopped (Both lanes are low) between bytes
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

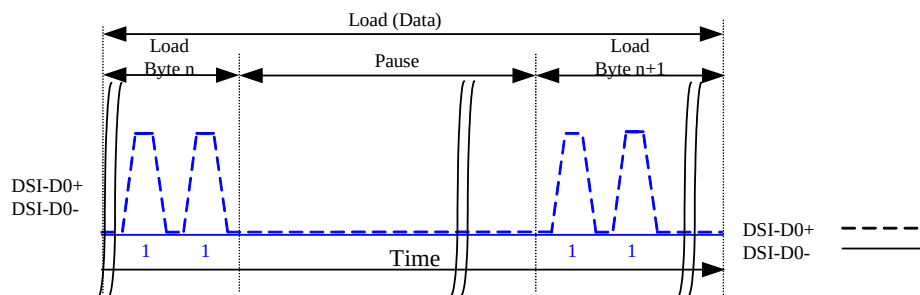
This sequence is illustrated for reference purposes below:



Note : Load (Data) is presenting that the first bit is logical 1 in this example

DSI-D0+ ---
DSI-D0- ———

Low-Power Data Transmission (LPDT)



Pause (Example)

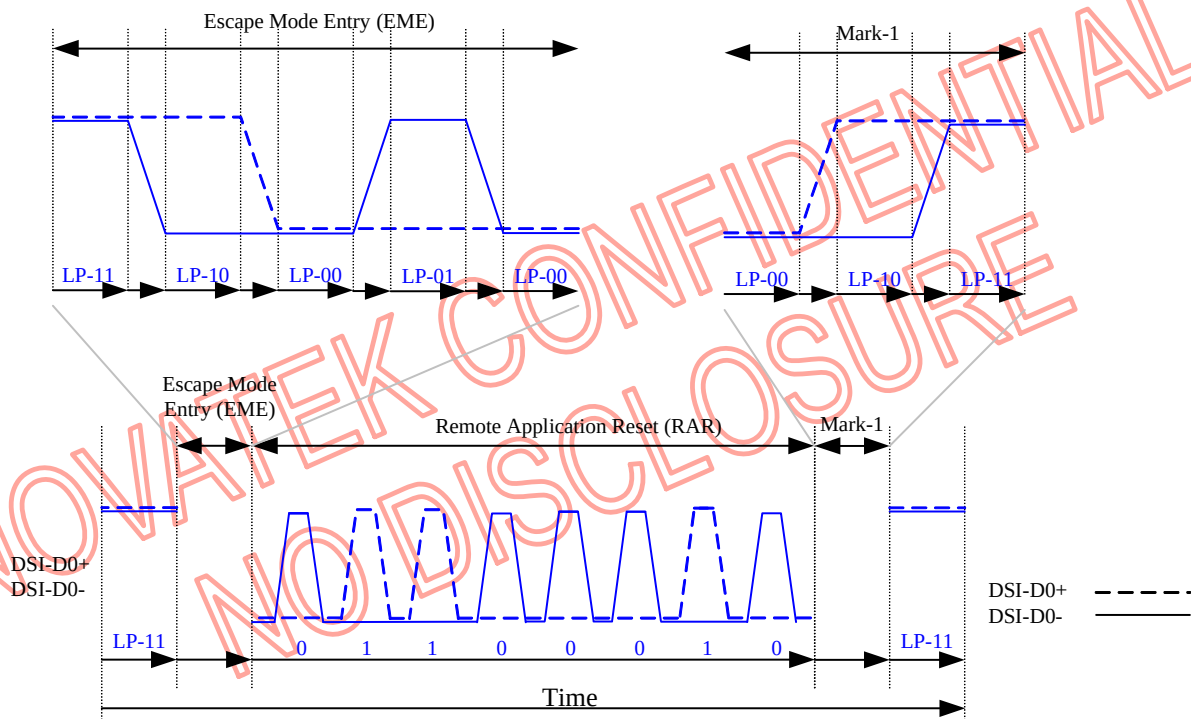
Remote Application Reset (RAR)

The MCU can inform to the display module that it should be reset in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode.

The Remote Application Reset (RAR) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Remote Application Reset (RAR) command in Escape Mode: 0110 0010 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Remote Application Reset (RAR)

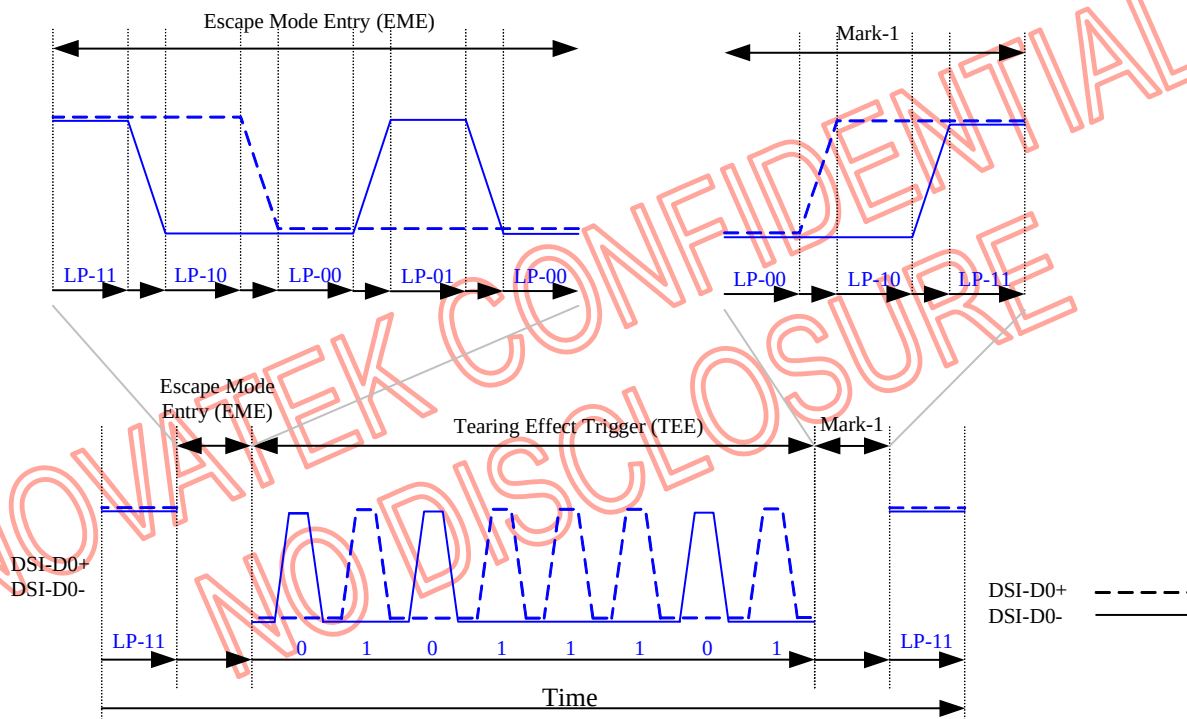
Tearing Effect (TEE)

The display module can inform to the MCU when a tearing effect event (New V-synch) has been happen on the display module by Tearing Effect (TEE).

The Tearing Effect (TEE) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Tearing Effect (TEE) trigger in Escape Mode: 0101 1101 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Tearing Effect (TEE)

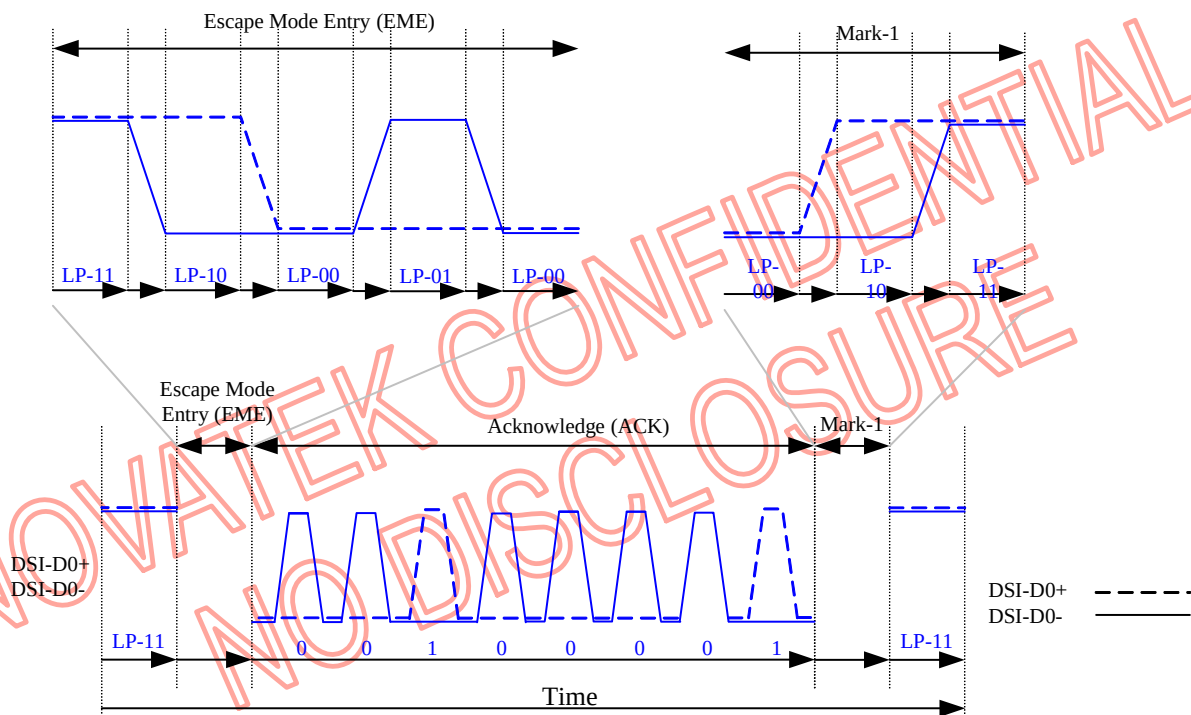
Acknowledge (ACK)

The display module can inform to the MCU when an error has not recognized on it by Acknowledge (ACK).

The Acknowledge (ACK) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Acknowledge (ACK) command in Escape Mode: 0010 0001 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Acknowledge (ACK)

5.3.2.2.3.3 HIGH SPEED DATA TRANSMISSION (HSDT)

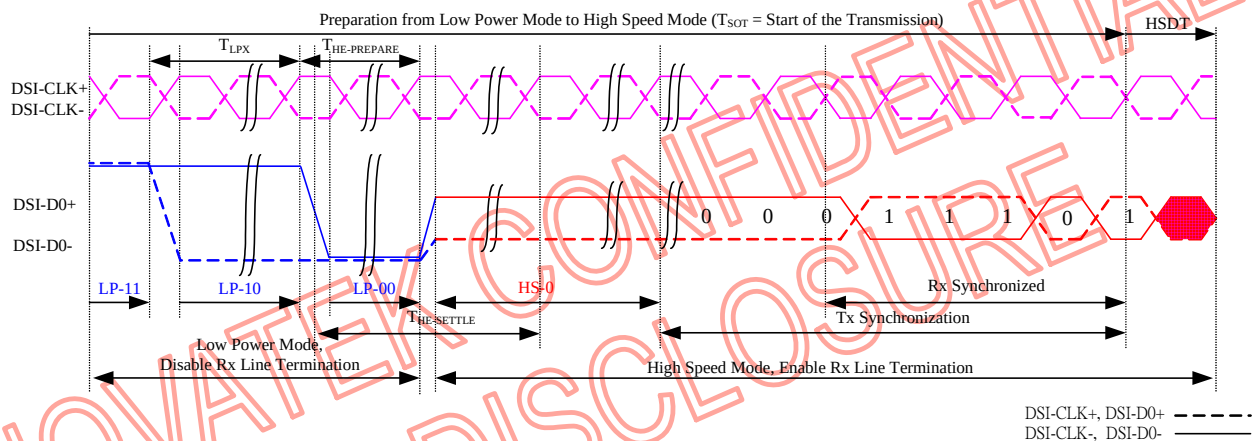
Entering High-Speed Data Transmission (T_{SOT} of HSDT)

The display module is entering High-Speed Data Transmission (HSDT) when Clock lanes DSI-CLK+/- have already been entered in the High-Speed Clock Mode (HSCM) by the MCU. See more information on chapter "5.3.2.2.2.3 High-Speed Clock Mode (HSCM)".

Data lanes of the display module are entering (T_{SOT}) in the High-Speed Data Transmission (HSDT) as follows

- Start: LP-11
- HS-Request: LP-01
- HS-Settle: LP-00 => HS-0 (Rx: Lane Termination Enable)
- Rx Synchronization: 011101 (Tx (= MCU) Synchronization: 0001 1101)
- End: High-Speed Data Transmission (HSDT) – Ready to receive High-Speed Data Load

This same entering High-Speed Data Transmission (T_{SOT} of HSDT) sequence is illustrated below



Entering High-Speed Data Transmission (T_{SOT} of HSDT)

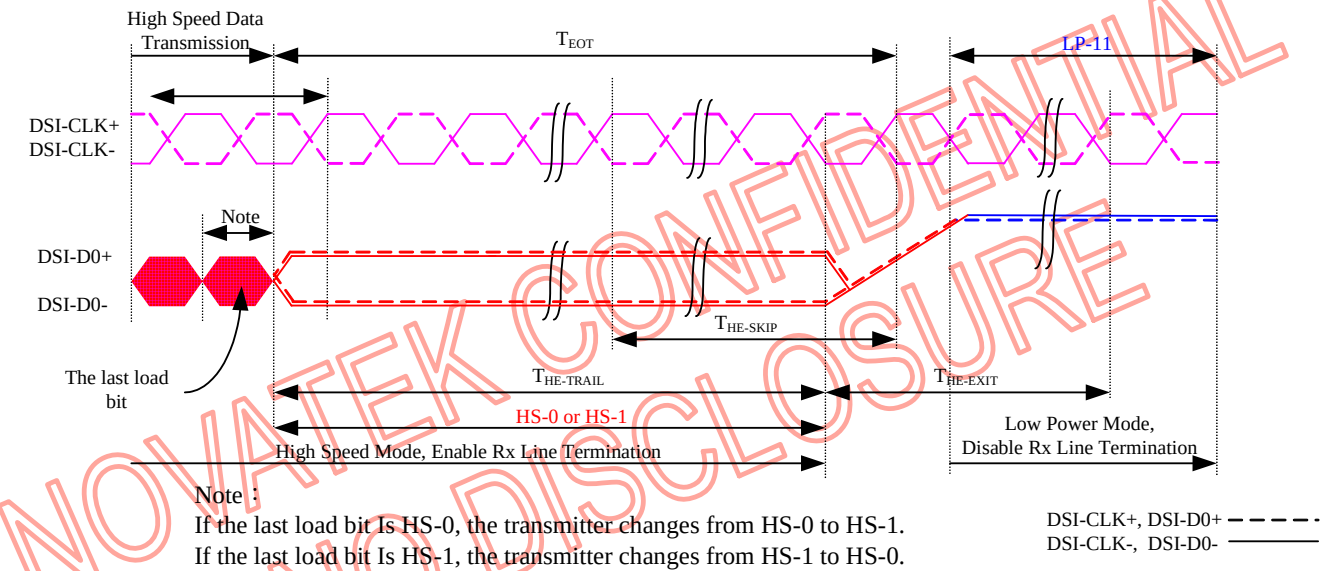
Leaving High-Speed Data Transmission (T_{EOT} of HSDT)

The display module is leaving the High-Speed Data Transmission (T_{EOT} of HSDT) when Clock lanes DSI-CLK+/- are in the High-Speed Clock Mode (HSCM) by the MCU and this HSCM is kept until data lanes are in LP-11 mode. See more information on chapter "5.3.2.2.3 High-Speed Clock Mode (HSCM)".

Data lanes of the display module are leaving from the High-Speed Data Transmission (T_{EOT} of HSDT) as follows

- Start: High-Speed Data Transmission (HSDT)
- Stops High-Speed Data Transmission
 - MCU changes to HS-1, if the last load bit is HS-0
 - MCU changes to HS-0, if the last load bit is HS-1
- End: LP-11 (Rx: Lane Termination Disable)

This same leaving High-Speed Data Transmission (T_{EOT} of HSDT) sequence is illustrated below

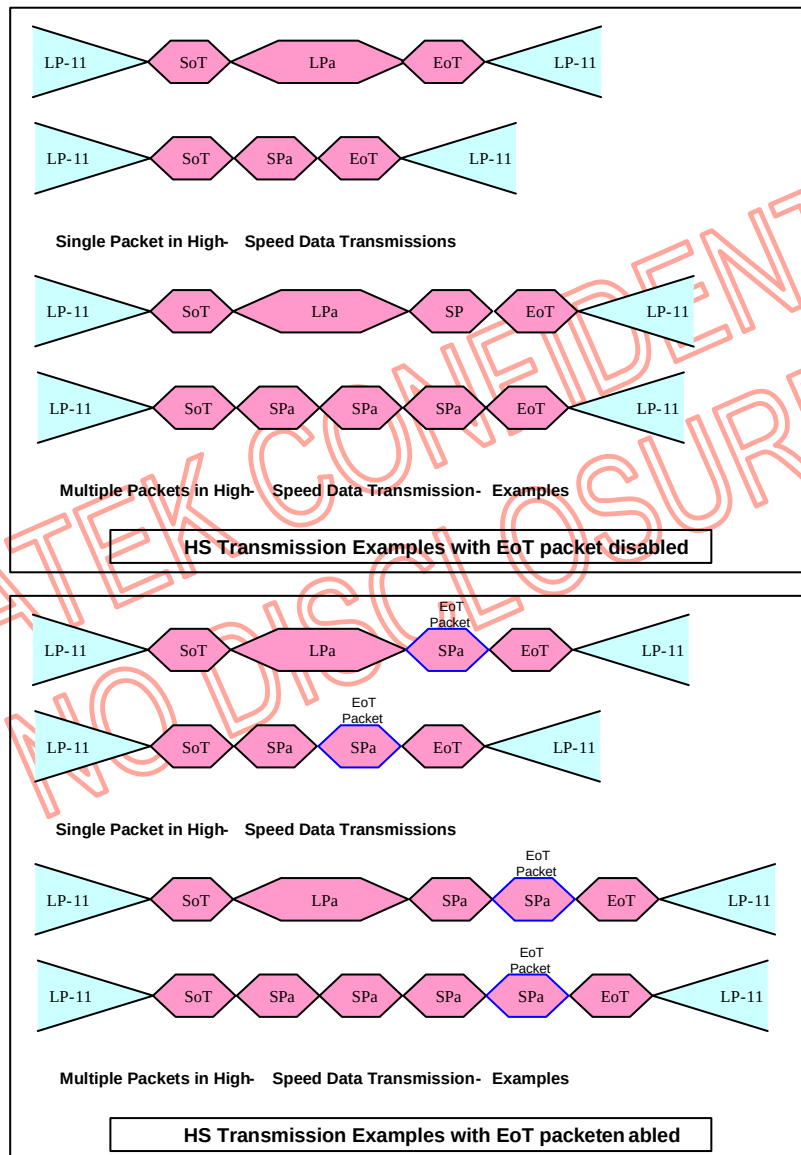


Leaving High-Speed Data Transmission (T_{EOT} of HSDT)

Burst of the High-Speed Data Transmission (HSDT)

The burst of the high-speed data transmission (HSDT) can consist of one data packet or several data packets. These data packets can be Long (LPa) or Short (SPa) packets. These packets are defined on chapter “5.1.9.2.3.1 Short Packet (SPa) and Long Packet (LPa) Structures”.

These different burst of the High-Speed Data Transmission (HSDT) cases are illustrated for reference purposes below.



Abbreviation	Explanation
EoT	End of the Transmission
LPa	Long Packet
LP-11	Low Power Mode, Data lanes are '1's (Stop Mode)
SPa	Short Packet
SoT	Start of the Transmission

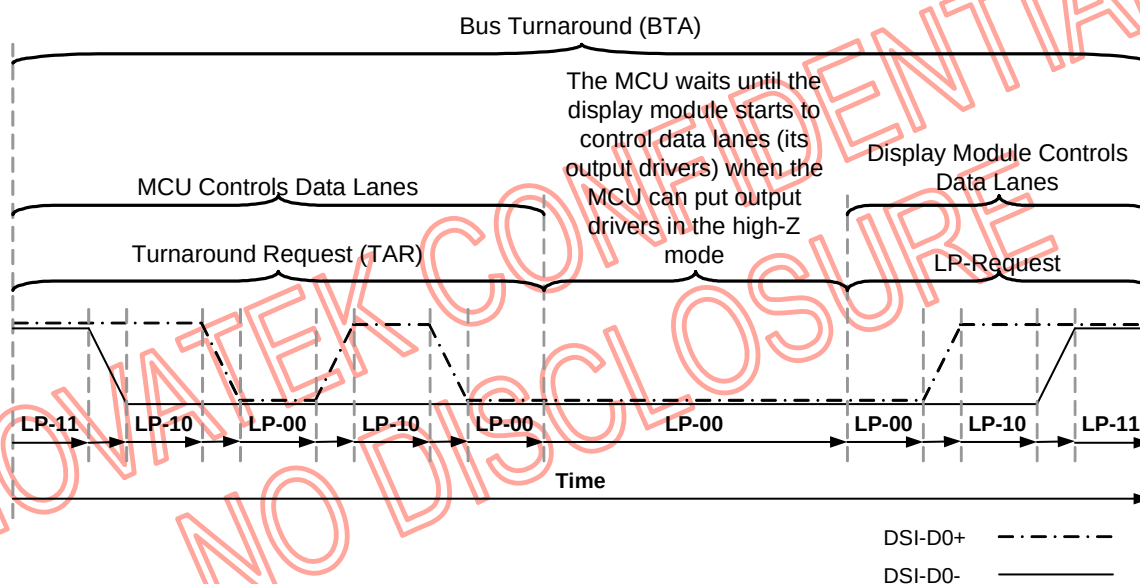
Bus Turnaround (BTA)

The MCU or display module, which is controlling DSI-D0+/- Data Lanes, can start a bus turnaround procedure when it wants information from a receiver, which can be the MCU or display module.

The MCU or display module are using the same sequence when this bus turnaround procedure is used. This sequence is described for reference purposes, when the MCU wants to do the bus turnaround procedure to the display module, as follow.

- Start (MCU):LP-11
- Turnaround Request (MCU): LP-11 → LP-10 → LP-00 → LP-10 → LP-00
- The MCU wait until the display module is starting to control DSI-D0+/- data lanes and the MCU stop to control DSI-D0+/- data lanes (=High-Z)
- The display module changes to the stop mode: LP-00 → LP-10 → LP-11

The same bus turnaround procedure (From the MCU to the display module) is illustrated below.



Bus Turnaround Procedure

MCU and the display module terms are switched on above figure, if the Bus Turnaround (BTA) is from the display module to the MCU..

5.3.2.3 PACKET LEVEL COMMUNICATION

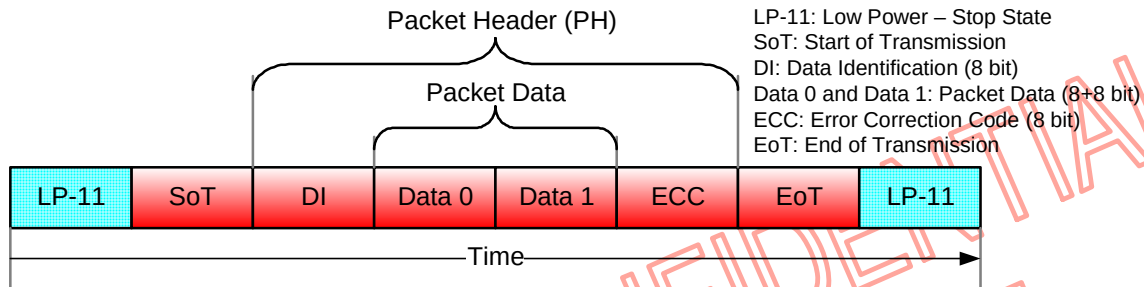
5.3.2.3.1 SHORT PACKET (SPa) AND LONE PACKET (LPa) STRUCTURE

Short Packet (SPa) and Long Packet (LPa) are always used when data transmission is done in Low Power Data Transmission (LPDT) or High-Speed Data Transmission (HSDT) modes.

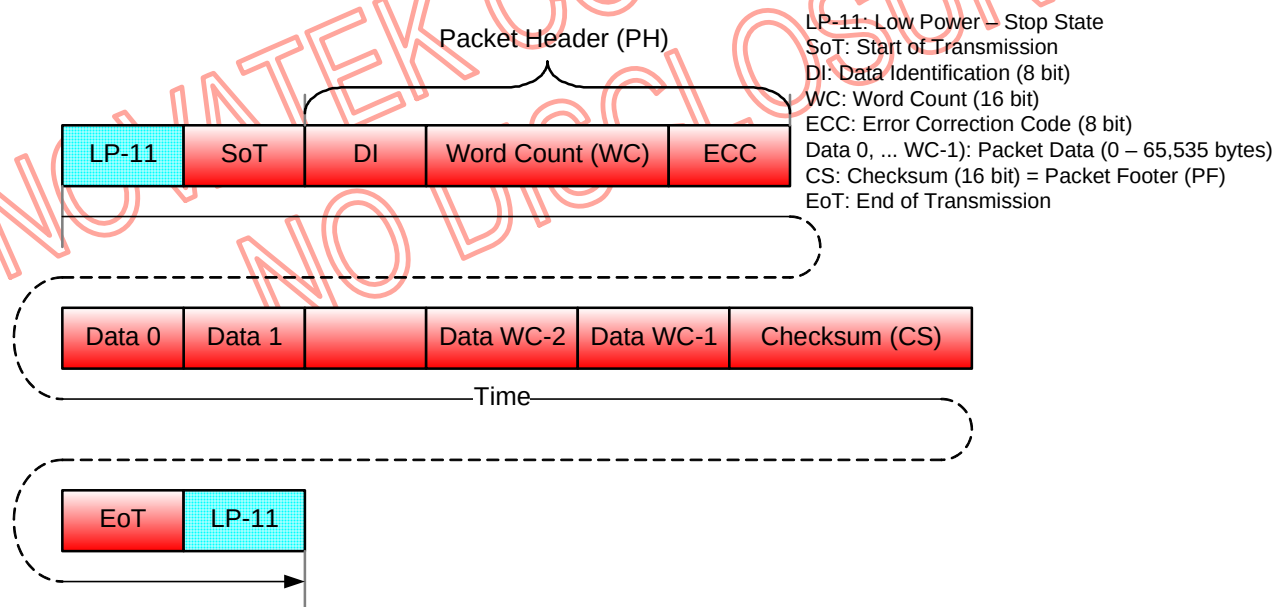
The lengths of the packets are

- Short Packet (SPa): 4 bytes
- Long Packet (LPa): From 6 to 65,541 bytes

The type (SPa or LPa) of the packet can be recognized from their package headers (PH).



Short Packet (SPa) Structure



Long Packet (LPa) Structure

Note:

Short Packet (SPa) Structure and Long Packet (LPa) Structure are presenting a single packet sending (= Includes LP-11, SoT and EoT for each packet sendings).

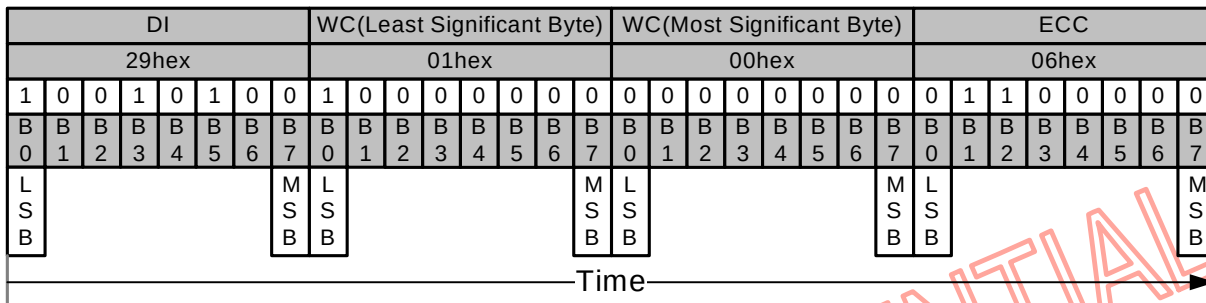
The other possibility is that there is not needed SoT, EoT and LP-11 between packets if packets have sent in multiple packet format e.g.

- * LP-11 =>SoT =>SPa =>LPa =>SPa =>SPa =>EoT =>LP-11
- * LP-11 =>SoT =>SPa =>SPa =>SPa =>EoT =>LP-11
- * LP-11 =>SoT =>LPa =>LPa =>LPa =>EoT =>LP-11

5.3.2.3.1.1 BIT ORDER OF THE BYTE ON PACKETS

The bit order of the byte, what is used on packets, is that the Least Significant Bit (LSB) of the byte is sent in the first and the Most Significant Bit (MSB) of the byte is sent in the last.

This same order is illustrated for reference purposes below.

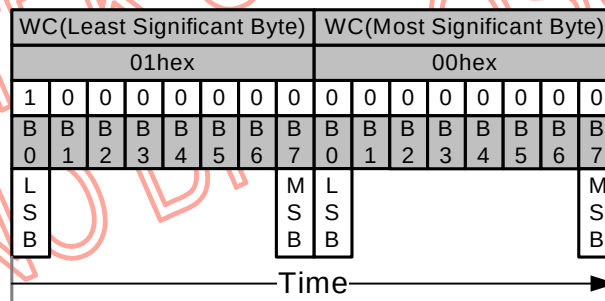


Bit Order of the Byte on Packets

5.3.2.3.1.2 BIT ORDER OF THE MULTIPLE BYTE INFORMATION ON PACKETS

Byte order of the multiple bytes information, what is used on packets, is that the Least Significant (LS) Byte of the information is sent in the first and the Most Significant (MS) Byte of the information is sent in the last e.g. Word Count (WC) consists of 2 bytes (16 bits) when the LS byte is sent in the first and the MS byte is sent in the last.

This same order is illustrated for reference purposes below.



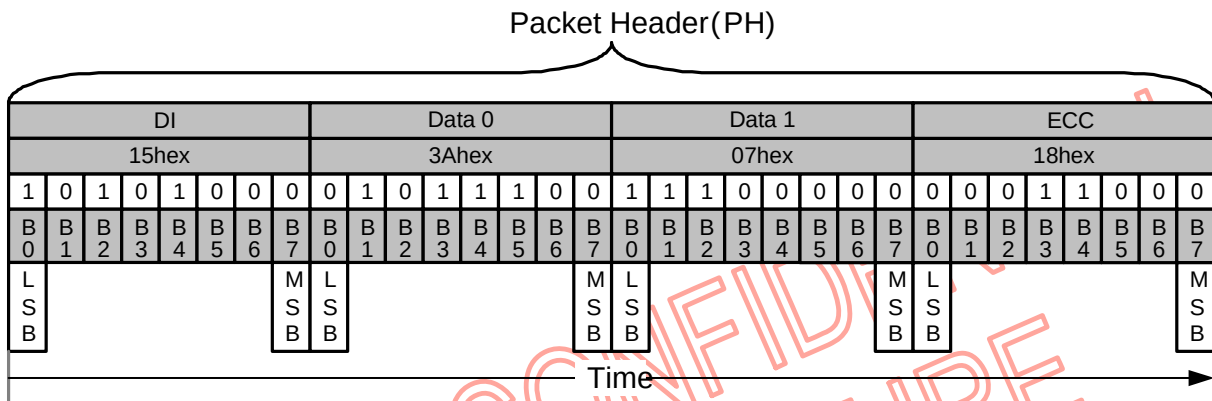
Byte Order of the Multiple Byte on Packets

5.3.2.3.1.3 PACKET HEADER (PH)

The packet header is always consisting of 4 bytes. The content of these 4 bytes are different if it is used to Short Packet (SPa) or Long Packet (LPa).

Short Packet (SPa):

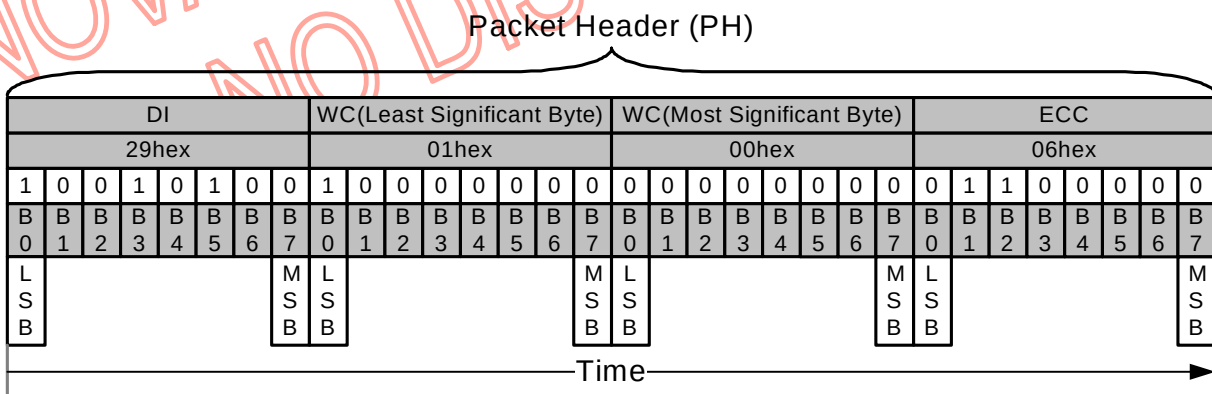
- 1st byte: Data Identification (DI) => Identification that this is Short Packet (SPa)
- 2nd and 3rd bytes: Packet Data (PD), Data 0 and 1
- 4th byte: Error Correction Code (ECC)



Packet Header (PH) on Short Packet (SPa)

Long Packet (LPa):

- 1st byte: Data Identification (DI) => Identification that this is Long Packet (LPa)
- 2nd and 3rd bytes: Word Count (WC)
- 4th byte: Error Correction Code (ECC)



Packet Header (PH) on Long Packet (LPa)